MUHS final-year BPTh (IV BPTh), 2012 syllabus

Neuro Physiotherapy

Short notes for revision: one sheet per topic, in syllabus order, with the page of the full notes for each.



Sheets	45 sheets: every topic in the syllabus, then the exam skills
Built from	The full notes and the 20 MUHS question papers, Summer 2016 to Summer 2026 (code 51412)
Content checked	20 October 2026
Online version	https://hrushikesh-shinde7.github.io/bpt-study-material/neuro/

Educational notes for exam preparation. Not medical advice. Sources for every fact are listed in the full notes.

How to use the short notes

These sheets are for revision, after you have read a topic once in the full notes. Each topic has its own sheet of one to three pages, in the same order as the full notes.

1. **Read the yellow line first.** It says how the topic is asked in the MUHS paper.
2. **Cover a box and recall it,** then check. Bold words and numbers are the ones to write in the answer.
3. **Tick a square at the top right of a sheet** each time you revise it. Aim for day 1, day 7 and day 21.
4. **Need more?** Each sheet gives the page of the same topic in the full notes.

What the colours mean

Grey
Definitions, types, medical and surgical management

Orange
Disease process, clinical features and the problems they cause

Blue
Investigations, scales and normal values

Green
Physiotherapy: assessment, ICF, goals, treatment and doses

Amber
Precautions, contraindications, stop criteria and complications

Yellow
Exam plan: answer structure, what to draw, keywords

The paper (20 papers, 2016–2026): 80 marks in 3 hours. Short answers: any 5 of 6 at 3 marks and any 5 of 6 at 7 marks. Long answers: two questions, each any 1 of 2 at 15 marks. Almost every long case is a patient with a neurological condition; most ask for ICF, a functional diagnosis or impairments, goals and management.

Contents

Part 1: Application of the ICF model

1 Application of the ICF model in neurological conditions 3

Part 2: Theoretical basis of motor control and learning

2 Motor control: theories behind neurological physiotherapy 4

Part 3: Plasticity and recovery, strength training and skill acquisition

3 Neuroplasticity, recovery, motor learning and skill acquisition 5

Part 4: Quality of life scales and independence measures

4 Quality of life, independence and other outcome measures 6

Part G: Assessment for every condition

G Neurological assessment and investigations 7
G Cognition, perception, apraxia and aphasia 9
G Movement dysfunction in daily tasks: bed mobility to reaching 10

Part 5A: Physiotherapy management: adult conditions

5A(a) Stroke 11
5A(b) Acquired brain injury: trauma and space-occupying lesions 13
5A(c) Spinal cord disorders: traumatic and non-traumatic 15
5A(d) Peripheral nerve injuries and the brachial plexus 17
5A(d) Peripheral neuropathies and nerve root lesions 19
5A(e) Vestibular disorders 21
5A(f) Facial (VII) nerve palsy and other cranial nerves 22
5A(g) Multiple sclerosis 23
5A(g) Guillain–Barré syndrome 25
5A(h) Cerebellar disease and ataxia 27
5A(i) Parkinson's disease and other extrapyramidal disorders 28
5A(j) Anterior horn cell diseases: MND, SMA and poliomyelitis 30
5A(k) Myopathies and myasthenia gravis 31
5A(l) Disorders of the autonomic nervous system 32
5A(m) Psychosomatic pain and paralysis 33

Part TP: Physiotherapy management: the adult treatment programme

TP1 Electrotherapy in neurological rehabilitation 34
TP2 Neurotherapeutic approaches: Brunnstrom, Rood, Bobath, MRP, PNF and CIMT 35
TP2 Managing tone: spasticity, rigidity and hypotonia 37
TP2 Balance and coordination training 38
TP3 Tools and adaptive equipment for neuro-rehabilitation 39
TP4 Transfers, functional re-education, posture and gait training 40
TP5 Bladder and bowel training 41
TP6 A philosophy of caring 42
TP7 Orthoses and splints 43
TP8 Lifting, wheelchairs and adaptive devices 44
TP9 Ergonomic advice and caregiver education 45

Part 5B: Physiotherapy management: paediatric conditions

5B Developmental neurology: milestones, reflexes and reactions 46
5B(1) Cerebral palsy 47
5B(2) Down's syndrome 49
5B(3) Neural tube defects: spina bifida and hydrocephalus 50
5B(4) Obstetric brachial plexus injury 52
5B(5) Infections of the nervous system 53
5B(6) Post-polio residual paralysis 54
5B(7) Duchenne muscular dystrophy and other myopathies 55
5B(8) Spinal muscular atrophy and hereditary neuropathies 57
5B(9) Paediatric extrapyramidal disorders 58

Exam skills

Answer writing for the Neuro paper 59
The Neuro practical exam: long case, short case, spots and journal 60
Abbreviations 61

Application of the ICF model in neurological conditions

ICF is asked in almost every long case; domains of ICF as 7 marks (S19). Full notes p. 4

Asked: 7 marks: domains of ICF (S19). Inside almost every long case (2016–2026): ICF with clinical reasoning, functional diagnosis, or impairments. **3 marks:** not asked alone; expect capacity vs performance, facilitators and barriers.

The ICF model

ICF (International Classification of Functioning, Disability and Health; **WHO, 2001**): health and disability as an **interaction** between a health condition and contextual factors. **Biopsychosocial:** joins the medical model (disease → disability) and the social model (disability created by society and environment). **Functioning** (body functions, activities, participation) vs **disability** (impairments, activity limitations, participation restrictions). **Principles:** universality, parity, neutrality, environmental influence.

Components and domains (7m S19)

Body functions (b)	Mental; sensory, pain; voice, speech; cardiovascular, haematological, immunological, respiratory; digestive, metabolic, endocrine; genitourinary, reproductive; **neuromusculoskeletal and movement**; skin (power b730, tone b735, gait pattern b770)
Body structures (s)	Matching 8: nervous system; eye, ear; voice, speech; cardiovascular, respiratory; digestive; genitourinary; structures related to movement; skin (brain, cord, nerve)
Activities and participation (d)	Learning; general tasks; communication; **mobility**; self-care; domestic life; relationships; major life areas (work); community life (transferring d420, walking d450)
Environmental (e)	Products, technology; natural and built; support, relationships; attitudes; services, policies (aids e120, family e310)
Personal	**Not classified:** age, sex, education, coping, habits, motivation

Qualifiers: 0 no, 1 mild, 2 moderate, 3 severe, 4 complete (8 not specified, 9 not applicable); **b730.3** = severe weakness; **e120.2** moderate barrier, **e120+2** moderate facilitator.

Relationships, factors, capacity

Two-way: spasticity limits walking; not walking → weakness, contracture (secondary impairments); lost job → low mood → fatigue; ramp helps, stairs hinder; motivation changes practice. **Use:** treat at several levels at once; predict secondary problems.

Facilitators: walking aid, orthosis, rails, western commode; ground floor, lift, ramps; joint family, supportive employer; schemes, disability certificate; young, motivated. **Barriers:** Indian-style toilet; stairs, uneven roads; living alone; overprotection, stigma; cost, distance, lost income; older, depressed.

Capacity: best in a **standard environment**, no help (clinic). **Performance:** what is **actually done** at home with real aids. **Capacity above performance → environment is a barrier.** Walks 50 m in clinic, never leaves first-floor flat → stair training and a rail.

Goals and documentation

From what matters to patient and family; **SMART**; **short-term (2–6 weeks):** impairments, basic activities; **long-term (months):** activities, participation; ICF reason for each. "Improve balance" → **Berg 38 to 46 in 6 weeks** to cook safely; "improve walking" → **200 m outdoors with a stick in under 4 min within 8 weeks** to reach the bus stop. **Document:** each level with measures, functional diagnosis, goals, dates; **ICF Core Sets** (stroke, SCI, MS, CP).

Red flags: stop, refer

Face droop, speech loss → **stroke**; vomiting, falling GCS, unequal pupils → **raised ICP**; saddle numbness, new bladder loss → **cauda equina**; spreading weakness, weak cough → GBS, myasthenic crisis; headache, high BP in SCI T6 or above → **autonomic dysreflexia**; calf swelling → DVT; neck stiffness, fever → meningitis; night pain, weight loss → tumour.

Exam plan

Domains (7): definition (1) → model with two-way arrows (1) → components with chapters (4) → qualifiers, b735.2 (1). **Long case:** draw model → table: component | findings | reasoning → one-line **functional diagnosis** → goals.

Draw: ICF model. **Underline:** biopsychosocial, qualifiers, capacity and performance, facilitators and barriers, SMART.

Motor control: theories behind neurological physiotherapy

Theories of motor control as 7 marks in W19, S21 and S23. Full notes p. 11

Asked: 7 marks: theories of motor control, any two (W19, S21, S23). No 3-mark or long-case questions; expect dynamic systems theory or limitations of reflex theory as 3 marks.

Definition

Motor control: the ability to regulate or direct the mechanisms essential to movement. Movement emerges from **the individual** (perception, cognition, action systems), **the task** and **the environment** acting together; change any one and the movement changes. The theory chosen decides what the therapist treats: reflexes, tone, or the person practising a task in a real setting.

The six theories (7m W19, S21, S23)

- **Reflex (Sherrington):** reflexes are the building blocks; complex movement = **chain of reflexes** triggered by sensory input. *Limit:* cannot explain movement without sensory input (deafferented animals still move), fast sequences, new movements. *Use:* assess reflexes; sensory stimuli to bring out or inhibit movement (Rood).
- **Hierarchical (Hughlings Jackson, Magnus): top-down**: cortex → brainstem → spinal cord; higher lesion **releases primitive reflexes**; development in a fixed sequence. *Limit:* lower levels also control higher ones (spinal reflexes can dominate); not every adult movement goes through a hierarchy. *Use:* Bobath, Brunnstrom, Rood: inhibit abnormal reflexes, follow developmental sequences.
- **Motor programming:** stored central program or **central pattern generator** moves without sensory input (spinal stepping generator). *Limit:* cannot alone explain adaptation to body and environment. *Use:* retrain whole patterns, not single muscles; **treadmill stepping** uses spinal pattern generators.
- **Systems (Bernstein):** body as a mechanical system with many **degrees of freedom**; many systems work together, controlled through **synergies**, with forces from body and environment. *Limit:* less detail on how nervous system and environment interact. *Use:* assess and treat all systems (strength, range, sensation, perception); train functional synergies.
- **Dynamic systems:** movement **self-organises**; preferred patterns are stable **attractor states**; a **control parameter** (speed) shifts behaviour to a new pattern (walk to run). *Limit:* can underplay the nervous system. *Use:* find the parameter that shifts an abnormal pattern (speed, support, task demand); practise at transition points.
- **Ecological (Gibson):** perception and action linked; people perceive what the environment allows them to do (**affordances**) and act to achieve goals. *Limit:* less on the nervous system's organisation. *Use:* real tasks in varied real environments; teach reading the environment (kerbs, crowds).

Theory to approach

Theory	Approach and what the therapist does
Reflex, hierarchical	**Neurofacilitation**: Rood, Bobath, Brunnstrom, PNF; changes tone and reflexes with handling and sensory input; developmental sequences
Systems, dynamic, ecological; motor learning	**Task-oriented training**, MRP, CIMT; practises real tasks, adapts task and environment, treats impairments that limit the task

Current: practice draws on several theories at once; the systems view (person, task, environment) is the basis of task-oriented training. Hemiplegia: hierarchical view → inhibit reflexes and tone; systems view → treat all impairments, many repetitions of sit to stand, reaching, walking, varied task and setting, feedback. Evidence favours task-oriented practice, borrowing older techniques when they help. See approaches (p. 35), motor learning (p. 5).

Exam plan (any two, 7)

1. **Introduction (1):** define motor control; individual, task, environment.
2. **Theory 1 (3):** proponent, key idea, limitation, clinical implication.
3. **Theory 2 (3):** same four points. Pick a contrasting pair: **hierarchical and systems**. Close: current practice combines theories.

Draw: individual, task and environment circles; hierarchy cortex over brainstem over spinal cord. **Underline:** reflex chaining, hierarchy, central pattern generator, degrees of freedom, self-organisation, attractor state, affordance, task-oriented training.

Neuroplasticity, recovery, motor learning and skill acquisition

Motor learning stages and feedback in 6 papers; neuroplasticity in 3. Full notes p. 15

Asked: 7 marks: role and types of feedback in neuro-rehabilitation (S25). **3 marks:** define neuroplasticity (S16); factors that help motor learning (W17); brain changes in neuroplasticity (S18); stages of motor learning (S21, W24, W25); intrinsic and extrinsic feedback (S22); neuroplasticity and synaptogenesis (S26).

Neuroplasticity (3m S16, S18, S26)

Neuroplasticity: the ability of the nervous system to **change its structure and function** in response to experience, learning, training or injury. In the intact brain (learning) and after a lesion (reorganisation); **adaptive** or **maladaptive** (spasticity, pain, not using the weak arm).

Brain changes (S18): synapse: long-term potentiation and depression, unmasking silent synapses; **neuron:** synaptogenesis, new dendrites and spines, **axonal sprouting** into areas that lost input; **white matter:** myelination changes; **cortical maps:** trained part's area enlarges, areas next to a lesion take over, other hemisphere recruited.

Synaptogenesis (S26): formation of new synapses; fastest in development, continues with learning; after injury intact neurons sprout and form new synapses (**reactive synaptogenesis**), which therapy can shape.

Kleim and Jones (2008): use it or lose it; use it and improve it; **specificity**; **repetition**; **intensity**; time; **salience**; age; transference; interference (a compensation can block learning).

Recovery and muscle adaptation

- **Spontaneous:** swelling settles, penumbra reperfused, **diaschisis** resolves (distant areas switched back on); first weeks to months. **Experience-dependent:** practice-driven, months to years.
- **Recovery** = same movement as before; **compensation** = new way (trunk lean to reach); can block true recovery (Levin 2009). **Learned non-use**: brain area shrinks; CIMT reverses it. Enriched environment helps; ward inactivity slows.
- **Muscle:** motor units lost, fibres smaller, types change; shortened muscles lose sarcomeres → stiffness, contracture; non-neural stiffness adds to tone: treat length and strength.

Strength and conditioning

Progressive resistance, functional strengthening (sit-to-stand, step-ups), power training, load to fatigue in a set; aerobic conditioning improves fitness and walking. Watch BP, fatigue, heat (MS), dysreflexia (SCI), overwork (GBS, DMD). **Classic** "no resisted exercise in spasticity" vs **current**: strength training **does not increase spasticity** (Ada 2006); weakness limits function most.

Motor learning (3m S21, W24, W25; W17)

Motor learning: processes linked to practice or experience leading to **relatively permanent** change in capability for skilled movement; shown by **retention and transfer**, not performance during practice.

Stage (Fitts, Posner)	Learner; therapist
Cognitive	What to do; many errors, slow, full attention; clear instruction, demonstration, frequent feedback, quiet setting
Associative	Refining how; fewer errors; less feedback, more practice, vary the task
Autonomous	Automatic, can dual-task; real-world varied practice, little feedback

Factors (W17): enough, **distributed**, **variable and random** practice (contextual interference), whole task; feedback faded; meaningful goals, **external focus** ("reach the cup"); task- and context-specific; demonstration, mental practice, sleep. Blocked practice helps early performance only.

Feedback (7m S25; 3m S22)

Intrinsic: own senses (foot hitting the ground, seeing the cup move). **Extrinsic (augmented):** therapist's words, mirror, video, **EMG biofeedback**, force platform. **KR** = outcome ("10 m in 12 s"); **KP** = pattern ("knee locked back in stance"). Concurrent or terminal; immediate or delayed. **Schedule:** frequent early, then **fade**; summary or bandwidth (too much → dependence). **Role:** corrects errors, motivates, reinforces, **replaces lost sensation** (vision for poor proprioception, EMG for weak dorsiflexor).

Exam plan

Feedback (7): define (1) → types: intrinsic/extrinsic, KR/KP, timing, schedules (3) → role (2) → examples, fade (1). **3 marks:** definition + three points, three stages, or three factors.

Draw: learning curve with three stages. **Underline:** synaptogenesis, axonal sprouting, use it or lose it, repetition and intensity, learned non-use, cognitive-associative-autonomous, KR and KP, retention and transfer.

Quality of life, independence and other outcome measures

GCS, Modified Ashworth, ASIA and Hoehn and Yahr as 3-mark questions. Full notes p. 21

Asked: 3 marks only: Glasgow Coma Scale (W16 with two conditions, S19, S20, S26 interpretation); Modified Ashworth Scale (S19, W21); ASIA scale (S19); Hoehn and Yahr (W20). Expect independence measures as 7 marks; a named measure strengthens every long case.

GCS (3m W16, S19, S20, S26)

Eyes (E 1–4)	4 spontaneous; 3 to sound; 2 to pressure; 1 none
Verbal (V 1–5)	5 oriented; 4 confused; 3 words; 2 sounds; 1 none
Motor (M 1–6)	6 obeys; 5 localises; 4 normal flexion; 3 abnormal flexion; 2 extension; 1 none

Total **3–15**: **severe 3–8**, moderate 9–12, mild 13–15. Report parts ("E2 V2 M4 = 8"); **motor score predicts outcome best**. Used in **head injury, stroke**, meningitis, overdose. See brain injury (p. 13).

MAS (3m S19, W21), ASIA (S19), Hoehn and Yahr (W20)

MAS: 0 no increase; **1** catch and release or minimal resistance at end of range; **1+** catch then minimal resistance through **less than half** the remaining range; **2** more marked through most of range, part moves easily; **3** considerable, passive movement difficult; **4** rigid. Patient relaxed; full range in about **1 s**; grade the first movement.

ASIA (AIS): A complete, no function in **S4–S5**; **B** sensory incomplete (sensation, no motor below level, including S4–S5); **C** motor incomplete, **less than half** key muscles below level grade 3; **D** half or more grade 3 or more; **E** normal in someone who had deficits.

Hoehn and Yahr: 1 one side; **2** both sides, balance normal; **3** both sides, some postural instability, independent; **4** severe disability, still walks or stands unaided; **5** wheelchair or bed unless helped. Modified adds 1.5 (one side + trunk), 2.5 (recovers on pull test).

Independence and quality of life

Barthel: 10 items (feeding, bathing, grooming, dressing, bowels, bladder, toilet use, transfers, level mobility, stairs); steps of 5; **0–100**. Shah: 0–20 total, 21–60 severe, 61–90 moderate, 91–99 slight dependence; 100 independent (not free of disability). Ceiling effect. **FIM:** 18 items (13 motor, 5 cognitive), each 1–7, **18–126**; includes cognition, needs training. **Modified Rankin** (stroke): 0 no symptoms; 1 no significant disability; 2 slight, looks after own affairs; 3 moderate, needs some help, walks unaided; 4 cannot walk or attend to bodily needs without help; 5 bedridden, incontinent, constant care; 6 dead.

QoL (the person's own view; asked, not observed): **SF-36** 36 questions, **8 domains** (physical functioning, physical role, pain, general health, vitality, social, emotional role, mental health), each 0–100; **EQ-5D** 5 dimensions + 0–100 scale; disease-specific: Stroke Impact Scale, PDQ-39.

Other syllabus scales

Berg	14 tasks × 0–4 = **56**; below ~45 fall risk; 41–56 independent, 21–40 walks with help, 0–20 wheelchair
TUG; Functional Reach	Stand, walk 3 m, turn, sit: **13.5 s** or more fall risk; reach under ~15 cm high risk
DGI; MMSE	8 tasks × 0–3 = 24, **19 or less** fall risk; 30 points, 23 or less suggests impairment (Hindi version for rural non-literate)
STREAM; Fugl-Meyer	30 items, 70 points; arm 66, leg 34 (research standard)
MDS-UPDRS; EDSS	4 parts, items 0–4 (part III motor exam); 0–10 in half steps, 6.0 = stick for 100 m

Good measure: reliability (test–retest, inter-rater), validity, responsiveness, MCID, no floor or ceiling effect, feasibility. Match the ICF level; same tester and conditions on retest; put numbers in goals ("Berg 38 to 46 in 6 weeks").

By condition: stroke: Fugl-Meyer, MAS, Berg, TUG, Barthel, mRS; SCI: ISNCSCI, SCI Independence Measure; PD: UPDRS III, Hoehn and Yahr, PDQ-39; MS: EDSS; brain injury: GCS, Rancho, FIM, Glasgow Outcome Scale; CP: MAS or Tardieu, GMFCS, GMFM.

Exam plan (any scale, 3)
What it measures and in whom (one line) → grades or items with scoring (small table) → interpretation or one use. **Draw:** GCS table, MAS grades. **Underline:** GCS 3–15, MAS 1+, AIS A–E, Barthel 0–100, FIM 18–126, Berg 56, TUG 13.5 s.

Neurological assessment and investigations

Tone, reflexes, sensation, coordination; EMG and nerve conduction are practical spots.

Full notes p. 29

Asked: 3 marks in 6 papers: differences between UMN and LMN lesions (S16, S20, W21); signs or features of a UMN lesion (W16, W22, S24). No 7-mark question; expect sensory examination. The same skills are the assessment in every long case (S16 ALS case: sort UMN and LMN signs).

Order of assessment

1. **History:** chief complaints; onset (sudden or gradual), duration, progression; past, family, personal, social; drugs; red flags.
2. **General:** vitals, **GCS**, built, posture, limb attitude, deformities, skin and pressure areas, aids.
3. **Higher functions:** orientation, memory, speech, MMSE or MoCA, perception. **Cranial nerves.**
4. **Motor:** bulk (girth), tone, power, reflexes, coordination, involuntary movements. **Sensory:** superficial, deep, combined; dermatomes.
5. **Balance and gait; function:** bed mobility, lying to sitting, sit to stand, walking, reaching, ADL (Barthel). Outcome measures, investigations.
6. **Sum up:** problem list, ICF and functional diagnosis, goals, plan (ICF (p. 3)).

UMN vs LMN (3m S16, S20, W21)

UMN: motor cortex → corticospinal tract → anterior horn (or cranial nerve nuclei). **LMN:** anterior horn cell (or nucleus) → root → plexus → peripheral nerve → muscle. The **anterior horn cell** is where they meet.

Feature	UMN	LMN
Site	Cortex, internal capsule, brainstem, cord tracts	Anterior horn cell, root, plexus, nerve
Weakness	Groups, limb or one side; arm extensors, leg flexors; fine movement lost	Individual muscles of a root or nerve
Tone	**Spasticity** (clasp-knife)	**Flaccid**
Wasting	Slight, late (disuse)	**Marked, early**
Fasciculation	Absent	May be present (anterior horn cell)
Deep reflexes	**Brisk, clonus**	Reduced or absent
Plantar	**Extensor (Babinski)**	Flexor or absent
Abdominal	Lost	Present unless own segments involved
EMG, conduction	No denervation; normal conduction	Denervation; slowed or absent conduction
Examples	Stroke, SCI, MS, CP	Polio, GBS, nerve injury, Bell's palsy

> **UMN goes up, LMN goes down**

UMN: tone up, reflexes up, toes up. LMN: tone down, reflexes down, bulk down.

Features of a UMN lesion (3m W16, W22, S24)

- **Negative** (lost): weakness in a **pyramidal pattern**; loss of fine and skilled movement; slowness, early fatigue.
- **Positive** (released): **spasticity**; brisk reflexes with spread and **clonus**; **Babinski and Hoffmann's signs**; flexor or extensor spasms; associated reactions, mass synergies.
- Abdominal and cremasteric reflexes lost; little wasting, no fasciculation.
- **Exception:** acute lesion (cerebral or spinal **shock**) → flaccid, reflexes absent; spasticity appears over days to weeks.

Why it matters: UMN: manage tone, train movement control and tasks, prevent contracture, strengthen (does not increase spasticity). **LMN**: strengthen each muscle by its grade, splint to protect denervated muscle from overstretch, care for numb skin, watch for reinnervation.

Motor examination

Bulk: wasting, fasciculation; girth at fixed distances from a bony point, both sides. **Tone:** passive movement slow and fast; spasticity (velocity-dependent, clasp-knife), rigidity (lead-pipe, cogwheel), hypotonia; MAS (tone (p. 37)). **Power:** MRC for LMN and muscle disease; in UMN record voluntary control and synergy (Brunnstrom, Fugl-Meyer). **Coordination:** finger–nose, heel–shin, rapid alternating, rebound. **Involuntary:** tremor (rest in PD; intention in cerebellar), chorea, athetosis, dystonia, ballismus, myoclonus, tics.

MRC: 0 no contraction; **1** flicker; **2** movement with gravity eliminated; **3** against gravity; **4** against gravity and resistance; **5** normal.

Reflexes

Deep reflex	Root; nerve
Biceps	**C5, C6**; musculocutaneous
Brachioradialis	C5, C6; radial
Triceps	**C7** (C6–C8); radial
Knee	**L3, L4** (L2–L4); femoral
Ankle	**S1** (S1, S2); tibial

> **1, 2, buckle my shoe**

S1–2 buckle my shoe (ankle); L3–4 kick the door (knee); C5–6 pick up sticks (biceps); C7–8 lay them straight (triceps).

Grading: 0 absent; 1+ reduced; 2+ normal; 3+ brisk; 4+ with clonus. Compare sides; **Jendrassik's manoeuvre** before calling one absent. **Superficial:** abdominal (T8–T12), cremasteric (L1–L2), plantar (S1), anal (S2–S4, important in SCI). **Pathological:** Babinski (big toe up, toes fan; normal in infants); Hoffmann's (flick middle fingernail → thumb and index flex); sustained clonus = UMN; grasp, sucking, palmomental in adults = frontal lobe damage.

Sensory examination

Eyes closed, compare sides, numb → normal skin. **Light touch** (cotton; dorsal columns and spinothalamic); **pain** (pin; spinothalamic, crosses in cord); **temperature** (warm and cold tubes); **joint position** (hold digit's sides; dorsal columns, cross in medulla); **vibration** (**128 Hz** fork, distal first); **combined** (parietal): two-point discrimination, stereognosis, graphaesthesia, localisation, extinction.

Patterns: glove and stocking → polyneuropathy; one dermatome → root; nerve territory → nerve injury; level on trunk → cord; half body → brain; pain and temperature lost, touch kept → syringomyelia, anterior cord; Brown-Séquard; **saddle anaesthesia + bladder change → cauda equina (emergency)**.

Key points: C5 outer elbow crease; C6 thumb; C7 middle finger; C8 little finger; T1 inner elbow; T4 nipple; T6 xiphisternum; **T10 umbilicus**; T12 mid inguinal ligament; L2 mid front thigh; L3 inner knee; L4 medial malleolus; L5 dorsum at third toe joint; S1 outer heel; S2 back of knee; S3 ischial tuberosity; S4–5 perianal.

Cranial nerves

I smell each nostril (lost in head injury). **II** acuity, fields by confrontation, light reflex (hemianopia). **III, IV, VI** H pattern (diplopia, squint; III: ptosis, large pupil). **V** face sensation 3 divisions, clench, corneal (jaw deviates to weak side). **VII** eyebrows, eyes tight, teeth, cheeks (**LMN whole half face; UMN lower face only**). **VIII** whisper, Rinne, Weber, head impulse. **IX, X** "aah", gag, swallow (uvula pulled away from weak side). **XI** shrug, head turn. **XII** tongue out (deviates to weak side; wasting, fasciculation in LMN).

Investigations

Conduction: demyelinating	Slow velocity, long latencies, conduction block, absent F-waves: GBS (AIDP)
Conduction: axonal	Low amplitude, near-normal velocity: diabetic, alcoholic neuropathy
Normal sensory + weakness	Lesion before dorsal root ganglion: avulsion, MND, polio
EMG neurogenic	Fibrillations, positive sharp waves (from 2–3 weeks); large, long, polyphasic units, fewer recruited
EMG myopathic	**Small, short, polyphasic**, recruited early: DMD, polymyositis
Repetitive stimulation	Decrement in myasthenia; increment after exercise in Lambert–Eaton
Imaging, labs	CT: bleeding, quick first scan; MRI: infarcts, MS plaques, cord; evoked potentials (MS); CSF: high protein, normal cells in GBS; very high CK in DMD

Read reports for level and side, nerve recovery, prognosis and plan.

Exam plan

UMN vs LMN (3): one line where each runs → table of **six** differences (weakness pattern, tone, wasting, fasciculation, deep reflexes, plantar) → example of each and the shock-phase exception. **UMN features (3):** negative, positive, lost superficial reflexes, shock exception. **Sensory examination (7):** preparation → superficial → deep → combined → chart on dermatome map, name pattern → PT meaning (skin protection, balance with vision, sensory re-education).

Draw: motor pathway with UMN and LMN marked; reflex arc; dermatome map. **Underline:** corticospinal tract, anterior horn cell, clasp-knife, clonus, Babinski, flaccid, wasting, fasciculation, denervation.

Cognition, perception, apraxia and aphasia

Apraxia, aphasia, neglect and cortical sensations in 9 papers.

Full notes p. 38

Asked: 7 marks: apraxia and its types (S23 with examples, S24, W24). **3 marks:** figure-ground discrimination and combined cortical sensations (W18); assessment of unilateral visual neglect (S19); apraxia types (S22); Broca's and Wernicke's aphasia (S23); aphasia and types (S25).

Apraxia (7m S23, S24, W24; 3m S22)

Apraxia: inability to carry out **learned, purposeful movements on command** despite adequate strength, sensation, coordination, comprehension and willingness; usually **left parietal or frontal**.

Ideomotor	Knows what to do, cannot do on command or imitation (left parietal): cannot show combing, combs spontaneously
Ideational	Loses idea or sequence; misuses objects: **toothpaste on the comb**
Limb-kinetic	Clumsy fine finger movement (premotor): cannot button, pick up a coin
Constructional	Cannot copy or build 2- or 3-dimensional shapes (parietal, either side): cube, blocks
Dressing	Shirt inside out or one arm only (right parietal, with neglect)
Buccofacial	Cannot blow out a candle on command, does when lit (left frontal, with Broca's)
Gait	Feet "glued to the floor"; leg movements normal in lying (frontal, hydrocephalus)

Assess: pantomime ("show me how you brush"), imitate gestures, real objects, multistep task (making tea); note wrong movement, body part as tool, wrong order, perseveration. **PT: real tasks with real objects** in natural setting; strategy training (talk through steps, step cards); **errorless learning, chaining**; physical guidance not long instructions; fade cues; gesture training (ideomotor).

Aphasia (3m S23, S25)

Loss of **language** (speak, understand, read, write), usually left hemisphere; **dysarthria** = motor speech, language normal.

- **Broca's** (expressive, non-fluent): slow, effortful, **telegraphic**; comprehension fairly good; aware, frustrated; **left inferior frontal**; right arm weak.
- **Wernicke's** (receptive, fluent): fluent but empty, made-up words; **poor comprehension**; unaware; **left posterior superior temporal**. Repetition poor in both.
- **Conduction**: repetition very poor (arcuate fasciculus); **global**: all lost (large left MCA); **anomic**: word-finding; **transcortical**: repetition kept.

PT: short simple sentences, one instruction, yes or no questions; gestures, demonstration, pictures, guidance; check by action not "yes"; quiet room, time to respond; work with speech therapist.

Unilateral neglect (3m S19)

Failure to report, respond or orient to the side **opposite the lesion**, not from weakness or sensory loss; commonest after **right parietal** lesion. Eats half the plate, bumps left door frames, denies the arm. **Vs hemianopia:** hemianopia is aware and turns the head; neglect is unaware.

Tests: line bisection (shifted to good side); **cancellation** (star, letter, Albert's: targets missed); clock drawing, copying (half missing); reading (starts mid-line); observing ADL, **Catherine Bergego Scale**. **PT:** visual scanning with a coloured **anchor** on the neglected edge; approach from that side as tolerated; limb activation, trunk rotation to that side; prism adaptation, mirror (limited evidence); check limb in wheels and footplate.

Perception and cognition (3m W18)

Figure-ground: cannot pick an object from its background (white shirt on white bed, wheelchair brake, step edge; right parietal-occipital). Find item in cluttered drawer; reduce clutter, **colour contrast** tape, scanning, busier settings gradually.

Combined cortical sensations (primary sensation + parietal): **stereognosis** (coin, key, eyes closed); **graphaesthesia** (number on skin); **two-point discrimination**; localisation; barognosis, texture; bilateral simultaneous stimulation (extinction).

Others: body scheme (somatoagnosia, right-left, finger agnosia); spatial (depth, getting lost); agnosias, prosopagnosia; **anosognosia**. **Cognition:** MMSE (30; adjust for education), MoCA (30; better for mild, executive); short structured sessions, one-step instructions, errorless learning, memory aids, family.

Exam plan (apraxia, 7)
Define (1) → types with lesion and example (4) → assessment and PT (2). **Draw:** line bisection and cancellation with left omissions; Broca's, Wernicke's, arcuate fasciculus. **Underline:** ideomotor, ideational, fluent vs non-fluent, line bisection, figure-ground, stereognosis.

Movement dysfunction in daily tasks: bed mobility to reaching

The functional tasks the syllabus lists for every condition.

Full notes p. 44

Asked: not asked on its own (2016–2026). It sits inside motor relearning questions (S18, W18), long cases ("management with emphasis on trunk control and ambulation"), hemiplegic gait and the spastic hand. Expect 3 marks on the components of sit to stand or compensations in reaching.

How movement fails

Impairments: weakness, slowness, loss of dexterity and coordination, altered tone, sensory loss, shortened soft tissue, poor balance. **Adaptations (compensations):** uses the stronger side, the arms, a wider base or a stiff trunk; some are useful, others block recovery (**learned non-use**, all weight on the strong leg).

Training rule: analyse the task → find missing components → practise them → **whole task** many times in varied real settings → measure (MRP (p. 35), motor learning (p. 5)).

Bed, sitting up, standing up

- **Bed mobility:** head turn, arm across, top hip and knee flex, trunk rotation; **bridging** to move up and onto a bedpan. *Problems:* weak limbs not brought across; pulls on the rail; weak hip extensors. *Train:* roll both ways with weak limbs placed to help; bridge with the weak foot loaded; less rail.
- **Lying to sitting:** roll to side lying, legs over the edge, push up through the lower arm while side-flexing neck and trunk up. *Problems:* pulls straight up with the strong arm; legs stay on the bed; poor sitting balance. *Train:* each part, then whole; vary bed height and side; lie down with control.
- **Sit to stand: feet back** behind the knees; **trunk forward** by hip flexion, straight spine; **knees forward** over the feet; hips and knees **extend together**; sit down reverses with control. *Problems:* weight on the strong leg, weak foot forward, too little lean, arm pushing, drops into the chair. *Train:* weak foot back; higher seat, then lower; repetitions in sets; step-ups; vary chair, bed, toilet, car; five times sit-to-stand test.

Walking and balance

- **Walking:** stance: hip extension, small knee bend at loading then extension, roll over the ankle, push-off; swing: knee bend, hip flexion, **dorsiflexion** to clear; small pelvic shift. *Problems:* short stance on the weak leg, knee hyperextension or buckling, little push-off, stiff knee, foot drop, **circumduction**, slow. *Train:* intensive varied walking, treadmill, stepping drills, strengthening, AFO when needed; speed and distance.
- **Balance:** centre of mass over base, still and moving; **anticipatory** adjustments; **reactive** steps; vision, vestibular, body sensation. *Problems:* wide base, weight on the strong side, stiffening, holding on, fear of falling. *Train:* reaching beyond arm's length sitting and standing, stepping all directions, varied surfaces, perturbations, dual tasks; Berg, TUG, Mini-BESTest.

Reaching and manipulation

- **Reaching:** shoulder flexion with **scapular upward rotation** and protraction, elbow extension, wrist ready; trunk only beyond arm's length. *Problems:* **shoulder hitch**, elbow stays bent, **trunk leans forward**, uses the other hand. *Train:* targets at varied heights and distances; **trunk restraint** (Michaelsen 2006); strengthen shoulder and elbow; real objects.
- **Manipulation: wrist extension during grasp**; hand opens to the object; thumb opposition; grip force scaled; release; in-hand movement. *Problems:* weak wrist extensors → weak grip; hand will not open; poor opposition, force, release. *Train:* graded objects (size, weight, texture), wrist extension in function, **CIMT** or intensive bimanual practice, splints only for a stated aim.

Exam plan (training any task)

Essential components → what is missing and how the patient compensates → practise missing parts, then whole task with repetition and variety, feedback → measure. **7m sit to stand in hemiplegia:** analyse, problems, training list above, timed test.

Draw: sit to stand stick figures (feet back, trunk forward, knees forward, extension). **Underline:** task analysis, essential components, compensation, learned non-use, feet back, scapular upward rotation, trunk restraint, wrist extension in grasp, whole-task practice.

Stroke

11 long cases and 13 seven-mark questions in 20 papers.

Full notes p. 50

Asked: a stroke long case in 11 of 20 papers (ICF with reasoning, goals, management; MCA, ACA, haemorrhagic, after craniotomy). **7 marks:** hemiplegic gait (5 times), hemiplegic shoulder, spastic hand, acute stage, flaccid arm, MRP for the arm, circle of Willis. **3 marks:** pusher syndrome, flexion synergy, ACA syndrome, types of stroke.

Definition and types

Stroke (WHO): rapidly developing focal (or global) loss of brain function lasting **more than 24 hours** or causing death, of vascular origin. **TIA:** brief focal deficit without infarction on imaging: a warning, highest stroke risk in the next days. Hemiplegia is on the side **opposite** the lesion.

Type	Clue
Thrombotic	At rest or in sleep; stepwise; earlier TIAs
Embolic	Sudden, maximal at onset (AF, rheumatic valve)
Lacunar	Pure motor or sensory; no cortical signs
Intracerebral bleed	Headache, vomiting, falling GCS, high BP
Subarachnoid	"Worst headache of my life", neck stiffness

About 85% ischaemic in the West; a fifth to a third haemorrhagic in India. **Phases:** hyperacute 0–24 h, acute 1–7 days, early subacute to 3 months, late subacute 3–6 months, chronic after 6 months.

Arterial syndromes (opposite side)

MCA	**Face and arm more than leg**; hemianopia; aphasia (left) or neglect (right)
ACA	**Leg more than arm**; incontinence; abulia; grasp reflex; gait apraxia
PCA	Hemianopia; visual agnosia; memory loss; thalamic pain
Vertebrobasilar	**Crossed signs**; vertigo, diplopia, dysarthria, dysphagia, ataxia; locked-in
Lacunar	Pure motor, pure sensory, ataxic hemiparesis

Circle of Willis (7m S20): ACAs + ACoA in front; ICA → ACA + MCA; PCoA joins ICA to PCA; vertebrals → basilar → PCAs. A safety link right–left and front–back; aneurysms at its junctions (ACoA).

Risk factors and pathophysiology

INTERSTROKE: 10 factors ≈ 90% of risk: **hypertension** (top), smoking, diabetes, inactivity, diet, stress, abdominal obesity, alcohol, AF, lipids. Young: RHD, venous thrombosis after childbirth, TB meningitis, dissection. India: 1 in 7 under 45.

- **Core** dies in minutes; **penumbra** salvageable for hours ("time is brain").
- **UMN syndrome:** negative (weakness, poor dexterity, fatigue); positive (spasticity, brisk reflexes, clonus, Babinski, associated reactions, **synergies**); adaptive (contracture, **learned non-use**, deconditioning).
- Recovery: **use-dependent plasticity**; most spontaneous recovery in the first 3 months.

Features, synergies, pusher

BE-FAST: Balance, Eyes, Face, Arm, Speech, Time. Flaccid stage first, then spasticity in antigravity muscles; sensory loss, hemianopia, aphasia, dysphagia, neglect, apraxia, shoulder pain, depression.

Synergy	Components (strongest in bold)
UL flexion	Scapular retraction, elevation; shoulder abd, ER; **elbow flexion**; supination; wrist and finger flexion
UL extension	Protraction; **shoulder add, IR**; elbow extension; pronation
LL flexion	**Hip flexion**, abd, ER; knee flexion; DF, inversion
LL extension	Hip ext, add, IR; knee extension; PF, inversion

Pusher syndrome (3m S20, W24): pushes with the strong side towards the weak side and resists correction (right hemisphere, thalamus). Treat with visual vertical (mirror, door frame), reaching to the strong side, strong side against a wall.

Investigations and scales

CT first (excludes bleeding); **MRI diffusion** (early infarct); angiography, carotid Doppler; ECG, echo (AF); sugar, lipids, clotting. **NIHSS** 0–42; GCS.

Brunnstrom stages: 1 flaccid; 2 synergies begin; 3 voluntary synergy, spasticity peaks; 4 some movement out of synergy; 5 complex movement; 6 isolated, near normal. **Scales:** MAS, Fugl-Meyer (arm 66, leg 34), Berg (out of 56, below ~45 = fall risk), TUG, 10 m walk, 6MWT, Barthel, FIM.

Medical management

Stroke unit for all. **Clot-busting** (tenecteplase or alteplase) within **4.5 h** (selected to 9 h); **thrombectomy** for large vessels within 6 h (selected to 24 h). Bleed: BP control, reverse anticoagulants, surgery. Nothing by mouth until a swallow screen. Craniectomy: no pressure on the bone flap side; helmet.

ICF (S23 worked case: 55, auto driver, left MCA, 3 months, diabetic)

- **Structure/function:** left fronto-parietal infarct; right arm weakness (Brunnstrom 3–4), flexor spasticity MAS 1+–2, hand sensory loss, balance and gait deficit, possible aphasia, shoulder pain risk.
- **Activity:** self-care with the right hand, community walking, climbing into the auto.
- **Participation:** cannot drive his auto, income lost.
- **Environment:** family, stick, rehab (facilitators); income loss, stairs, Indian toilet (barriers). **Personal:** sole earner; motivated by work.

Goals: short-term (4 weeks): cup to mouth; keep wrist and finger extension range; Berg +5; walks 200 m with stick. Long-term (3 months): bimanual self-care; community walking without stick; driving assessment.

Acute stage (7m S17)

- Start when stable, short frequent sessions (**AVERT**: very early high-dose mobilisation in 24 h worsened outcome).

- **Positioning every 2 h:** on the affected side (shoulder forward), on the unaffected side (arm on pillow), supine (pillow under shoulder and hip); head up 30° if swallow or chest risk.
- Chest care; shoulder support, **never pull the arm**; daily pain-free ROM; rolling, sitting, sit to stand, standing; DVT: early mobility, IPC (stockings alone do not work).

Subacute and chronic

- **Principles:** task-specific, repetitive, intensive, progressive, feedback, practice outside therapy; NICE: 3 h a day, 5 days a week if able.
- **Balance:** reaching beyond arm's length, weight shift, stepping, perturbation; falls programme.
- **Gait:** overground and treadmill (± body-weight support); AFO for foot drop.
- **Arm:** task practice; **CIMT** if ≥20° wrist and ≥10° finger extension (EXCITE: mitt 90% of waking hours, 6 h practice, 2 weeks); mirror therapy; FES; bilateral training.
- **Spasticity:** positioning, stretch, active use; **botulinum toxin + therapy** (Class 1).
- **Exercise dose (AHA 2014):** aerobic 40–70% HRR (RPE 11–14), 3–5 days, 20–60 min; strength 1–3 × 10–15, 8–10 exercises, 2–3 days; flexibility 10–30 s holds.

Gait, shoulder, hand (the 7-mark trio)

Hemiplegic gait: stance: short, knee snaps back or stays bent, no hip extension, weak push-off; swing: stiff knee, foot drop and inversion, **circumduction** or hip hiking. Train: treat the cause (strengthen hip extensors, quads, DF; stretch calf) → part practice → whole walking overground and treadmill → AFO or peroneal FES → outdoors. Phases: parallel bars with knee support → break extensor synergy → community.

Shoulder: subluxation, pain, IR/add spasticity, shoulder-hand syndrome. Arm trough, scapula forward, never pull; pain-free ROM with scapula mobilised first; FES to supraspinatus and deltoid; botulinum toxin for spastic painful shoulder; **no overhead pulleys** (harmful, AHA/ASA 2026).

Spastic hand: slow stretch, weight-bearing on the open hand, prolonged icing, grasp-release practice, FES to extensors, mirror, CIMT, botulinum toxin; resting splints do not prevent spasticity. **Flaccid arm:** protect shoulder; Rood facilitation; weight-bearing; placing and holding; FES; mirror; bilateral. **MRP arm (W18):** analyse → practise missing parts → whole task → transfer.

Precautions and stop

Unit BP limits (avoid exercise if resting BP above ~200/110); swallow screen before food; protect the shoulder; craniectomy helmet; neglect and sensory loss: check limb position; anticoagulants: bruising; diabetes: sugar and feet. **Stop:** new weakness, speech or vision change, sudden headache, falling GCS, chest pain, seizure, DVT signs.

Complications: aspiration pneumonia, DVT, shoulder pain, contracture, pressure injury, falls, depression, CRPS, learned non-use.

Education and prognosis

Secondary prevention: daily BP, sugar, lipid and antiplatelet medicines; stop all tobacco (bidi, gutka); less salt; exercise; BE-FAST. Carer: transfers, shoulder support; commode or raised seat for an Indian toilet; picture home programme. **EPOS:** finger extension + shoulder abduction at 72 h → 98% regain some dexterity (25% without). Poor outcome: older, severe, early coma, neglect, poor sitting balance, incontinence.

Exam plan (long case)

1. Restate: age, job, side and artery, type, time since onset (phase), risk factors, function.
2. **ICF (5)** with reasoning (artery → deficits) and a one-line functional diagnosis.
3. **Goals (3–5)**: SMART, short and long term, tied to job and home.
4. **Management (7–8)**: principles; by phase or problem; doses; education and secondary prevention.

Draw: circle of Willis; ICF filled from the case; spastic posture stick figure. **Underline:** contralateral, arm more than leg (MCA), synergy, Brunnstrom, task-specific, learned non-use, circumduction, no overhead pulleys.

Acquired brain injury: trauma and space-occupying lesions

5 long cases; GCS and raised intracranial pressure as 3 marks.

Full notes p. 66

Asked: a brain injury long case in 5 papers (S17 left bleed, right paralysis; S23, S24 right frontal subdural, GCS 12, S24 truck driver, poor vision; W24 Rancho IV in ICU; S26 diffuse axonal injury, GCS 5): ICF, goals, techniques. **7 marks:** PT in the acute stage of TBI (S18, S20). **3 marks:** GCS (W16, S19, S20, S26), raised ICP (W22, S25), classification of head injury (S24).

Definition and classification (3m S24)

Acquired brain injury: damage after birth, not hereditary, congenital or degenerative (trauma, stroke, tumour, infection, anoxia). **TBI:** altered brain function or pathology from an **external force** (blow, acceleration–deceleration, penetration, blast). **SOL:** a mass in the skull (tumour, abscess, haematoma, cyst) that compresses the brain and raises ICP.

Severity (GCS)	Mild 13–15; moderate 9–12; **severe 3–8**
Mechanism	Closed or open (penetrating); acceleration–deceleration; blast
Skull	Linear, depressed, basal (CSF leak, raccoon eyes)
Focal	Extradural, subdural, intracerebral haemorrhage, contusion
Diffuse	Concussion; diffuse axonal injury; diffuse swelling
Timing	**Primary** at impact; **secondary** after: low O_2, low BP, raised ICP, seizures, infection

Extradural: arterial (middle meningeal, temporal fracture), stops at sutures → **lens-shaped**; **lucid interval**. **Subdural:** torn bridging veins → **crescent**; acute after major trauma, chronic in older people after a minor fall. **Causes:** road accidents (two-wheelers, no helmet), falls, assault; SOL: glioma, meningioma, metastases, abscess, **tuberculoma**.

GCS (3m W16, S19, S20, S26)

E 1–4: spontaneous, to sound, to pressure, none. **V 1–5:** oriented, confused, words, sounds, none. **M 1–6:** obeys, localises, normal flexion (withdraws), **abnormal flexion (decorticate)**, **extension (decerebrate)**, none. Total **3–15**; **8 or less = coma**, airway needs protection.

Interpreting (S26): report the parts ("**E2 V2 M4 = 8**"); **motor predicts outcome best**; "T" for verbal when intubated; best response, better side for motor. **Used in (W16):** head injury, stroke, meningitis, encephalitis, overdose, any reduced consciousness.

Rancho Los Amigos levels (behaviour → PT)

I No response	None → positioning, chest, range, splints, family
II Generalised	Same response to any stimulus → as I; structured sensory input
III Localised	Turns to voice; inconsistent commands → one-step commands, tilt table
IV Confused, agitated	Aggressive, no short-term memory → **calm, familiar activity; no new learning**
V Confused, inappropriate	Not agitated, distractible → structured repeated simple tasks
VI Confused, appropriate	Goal-directed with cues → routine, cues, memory aids
VII Automatic	Poor judgement, insight → community skills, dual tasks, safety
VIII Purposeful	Subtle problems → fitness, work, sport, driving assessment

Pathophysiology and raised ICP (3m W22, S25)

- **Primary: coup and contrecoup** contusions, haematomas; **diffuse axonal injury** (rotation shears axons): coma from impact, near-normal CT, MRI lesions in corpus callosum, internal capsule, brainstem.
- **Secondary:** swelling, bleeding → ICP rises; skull cannot expand → **CPP = MAP – ICP** falls → more ischaemia; low O_2 and BP worsen it. **Herniation:** uncus on III → fixed dilated pupil → brainstem failure.
- **Raised ICP symptoms:** headache worse on waking, coughing, bending; vomiting (projectile); blurred vision. **Signs:** falling consciousness, **papilloedema**, dilating pupil on the mass side, VI palsy, seizures, bulging fontanelle (infants). **Cushing's triad** (late): rising BP with wide pulse pressure, **slow HR**, irregular breathing.
- **Numbers:** normal ICP **5–15 mmHg**; treat above **22 mmHg**; CPP target **60–70 mmHg** (Brain Trauma Foundation 2016).

Features: coma → vegetative or minimally conscious; post-traumatic amnesia; hemiparesis (often both sides), spasticity, ataxia; attention, memory, executive loss, agitation. **Frontal (S23, S24):** poor planning, judgement, safety, disinhibition or apathy. Vision loss (S24), vertigo, swallow, seizures. **SOL:** slow headache, seizures, focal signs, personality change.

Investigations and medical management

CT first (haematoma, midline shift); **MRI** (diffuse axonal injury, tumour); ICP monitoring, EEG. **Prevent secondary injury:** airway, O_2, BP; head up, sedation, osmotic therapy, CSF drainage. **Surgery:** evacuate haematoma, elevate depressed fracture, **decompressive craniectomy** (flap out: protect soft area), burr holes (chronic subdural); later botulinum toxin, baclofen.

ICF (W24 worked case: 30, skull fractures, ICU, 2 weeks, Rancho IV)

- **Structure/function:** severe TBI; agitation, poor attention, no short-term memory, poor safety; weakness, altered tone, poor balance; secretions, swallow; contracture, sore, ossification risk; protect skull.
- **Activity:** dependent for bed mobility, sitting, transfers, walking, feeding, hygiene; cannot follow complex instructions (cognition limits, not only strength).
- **Participation:** off work; family roles suspended.
- **Environment:** ICU care, family visits (facilitators); noise, alarms, many staff, **restraints** (raise agitation) (barriers). **Personal:** young, likely earner.

Goals: short-term (2–4 weeks): calm in short sessions, no falls; clear chest, no sores; full range; sits out, stands with help; washes face. Long-term (3–6 months): follows routines, safe at home; walks indoors and outdoors; independent self-care; planned return to work.

Acute stage in ICU (7m S18, S20)

- **Assess first:** GCS, pupils, ICP, CPP, BP, O_2, ventilator, craniectomy, other fractures; Rancho, post-traumatic amnesia; chest, MAS.
- **Positioning:** head of bed **up about 30°**, **head midline**; no neck flexion or rotation (jugular compression); side lying against abnormal postures; turn regularly.
- **Chest:** positioning, gentle manual techniques, **short pre-oxygenated suction**; **no head-down drainage** while ICP raised.
- **Range:** daily passive range all joints; splints or **serial casts** for ankles, wrists when tone rises.
- **Early sitting, tilt table** once ICP controlled. Structured sensory input (family voices, music, touch) in short periods; **"coma stimulation" evidence weak: avoid overstimulation**. Monitor ICP, CPP, BP, O_2 throughout.

By level of cognition

- **Unresponsive (GCS 3–8, Rancho I–II; S26):** positioning, turning, chest care, range, splints, casting, tilt table and sitting for arousal, sensory input, teach family. Goals: tolerates sitting, more responses → sits supported, follows commands.
- **Agitated (IV; W24):** quiet room, one therapist, short sessions; familiar automatic tasks (walking, sit to stand, washing face); low bed, mats, **no restraint**; do not argue or test memory; stop and retry later.
- **Confused, non-agitated (V–VI):** same time daily, one-step instructions, **errorless learning**, memory aids, repeated functional practice.
- **VII–VIII:** balance, dual tasks, community walking, fitness, work and driving assessment. Graded aerobic exercise with pacing once stable.

Precautions, complications

Stop: ICP stays above limit, CPP falls, **GCS drops**; report a new dilated pupil at once. No head-down or tight collars; short suctioning. **Craniectomy helmet**, no pressure on the soft area; clear fractures and spine first; seizure plan; tumours: fatigue, steroid effects.

Complications: aspiration; contractures (ankle, elbow, wrist); **heterotopic ossification** (warm swollen joint losing range: gentle range, report); sores (sacrum, heels, occiput); falls in agitation.

Education and prognosis

Agitation is a stage that passes; positioning, range, transfers, calm routine, memory aids, no alcohol; helmet; **driving only after clearance** (vision, cognition, seizures); graded return to work. **Worse:** older, **low GCS motor**, non-reacting pupils, low O_2 or BP, large lesions. Longer amnesia = more severe; cognition and behaviour limit return to work more than physical problems.

Exam plan (long case)

1. Restate: age, job, lesion, days since injury, **GCS and Rancho**, ICU or ward.
2. **ICF (5)**: six boxes, table with reasoning, one-line functional diagnosis.
3. **Goals (5–7)**: short-term for the present level; long-term walking, self-care, cognition, work.
4. **Management (3–8)**: ICU care within ICP limits → by Rancho level → motor and balance → family → precautions.

Draw: extradural lens vs subdural crescent; GCS table; ICF filled. **Underline:** secondary injury, Rancho, CPP, head up 30° midline, low stimulation, serial casting.

Spinal cord disorders: traumatic and non-traumatic

9 long cases; autonomic dysreflexia and bladder in almost every paper.

Full notes p. 77

Asked: an SCI long case in 9 papers (T9–T10, T10, L1 fracture; transverse myelitis; Pott's spine; cervical level and grade): ICF with reasoning, goals, PT, expected ambulation, bladder type. **7 marks:** autonomic dysreflexia (W18, W24), ambulation after an L3 fracture (S18). **3 marks:** autonomic dysreflexia (6 times), spinal shock, Brown-Séquard, sacral sparing, ASIA scale, paraplegia in flexion and extension, cauda equina, complete vs incomplete.

Definition, level, ASIA scale (3m S19)

Tetraplegia: cervical segments, arms affected; **paraplegia:** thoracic, lumbar, sacral, cauda equina, arms spared. **Key muscles** (0–5): C5 elbow flexors, C6 wrist extensors, C7 elbow extensors, C8 finger flexors, T1 little-finger abductor; L2 hip flexors, L3 knee extensors, L4 DF, L5 long toe extensors, S1 PF. 28 sensory points (light touch, pinprick 0/1/2); anal VAC, DAP. **Motor level:** lowest key muscle grade 3+ with those above 5. **NLI:** the higher of motor and sensory levels.

A complete	No sensory or motor function at **S4–S5**
B	Sensation but no motor below the level, incl. S4–S5
C	Motor: less than half the key muscles below NLI grade 3+
D	Motor: half or more grade 3+
E	Normal now, had deficits

Sacral sparing (3m S16); complete vs incomplete (3m W25): any S4–S5 sensation, deep anal pressure or anal contraction → **incomplete** (sacral fibres lie on the outer edge of the tracts); none → AIS A. **S16:** triceps 4/5, nothing below; sensation to T4; anal sensation intact → **NLI C7, AIS B**.

Cord syndromes (Brown-Séquard 3m W17, S23; cauda equina 3m W24)

Central	Commonest; older, hyperextension; **arms worse than legs**; walks, hands recover least
Anterior	Lost power, pain, temperature; posterior columns kept; poor
Brown-Séquard	Hemisection. **Same side:** UMN weakness, lost position, vibration. **Opposite:** lost pain, temperature 1–2 segments below. **Best: most walk**
Posterior	Rare; lost position, vibration; power kept; ataxia worse eyes closed
Conus	T12–L1; saddle anaesthesia, areflexic bladder early; mixed UMN/LMN
Cauda equina	Below L1–L2; **LMN**, asymmetrical, lost knee and ankle jerks, root pain, saddle anaesthesia; better (roots regrow)

Paraplegia in extension vs flexion (3m W22): extension: pyramidal only (vestibulospinal works), legs extended, adducted; incomplete, better. Flexion: pyramidal + extrapyramidal, hips and knees drawn up, spasms, **mass reflex**; poorer.

Causes, pathophysiology, spinal shock (3m S17, W20)

Causes: falls from height (India **53%**, road accidents 28%), diving; Pott's spine, transverse myelitis, tumour, myelopathy, anterior spinal artery infarct.

- Primary injury → **secondary** over hours–days (swelling, poor flow).
- Cord → **UMN** once shock ends; cauda equina → **LMN**. **T6 and above:** low BP, OH, AD, poor temperature control.
- **Vertebra → segment:** cervical +1; T1–T6 +2; T7–T9 +3; T10 → L1–L2; T12 → L5; L1 → conus.

Spinal shock: flaccid, all reflexes lost incl. **bulbocavernosus**, retention, ileus. Bulbocavernosus returns days 1–3 → hyperreflexia 1–4 weeks → spasticity 1–12 months. **Not neurogenic shock** (low BP, slow HR). **Transverse myelitis (S17, S21):** legs, sensory level, bladder over hours–days; steroids; a third each recover well, partly, poorly. **Pott's (W22):** gibbus, evening fever; anti-tubercular drugs, brace.

Medical: ISNCSCI within 72 h; **early stabilisation**; baclofen. **Classic vs current:** not 6–12 weeks' bed rest; mobilise early within precautions.

Autonomic dysreflexia (7m W18, W24; 3m S17, W19, S21, W21, S22, S24)

Systolic BP more than 20 mmHg above usual (15 in children), SCI **at or above T6**; emergency. **Mechanism:** stimulus below → sympathetic discharge → splanchnic vessels (T6–L2) narrow → BP up → vagus slows heart, flushing above; calming signals cannot pass down. **Causes: full bladder** (blocked catheter), bowel, pressure injury, ingrown toenail, tight straps. **Signs:** pounding headache, slow pulse, flushing, sweating above, blocked nose.

1. **Sit upright, lower the legs**; loosen clothes, straps, binder.
2. BP and pulse every **1–2 min**.
3. **Bladder first:** catheterise, or unkink, flush, change.
4. Systolic still **150 or more** → doctor's short-acting drug (nitrate paste, nifedipine).
5. Bowel (anaesthetic gel) → other triggers; monitor **2 h**; teach; wallet card.

Prevent in PT: ask about the bladder each session; know usual BP (tetraplegia 90–110, so 130 may be high); stop exercise or stimulation if headache starts.

Other complications

Orthostatic hypotension (fall of 20 systolic or 10 diastolic): tilt table in small steps, binder, stockings. **Pressure injury:** turns **every 2–6 h**; relief **every 15–30 min for 30–90 s**; mirror skin checks twice a day. **DVT** (painless swollen leg): stop leg exercise, report. **Heterotopic ossification** (hip): no forceful stretch. **Shoulder pain:** strengthen, level transfers.

ICF (S22: 42, photographer, complete T9–T10, 1 month)

Assess: ISNCSCI, VC, cough, MAS, skin, sitting, transfers, **SCIM III**, **WISCI II**, BP lying and sitting.

- **Structure/function:** paralysis and sensory loss below (AIS A; UMN, shock over); spasticity; **reflex bladder** (sacral centre intact, cut off); weak lower abdominals; skin, DVT risk; below T6, so AD unlikely.

- **Activity:** cannot stand, walk, climb stairs; **already sits unsupported and transfers**.
- **Participation:** cannot reach his **first-floor studio**; outdoor work.
- **Environment:** good finances, studio at home, family (facilitators); stairs, Indian toilet (barriers). **Personal:** self-employed, seated job.

Goals: short-term (4–6 weeks): own pressure relief; push-ups clear of seat; stands in KAFOs in bars; self-catheterises, volumes under 400–500 mL. Long-term (3–6 months): floor transfer; kerbs; **household walking, KAFOs, swing-through**; back at work, hand controls.

Expected outcome and PT management

Expected (complete): C1–C4 power chair (C1–C3 ventilator); C5 power chair, feeds with set-up; **C6 tenodesis**, manual chair on level; C7–C8 manual chair, independent; **T1–T12 household walking with KAFOs**; L1–S1 KAFO or AFO and crutches.

- **Acute:** deep breathing, assisted cough, positioning; turns; daily ROM within precautions; ankle splints at 90°; strengthen working muscles; sit up in stages watching BP.
- **Rehab:** tilt table → rolling with momentum → **long sitting** (SLR about 100–110°, do not overstretch) → sitting balance → **push-ups on blocks** → sliding board → lifting → floor transfers; wheelchair skills.
- **Dose (Martin Ginis 2018):** arm cycling, wheeling **20 min moderate–vigorous 2 times a week** (30 min 3 times for heart health); resistance **3 sets** per group 2 times a week; daily stretch, prone lying. AIS C, D: treadmill ± body-weight support.
- **Tetraplegia:** assisted cough (push in and up under ribs); **tenodesis (C6–C7):** stretch finger flexors only with wrist flexed.

Ambulation (S22, S26; L3 fracture 7m S18)

- **Before:** strong shoulder depressors, latissimus, triceps; full hip extension, DF to neutral; standing without dizziness.
- **KAFOs, drop-lock knees**, ankles slight DF; higher thoracic: HKAFO or **RGO**.
- **Bars:** sit to stand knees locked; hips forward (Y-ligaments), shoulders behind; **swing-to → swing-through**; walker → elbow crutches; floor rise, kerbs, stairs. KAFO walking costs far more energy: chair stays main mobility.
- **L3 fracture:** below the cord → **cauda equina, LMN**, often incomplete, uneven: strengthen hip extensors, abductors, quads; **AFOs** (KAFO only if quads weak); crutches four-point → two-point → canes; stairs, kerbs, slopes; expect **community walking**; protect insensate feet.

Education, prognosis, bladder (S26)
Skin checks, AD card; ramps, commode, ground floor or lift. **Prognosis:** most recovery in **6–9 months**; **20–30% of AIS A** improve; walking likelier with AIS D, youth, strong quads and calves, preserved pinprick.

S26 bladder: T9–T10 is above the sacral centre (S2–S4 at L1 vertebra) → **reflex (UMN) bladder**; **clean intermittent catheterisation 4–6 times a day**. See bladder and bowel (p. 41).

Exam plan (long case)

1. Restate: age, job, level, complete or not, time since surgery, bladder, home.
2. **ICF (5)** with reasoning; functional diagnosis.
3. **Goals (3–5)** matching the expected outcome for the level.
4. **Management (7–10)**: phase; chest, skin; strength; mat, transfers, wheelchair; orthotic walking; bladder; AD if T6 or above; home, work.

Draw: four cord syndromes; AD flowchart; ICF from the case. **Underline:** AIS grade, sacral sparing, autonomic dysreflexia, pressure relief, tenodesis, KAFO, swing-through.

Peripheral nerve injuries and the brachial plexus

Nerve injury classification 5 times; radial, Erb's, Klumpke's, peroneal.

Full notes p. 94

Asked: long cases W16 (Erb's palsy, shopkeeper: ICF, goals), S22 (fall from a tree: plexus level, goals, PT with orthoses), S23 (fall from a train, paralysed arm: ICF, PT). **7 marks:** Erb's palsy (S17, S21, W21, W24), radial palsy (S18, W18), claw hand (W19), carpal tunnel (S22), Klumpke's (S25), common peroneal (W25), Sunderland + foot drop (S16). **3 marks:** classification (5 times), orthosis for radial injury (W17), dynamic cock-up (S25), cock-up splint (W25).

Definition and classification (3m S21, W21, S22, S23, S25; 7m S16)

Peripheral nerve injury: LMN loss in its area: weakness, wasting, lost reflexes, sensory loss, lost sweating; can regrow, slowly, from the injury site down. **Plexus injury:** C5–T1 roots, trunks or cords, usually traction.

Sunderland (Seddon)	Damage; recovery
I (neurapraxia)	Myelin, **conduction block**, axon intact, no Wallerian degeneration; full in days to ~3 months
II (axonotmesis)	Axon cut, **endoneurium intact**; full at **1 mm a day**
III	+ endoneurium; perineurium intact; incomplete (misdirection)
IV	Only **epineurium** intact (neuroma in continuity); needs surgery
V (neurotmesis)	Nerve divided; none without repair

Causes: radial: **shaft of humerus**, Saturday night palsy, crutch; ulnar: elbow fracture, leprosy; median: wrist cuts, carpal tunnel; axillary: shoulder dislocation; peroneal: **fibular neck**, tight plaster, squatting; sciatic: **gluteal injection**.

Pathophysiology

- Axon cut → **Wallerian degeneration** below → flaccid, wasted muscle; dry, shiny, numb skin.
- Regrows **1 mm a day (2.5 cm a month)**: proximal muscles first; **Tinel's sign** moves down the limb.
- Long delay → end-plates degenerate, so proximal injuries recover less; misdirected axons → poor localisation.

Nerve by nerve

Radial, spiral groove (7m S18, W18)	**Wrist, finger, thumb drop**; triceps spared; weak grip; patch on back of first web
Ulnar, wrist	Claw of ring and little; hypothenar, interossei wasting; **Froment's sign**, card test
Ulnar, elbow	Less clawing (long flexors weak too): **ulnar paradox**
Median, wrist	**Ape-thumb**: thenar wasting, no opposition; pen test
Median, elbow	+ **hand of benediction**; Ochsner's clasp test
Common peroneal (7m W25)	**Foot drop**: weak DF, eversion, toe extension; high-stepping, slapping gait; outer leg, top of foot numb
Axillary; sciatic	Weak abduction, regimental badge; foot drop + hamstrings, calf

Claw hand (7m W19): MCP hyperextension + IP flexion: **lumbricals and interossei** paralysed, long extensors and flexors unopposed. Ulnar: ring and little; median + ulnar: all four. Also Klumpke's, leprosy, Volkmann's.

Brachial plexus (Erb's 7m S17, S21, W21, W24; Klumpke's 7m S25)

- **Erb's (C5–C6):** head and shoulder forced apart (two-wheeler fall). Lost: abduction (deltoid, supraspinatus), ER (infraspinatus), elbow flexion (biceps, brachialis, brachioradialis), supination. **"Waiter's tip"**: adducted, IR, elbow extended, pronated; outer arm numb; biceps jerk lost.
- **Klumpke's (C8–T1):** arm pulled overhead (grabbing a branch). Lost: intrinsics, finger and wrist flexors; **claw hand**; inner forearm numb; **Horner's** if T1 avulsed.
- **Root avulsion:** Horner's, winged scapula, raised hemidiaphragm, burning pain; **SNAP preserved** in the numb area (dorsal root ganglion still joined); no recovery without nerve transfer. Postganglionic: Tinel's in the neck, SNAP absent, may recover.

S22 level (2m): no fist (C8), elbow extension (C7) or thumb (C8–T1); inner arm and forearm numb → **lower plexus, C7–T1**; check Horner's.

Investigations and medical management

NCS and EMG: fibrillations from **2–3 weeks**; repeat to show reinnervation. MRI or CT myelography for avulsion. Degree I–II: splints, exercise, watch. Clean cut: early repair; gap: sural graft. Plexus: **nerve transfer** best within **6 months** (about 3 usual); late: tendon transfers.

Assessment and ICF (W16: 35, shopkeeper, right Erb's, 5 months)

Assess: dominant hand, job; wasting, skin; **MMT proximal to distal**, trick movements; sensory map, **Tinel's level**; Froment's, Phalen's; grip, Nine-Hole Peg, DASH.

- **Structure/function:** weak deltoid, spinati, biceps, brachialis, supinator; waiter's tip, subluxation risk; tight IR and adductors; outer arm numb.
- **Activity:** cannot lift to shelves, carry goods, comb, eat with right hand; the hand works.
- **Participation:** running his shop; family income at risk.
- **Environment:** family, own shop he can rearrange (facilitators); high shelves, surgery cost (barriers). **Personal:** sole earner, young (favours recovery).

Goals: short-term (4–6 weeks): full passive shoulder range, supported arm; flicker → gravity-eliminated movement; protects numb skin; **refer to a plexus surgeon now if no recovery**. Long-term (6–12 months): biceps and deltoid grade 4; lifts goods to chest height; runs the shop.

PT management by phase

- **Denervation:** functional splint; **gentle daily passive ROM**; prevent overstretch; oedema and skin care; strengthen working muscles.

- **Reinnervation** (Tinel's advancing, flicker): **one muscle at a time**, EMG biofeedback or stimulation; gravity-eliminated → active → resisted; few repetitions, short sessions; stop trick movements.
- **Functional:** grip and pinch, task practice, work simulation. **After nerve transfer:** contract the donor to fire the target (squeeze fingers to bend elbow).
- **Sensory re-education:** protective (look at the hand) → desensitisation (cotton → towel → rough) → discriminative (eyes open then closed, objects).

Splints (3m W17, S25, W25)

Static cock-up	Radial palsy (night); CTS (neutral); wrist **20–30° extension**, fingers free
Dynamic cock-up	Radial palsy by day: wrist support + **outrigger with elastic slings**
Knuckle-bender	Claw: blocks MCP hyperextension
Opponens	Median: thumb in opposition
AFO; sling	Foot drop; flail arm

Dynamic cock-up principle (S25): replaces paralysed extensors; wrist held up so flexors grip strongly; elastic re-extends fingers and thumb after each grasp. Uses: radial palsy, extensor tendon repair, tendon transfers. Prevents extensor overstretch and flexor shortening.

Specific lesions

- **Radial (7m S18, W18):** static cock-up at night, dynamic by day; ROM; re-educate brachioradialis and wrist extensors first, then fingers and thumb.
- **Claw (7m W19):** knuckle-bender; stretch MCPs into flexion; intrinsics (finger spreading, **table-top**); tendon transfer if no recovery.
- **Foot drop (3m S16), peroneal (7m W25):** AFO; daily calf stretch; DF and evertor re-education, FES; gait training; protect numb foot.
- **Erb's:** support arm; full range esp. **ER, abduction, supination**; re-educate deltoid, spinati, biceps **gravity-eliminated** (powder board, sling suspension), biofeedback; refer for transfer if no recovery by 3–6 months.
- **Klumpke's (S25):** anti-claw splint; finger and wrist range; intrinsic re-education; inner-hand sensory care.
- **Carpal tunnel (7m S22, APTA 2019): neutral night splint** (moderate evidence); education; manual therapy, heat (weak). **Do not use** laser, iontophoresis, magnets, thermal ultrasound. Refer if thenar wasting.

Complications and precautions

Contracture (thumb web, equinus) → splints, daily range; **overstretch** of paralysed muscles → cock-up, AFO, sling; burns on numb skin; oedema; CRPS; learned non-use. No heat or stimulation over numb skin unchecked; protect repairs; check splint pressure; no traction on a flail arm.

Education and prognosis

Regrowth **2.5 cm a month**: keep exercising for months; splint care; gloves, test water. **Better:** low degree, young, distal, early repair. **Worse:** traction, avulsion, old age, delay, proximal. Upper plexus recovers better than lower.

Exam plan (long case)

1. Restate: age, job, dominant side, mechanism, time; nerve or plexus level and likely degree.
2. **ICF (5–7)** with reasoning; one-line functional diagnosis.
3. **Goals (3–6):** range, protection, splints, first movement; strength, work; refer for surgery on time.
4. **Management (7–8):** splints; passive range; re-education with biofeedback; strengthening; sensory re-education; work; precautions.

Draw: brachial plexus with Erb's and Klumpke's levels; ICF; dynamic cock-up splint. **Underline:** Wallerian degeneration, Sunderland, 1 mm a day, Tinel's sign, waiter's tip, root avulsion, sensory re-education.

Peripheral neuropathies and nerve root lesions

Diabetic neuropathy 5 times; leprosy, alcoholic neuropathy, CRPS.

Full notes p. 107

Asked: long cases S16 (diabetic woman, numb feet, "walking on cotton", slippers slip off: investigations, goals, PT) and W25 (labourer, numb weak hands, red patches: leprosy diagnosis with reasoning, goals, PT). **7 marks:** diabetic neuropathy types and diabetic foot care (W17, W19, W22, S23), leprous neuritis and lepromatous leprosy (W22), reflex sympathetic dystrophy (S21), neuropathic pain (W21), alcoholic neuropathy (S25). No separate 3-mark questions.

Diabetic neuropathy: types (7m W17, W19, W22, S23)

India: about **101 million** adults with diabetes (11.4%). Long poor control damages nerves and their blood supply; **longest nerves fail first**, so it starts in the toes and rises.

Group (ADA 2017)	Types
Diffuse	**Distal symmetric polyneuropathy** (commonest, glove and stocking; small, large-fibre or mixed); **autonomic** (HR, BP, gut, bladder, sexual, sweating)
Mononeuropathy	One cranial or limb nerve (third cranial, median, ulnar, peroneal); mononeuritis multiplex
Radiculopathy	Lumbosacral radiculoplexus (**diabetic amyotrophy**: painful thigh weakness, wasting); thoracic

Diabetic: features and the diabetic foot

- **Sensory:** numbness, tingling, burning, "**walking on cotton wool**"; **loss of protective sensation**; poor position sense, balance worse in the dark.
- **Motor:** weak foot intrinsics → toes claw, **slippers slip off**, pressure under metatarsal heads.
- **Autonomic:** dry, cracked skin; orthostatic hypotension; resting tachycardia.
- **Diabetic foot:** neuropathy + PAD + deformity → ulcer, infection, amputation. Hot, swollen, painless foot = **Charcot foot**.

Investigations (3m in S16) and risk

Blood sugar, **HbA1c**; **10 g monofilament**, **128 Hz tuning fork**; NCS; foot pulses, **ankle-brachial index**; B12 (low on metformin), kidney, thyroid. **Wagner** ulcer grades 0–5 (0 at-risk, 1 superficial, 2 deep, 3 abscess or bone, 4 part gangrene, 5 whole foot). Control sugar, BP, lipids; stop smoking; plantar ulcer: **total contact cast** or non-removable walker.

IWGDF 2023	Findings; screen
0 very low	No LOPS, no PAD; yearly
1 low	LOPS or PAD; 6–12 months
2 moderate	LOPS + PAD, or either + deformity; 3–6 months
3 high	+ past ulcer, amputation, end-stage kidney disease; 1–3 months

ICF (S16: 50, diabetic 8 years, numb feet, slippers slip off)

- **Structure/function:** LOPS and lost position sense in both feet; weak intrinsics and toes; possible clawing, high-pressure areas; poor balance.
- **Activity:** unsteady walking, worse in the dark and on uneven ground; footwear falls off; fall risk.
- **Participation:** housework, market, religious outings, family roles.
- **Environment:** loose slippers, barefoot at home (barriers); family, diabetes clinic (facilitators).

Goals: short-term: daily foot-care routine, proper footwear, safer walking, no new skin damage. Long-term: no ulcers or falls, walks daily, good sugar control.

Diabetic foot PT

- **Ulcer risk: daily inspection with a mirror**; wash and dry between toes, moisturise, nail and callus care; protective footwear with a back strap; **no barefoot walking**, no socks-only or thin slippers; screen by risk category.
- **Foot-ankle exercises** (toe flexion and extension, ankle range and strength), preferably supervised: may lower plantar pressure.
- **Balance:** static → dynamic, with vision → varied surfaces; good home lighting.
- **Activity:** walking in proper footwear, building up to an **extra 1000 steps a day**; cycling or swimming if an ulcer is present. Foot risk is not a barrier if footwear is right (IWGDF 2023).
- **Ulcer:** offloading (cast, walker), crutches or wheelchair; keep the rest of the body active. **Pain:** graded activity, TENS trial.

Precautions: neuropathic feet

No heat, ice or electrical treatment on numb feet unchecked; check feet after every session, stop weight-bearing for a new red area or blister; **check sugar before exercise**, carry sugar; watch orthostatic hypotension; **hot swollen foot → refer the same day** (Charcot or infection).

Leprosy (W25 case; 7m W22)

Mycobacterium leprae infection of skin and peripheral nerves. India: about **59%** of world new cases (2023). **WHO:** paucibacillary (1–5 lesions, smear negative); multibacillary (more than 5, nerve involvement or smear positive). Ridley–Jopling: tuberculoid → borderline → lepromatous.

- **Features:** hypopigmented or reddish **anaesthetic patches**; **thickened nerves** (ulnar at elbow, common peroneal at fibular neck, posterior tibial, median, facial, great auricular). Claw hand, foot drop, clawed toes, plantar ulcers, lagophthalmos.
- **Lepromatous (W22):** many symmetrical lesions and nodules, **leonine face**, lost eyebrows, collapsed nose, glove-and-stocking anaesthesia, symmetrical nerve thickening, eye damage; smears strongly positive.
- **Leprous neuritis (W22):** painful, tender, swollen nerve with new sensory or motor loss, often in a reaction (type 1 reversal; type 2 erythema nodosum leprosum); **silent neuritis** = loss without pain.
- **Disability grade:** 0 none; 1 anaesthesia only; 2 visible deformity.
- **Medical (WHO 2018):** rifampicin, dapsone, clofazimine for **6 months** (PB) or **12 months** (MB); steroids and rest for neuritis; tendon transfers once controlled.

Leprosy: W25 reasoning and PT

W25 (42, labourer, sole earner, slum): leprosy, probably **multibacillary**, with ulnar and median neuropathy: patches + sensory loss + weak numb hands; check nerve thickening at elbows and wrists; confirm by smear; rule out diabetic, hereditary, cervical root. Impairments: intrinsic weakness, early claw, dry numb skin; activity: grip, tools, burns; participation: work, income, **stigma**; free MDT (facilitator), slum, hot or sharp tools (barriers).

- **Monitor:** voluntary muscle and monofilament testing of hands, feet, eyes every visit (catch new damage for steroids).
- **Acute neuritis: rest in a splint**, no exercise of the inflamed nerve; refer for steroids; then gentle range.
- **Claw hand:** knuckle-bender, finger stretch, intrinsics; before and after tendon transfer.
- **Self-care: soak, scrape, oil**; daily inspection; padded handles, gloves. **Foot:** AFO, microcellular rubber footwear, short walks, ulcer care.
- **Work and stigma:** tool changes, vocational training, family counselling; not spread by touch, treated patients soon non-infectious.

Alcoholic neuropathy (7m S25)

Length-dependent axonal sensorimotor polyneuropathy from heavy drinking + poor nutrition (**thiamine**). Burning feet, glove and stocking numbness, lost ankle jerks, **sensory ataxia** (worse in the dark), distal weakness, foot drop. Others: B12, kidney, hypothyroid, isoniazid, chemotherapy, HIV, CMT. Medical: stop alcohol, thiamine, diet; slow, partial recovery.

PT: balance with visual feedback, stick, lighting, **Frenkel's exercises**; progressive ankle and hand strengthening, AFO; graded activity, TENS; foot inspection and footwear; fitness, support to stay off alcohol. Watch withdrawal, low sugar, liver disease.

Neuropathic pain (7m W21) and CRPS (7m S21)

Neuropathic pain (IASP): pain from a lesion or disease of the somatosensory system: burning, shooting, **allodynia**, hyperalgesia (diabetes, post-herpetic, SCI, stroke, phantom). Drugs: gabapentin, pregabalin, duloxetine, amitriptyline. **PT:** explain pain (hurt is not harm); graded activity with pacing (no boom-and-bust); desensitisation; sensory discrimination, **graded motor imagery**, mirror; TENS; relaxation, sleep.

CRPS (RSD): pain out of proportion after injury or immobilisation (wrist fracture); type I no nerve injury, type II (causalgia) with one. **Budapest:** symptoms in 3 of 4, signs in 2 of 4 groups: sensory, vasomotor, sudomotor and oedema, motor and trophic. **PT:** early gentle use; graded motor imagery (left/right → imagined → mirror); desensitisation; oedema (elevation, pumping, retrograde massage); graded exercise, return to work; team. **No forced painful stretching; no long splinting.**

Exam plan

1. **Diabetic foot 7m:** classify (2): diffuse, mono, radiculopathy → assess (1): monofilament, fork, pulses, IWGDF → PT (4): inspection, footwear, exercises, balance, walking, offloading, precautions.
2. **Leprosy case:** diagnosis with reasoning (patch + lost sensation + thick nerve) → ICF → goals → monitoring, neuritis rest, splints, self-care, footwear, surgery rehab, stigma.

Draw: ICF from S16; sole with monofilament sites and ulcer sites (metatarsal heads, heel, toe tips). **Underline:** glove and stocking, loss of protective sensation, 10 g monofilament, offloading, thickened nerves, multidrug therapy, soak scrape oil, graded motor imagery.

Vestibular disorders

8 seven-mark questions: BPPV, adaptation, habituation, tests.

Full notes p. 116

Asked: 7 marks only, in 8 papers: BPPV physiotherapy (S20, W21), Dix-Hallpike + one treatment (S26), principles + two manoeuvres (S24), vestibular rehabilitation (S22), tests: enumerate, explain two (W24), signs + adaptation exercises (S17), substitution and habituation (W25).

System and lesion signs (7m S17)

3 canals (rotation), utricle and saccule (linear, gravity), nerve, nuclei, cerebellum. **VOR:** eyes move opposite to the head, vision stays clear. Balance = vision + somatosensation + vestibular.

Signs: vertigo; **nystagmus** (peripheral: horizontal-torsional, beats away from the bad side, suppressed by fixation); imbalance, veers towards the bad side; **oscillopsia** (bilateral); nausea; hearing loss, tinnitus (Ménière's, labyrinthitis); anxiety, avoids head movement. **Central** (stroke, MS): direction-changing or vertical nystagmus, normal head impulse, cannot stand, diplopia, dysarthria. **HINTS red flag:** normal head impulse, direction-changing nystagmus or skew deviation → emergency referral.

Others: neuritis (days of vertigo, no hearing loss: early gaze and balance); labyrinthitis (+ hearing loss); Ménière's (hours, fluctuating hearing, tinnitus); bilateral loss (gentamicin: substitution); persistent postural-perceptual dizziness (habituation).

Tests (7m W24)

- **Head impulse:** fix on your nose, quick 10–20° turns; **catch-up saccade** = weak VOR on that side.
- **Dix-Hallpike:** sit, head 45° to one side, lie back, neck **20° below horizontal**; **torsional upbeating nystagmus** after a delay, under a minute = posterior canal BPPV (lower ear).
- **Supine roll** (head up 30°, turn 90°): horizontal canal. **Dynamic visual acuity** (head moved 2 per second): loss of more than 2 lines. **Head-shaking** 20 s. **Fukuda** 50 steps, turn more than 45°. **Romberg**, **mCTSIB** (4 conditions, 30 s, foam).
- Dizziness Handicap Inventory; Motion Sensitivity Quotient (MSQ); Dynamic Gait Index; TUG.

BPPV (7m S20, W21, S24, S26)

Loose **otoconia** from the utricle into a canal, mostly **posterior**; brief vertigo (under a minute) on lying down, turning in bed, looking up; no hearing loss. Diagnose with Dix-Hallpike; if negative, other side; roll test if horizontal.

Epley (right): head 45° right → lie back, neck extended 20°, wait ~30 s → head 90° left → roll onto left side, nose down, ~30 s → sit up, chin down → repeat Dix-Hallpike. **Semont:** head 45° away → quickly onto affected side ~1 min → swing through to other side nose down ~1 min → sit slowly. Horizontal canal: **barbecue (Lempert) roll**, Gufoni. **Brandt–Daroff**: home, residual symptoms.

Classic vs current: no sleeping upright 48 h or collar: **no postural restrictions** (2017); no routine suppressants or imaging; reassess within 1 month.

Vestibular rehabilitation (7m S22, S24, S17, W25)

Start early (within 2 weeks of acute loss); mild symptoms in exercise are expected; **stop suppressants** early; supervised + home, progressed; add balance and gait.

Adaptation	**Retinal slip** drives cerebellum to raise VOR gain. **VOR ×1:** fixed target, head moves; **VOR ×2:** target opposite to head. Progress speed, background, sitting → walking. One-sided loss
Habituation	Repeated exposure to provoking movements (from the MSQ) a few times, twice a day; Brandt–Daroff. Motion sensitivity
Substitution	Vision and touch; look at target then turn head; stick, night lights. Bilateral loss

Dose (Hall 2022): acute: gaze **3 times a day, 12 min** total; chronic: **3–5 times, 20 min daily, 4–6 weeks**; bilateral: 20–40 min, 5–7 weeks, balance 6–9 weeks. **Classic vs current:** no eye movements with head still (Cawthorne–Cooksey start); use head movement. **Balance:** narrow base, foam, eyes closed; head turns while walking; dual task; crowds, stairs.

Precautions

Refer if headache, diplopia, slurred speech, weakness or central HINTS; before Dix-Hallpike screen neck and vertebral artery (side-lying test if unsafe); guard, vomit bowl; no driving until settled. Ménière's: not during attacks.

Exam plan (BPPV 7m)

What it is (1) → Dix-Hallpike with result (2) → Epley or Semont steps, retest, Brandt–Daroff (3) → no restrictions, recurrence, falls (1). **Draw:** Dix-Hallpike position; VOR ×1 and ×2 arrows. **Underline:** VOR, torsional upbeating nystagmus, canalith repositioning, adaptation, habituation, substitution.

Facial (VII) nerve palsy and other cranial nerves

Bell's palsy in 6 papers; trigeminal neuralgia twice.

Full notes p. 124

Asked: long case W19 (23-year-old teacher with Bell's palsy: ICF, goals). **7 marks:** Bell's palsy PT with education (W16, W17), features and management (S24), trigeminal neuralgia (S20, W22). **3 marks:** facial nerve assessment (S16), trigeminal assessment, synkinesis, Bell's phenomenon (W18), signs of Bell's palsy (W24).

Definition, causes, grading

Bell's palsy: acute, one-sided, **LMN** facial palsy, no cause found, over **less than 72 h**; viral swelling in the bony canal → mostly neurapraxia. Other LMN: **Ramsay Hunt** (zoster, ear vesicles, worse), trauma, ear, parotid, GBS (both sides), leprosy. **UMN** (stroke): **forehead spared** (bilateral cortical supply). **House-Brackmann:** I normal; II mild; III eye closes with effort; IV eye does not close; V barely moves; VI total.

Features (3m W24; 7m S24), Bell's phenomenon, synkinesis (3m W18)

Whole half face over hours to 3 days; pain behind ear; cannot raise eyebrow; **eye does not close**; flat nasolabial fold; mouth droops, pulls to the normal side; food in cheek, drinks dribble; unclear p, b, m; watery or dry eye; **taste lost front two-thirds**; loud sounds (stapedius).

Bell's phenomenon: on trying to close the eyes, the eyeball on the weak side **rolls up and out**; visible because the lid does not close. **Synkinesis:** involuntary movement of one part during voluntary movement of another (eye narrows on smiling) from **misdirected regrowth**, months after severe palsy.

Assessment (3m S16, W18) and medical

Facial nerve: raise eyebrows (frontalis); close eyes against resistance (orbicularis oculi); smile, show teeth; puff cheeks, whistle (buccinator, orbicularis oris); tense neck (platysma); taste, Schirmer's tears, sound. **Trigeminal:** touch and pinprick V1 forehead, V2 cheek, V3 jaw; clench (masseter, temporalis); jaw deviates to weak side on opening; corneal reflex, jaw jerk.

Medical: oral steroids within 72 h (complete recovery 64% → 83% at 3 months); antivirals not alone; eye drops, night ointment and taping, glasses; refer if incomplete at **3 months**.

ICF (W19: 23, teacher, right Bell's palsy)

- **Structure/function:** all right facial muscles weak; eye does not close (cornea at risk); mouth droops; later synkinesis.
- **Activity:** lip sounds, eating, drinking, expression. **Participation: teaching**, being seen by students; social outings.
- **Environment:** early care, family, school (facilitators); dust, wind, stigma (barriers). **Personal:** young (good outlook), anxious about appearance.

Goals: short-term (2–4 weeks): no corneal damage; flicker → visible movement, symmetry at rest; lip seal. Long-term (2–3 months): full eye closure; **House-Brackmann I–II**; normal speech for a teaching day; no troublesome synkinesis.

PT by stage (7m W16, W17, S24)

- **Acute:** reassurance; eye care; gentle massage (upward strokes to eye and ear); moist warmth with eye protected; cheek taping; chew on strong side.
- **Recovering: mirror**-guided active-assisted → active: eyebrows, eye closure, smile, purse, puff, flare nostrils; **small, slow, symmetrical**; hold the strong side back; a few repetitions, several times a day.
- **Late:** neuromuscular retraining with mirror or EMG biofeedback, relax unwanted muscle; stretch, massage.
- **Advice (W16):** most recover; eye drops, night taping, glasses; eat on strong side, clean cheek, straw; **no forceful grimacing or gum chewing**.

Classic vs current: no faradic or galvanic stimulation: **no better than placebo** (Cochrane 2011).

Trigeminal neuralgia (7m S20, W22)

Sudden **electric shock-like** one-sided pain, V2 or V3, seconds to ~2 min; **trigger zones** (touch, talking, chewing, brushing, cold wind); normal between attacks. Usually a vessel on the root; MS or tumour if young. **Carbamazepine**; microvascular decompression. **PT:** avoid triggers (scarf, lukewarm water, soft brush); relaxation; TENS away from trigger zones; **never stimulate trigger zones**.

Precautions and prognosis

No heat with the eye open; refer if weakness spreads, other cranial nerves, ear vesicles, no recovery by 3 months. Worse: complete palsy, older, Ramsay Hunt, diabetes, no recovery by 3 weeks.

Exam plan

Bell's 7m: features (2–3) → steroids, eye care (1) → education, massage, mirror exercises, taping, retraining, evidence note (3–4). **Draw:** LMN vs UMN faces; ICF. **Underline:** forehead spared in UMN, Bell's phenomenon, steroids within 72 hours, synkinesis.

Multiple sclerosis

6 long cases; the second-relapse executive case was asked 3 times.

Full notes p. 133

Asked: an MS long case in 6 papers: the 29-year-old executive after a second relapse (W19, S24, S25: ICF, goals, PT); medical representative with visual loss and imbalance (S23); student with poor fine motor control (S18: goals, PT); teacher with ataxia and diplopia (W17: functional diagnosis). **7 marks:** visual, cognitive and behavioural disturbances (W18), PT for MS (S19), stages + PT for one stage (S20), features + PT (W20). **3 marks:** clinical features (W16, S22), Uhthoff's phenomenon (W22), Charcot's triad (W25).

Definition and causes

MS: chronic immune-mediated CNS disease; inflammation strips myelin in brain, cord and optic nerves → plaques, later axonal loss; lesions **scattered in space and time**. 2.8 million worldwide; diagnosis at about **32**; women twice as often; less common in India. **Risks: Epstein–Barr virus** (strongest), low vitamin D and sunlight, smoking, teenage obesity, genes (HLA-DRB1*15:01). Relapses after infection, after childbirth.

Courses (7m S20) and EDSS

Clinically isolated	First attack (optic neuritis) that may become MS
Relapsing-remitting	**About 85% at onset**; relapses days to weeks, full or partial recovery
Secondary progressive	After relapsing years, steady worsening ± relapses
Primary progressive	Worse from the start; slow walking difficulty

Each: active or not, progressing or not (Lublin 2014). **EDSS:** 0 normal; 1–3.5 walks fully; 4–5.5 walks unaided, about **500 m at 4.0, 100 m at 5.5**; **6.0 stick for 100 m**; 6.5 support both sides for 20 m; 7–7.5 wheelchair; 8–9.5 bed or chair; 10 death. Disability stages: mild 0–4, moderate 4.5–6.5, severe 7+.

Pathophysiology and features (3m W16, S22; 7m W20)

- Immune cells cross the blood–brain barrier → plaques near ventricles, brainstem, cerebellum, cord, optic nerves → slow or blocked conduction; symptoms by site.
- Inflammation settles, some remyelination → **remission**; axon loss → **progression**.

Features: numbness, band feelings, **Lhermitte's sign** (shock down spine on neck flexion); **optic neuritis** (painful eye movement, blurred central vision, faded colours), diplopia, nystagmus; leg weakness, spasticity, UMN signs; ataxia, intention tremor, dysarthria; **fatigue** (commonest, most disabling); bladder urgency, constipation, sexual; slowed thinking, depression, lability; pain, trigeminal neuralgia; heat sensitivity.

Uhthoff's phenomenon (3m W22): temporary worsening (blurred vision, weakness) when body temperature rises (exercise, hot bath, weather, fever): heat slows conduction in demyelinated fibres; settles on cooling; **not a relapse**. Manage: pre-cooling, cool room, breaks. **Charcot's triad (3m W25): nystagmus, intention tremor, scanning (staccato) speech**: cerebellar and brainstem.

Visual, cognitive, behavioural (7m W18)

Visual	Optic neuritis, diplopia, nystagmus, jumping vision, poor colour → good lighting, **high-contrast step edges**, uncluttered space, eye referral, gaze exercises
Cognitive	Slow processing, memory, attention, planning → **short simple instructions**, written programme, quiet room, no dual tasks at first
Behavioural	Depression, anxiety, irritability, sudden laughing or crying, low motivation → agree goals, encourage, exercise lifts mood, counselling

Investigations and medical

MRI brain and spine; **McDonald 2024:** lesions in 2 of 5 regions (periventricular, cortical, brainstem-cerebellum, cord, now optic nerve); new lesions over time not always needed; **CSF oligoclonal bands** or kappa free light chain index; visual evoked potentials; bloods for mimics (B12, neuromyelitis optica antibodies). **Relapse:** methylprednisolone 3–5 days (speeds recovery, not the course). **Disease-modifying:** interferon beta, glatiramer, fingolimod, natalizumab, **ocrelizumab** (also primary progressive). Baclofen, tizanidine, fampridine, botulinum toxin.

PT assessment

Course, last relapse, **fatigue pattern**, heat sensitivity, bladder, falls, work; **EDSS**; strength, MAS, range, finger-nose, heel-shin; sensory, vision, cognition; **Berg, Timed 25-Foot Walk**, TUG, 6MWT, MS Walking Scale-12; **Modified Fatigue Impact Scale** or Fatigue Severity Scale; Nine-Hole Peg.

ICF (W19, S24, S25: 29, unmarried executive, 2nd relapse, discharged 1 week, moderately dependent)

- **Structure/function:** relapsing-remitting, recovering; leg weakness, spasticity; poor balance, possible ataxia; **fatigue**, heat sensitivity (Uhthoff's); possible sensory, visual, bladder; low endurance after hospital; possible low mood, slow thinking.
- **Activity:** moderate help for **transfers, bathing, toileting**; short walking distance; stairs, travel.
- **Participation:** off work; social life; worry about career and marriage.
- **Environment:** family, large employer (flexible or home working), medicines (facilitators); commuting, hot weather, long hours (barriers). **Personal:** young, educated: can self-manage.

Goals: short-term (2–4 weeks): independent bed-chair and toilet transfers, bathes with shower chair; stronger hips and knees, full calf and hamstring length; walks indoors with a stick; pacing without next-day exhaustion; plans phased return. Long-term (2–3 months): independent self-care; stairs; outdoor walking without aid, no falls; full working day; back at work, **150 min a week**.

PT by phase (7m S19, S20, W20)

- **Relapse:** rest the affected function; keep range, positioning, gentle activity; avoid overheating.
- **Recovery:** grade back up: transfers, balance, walking, strength, endurance; short sessions with rests.
- **EDSS 0–6.5:** aerobic and resistance doses below; balance and gait; fatigue; work.

- **EDSS 7–9:** short daily sessions; upper-body exercise; standing frame; transfers; seating and pressure care; breathing; carer training.

Kalb 2020	Aerobic	Resistance
EDSS 0–6.5	**2–3 times a week, 10–30 min**, moderate (40–60% max HR, RPE 11–13)	2–3 times a week; **1–3 × 8–15**; 5–10 exercises
EDSS 7–9	Up to 20 min a day, 3–7 days, with rests; arm intervals	Upper body 3 times a week, 3 × 10

Everyone: **150 min a week** of exercise or activity, built gradually. **Classic vs current:** not "avoid exertion and heat": exercise is safe at every level; **cool them rather than stop**.

Other techniques

- **Spasticity:** daily stretching, standing, positioning; check triggers (bladder, skin, infection).
- **Balance and gait:** static → dynamic, stepping, turning, dual task when safe; AFO for foot drop; aids.
- **Ataxia, intention tremor:** proximal stability, **Frenkel's**, slow practice with visual feedback; **weighted wrist cuffs**.
- **Fine motor (S18):** grip and pinch; writing, buttoning, eating with thick pens, adapted cutlery.
- **Fatigue: energy conservation** (plan, prioritise, pace, sit for tasks), rests, exercise at the coolest time.
- **Heat: pre-cooling** (cool drink, shower, fan), cooling vest, cool water swimming; no hot packs or hot baths.
- **Bladder:** timed voiding, pelvic floor (see bladder (p. 41)).

Precautions and complications

Avoid overheating; stop before exhaustion, **worse next day → reduce dose**; new symptom over **24 h** without fever or heat = possible relapse: refer; fever or UTI → **pseudo-relapse**; guard balance work; after steroids: weaker bones, sugar, mood. Complications: falls, deconditioning, contractures (calf, hamstrings, hip flexors, adductors), pressure sores, osteoporosis.

Education and prognosis

Not contagious; most keep working; home exercise towards 150 min; energy conservation; keep cool; workplace changes (flexible hours, home working, seat near toilet, cool office, less travel); stop smoking; take disease-modifying medicines. **Better:** female, young, sensory or optic onset, good first recovery. **Poorer:** older, motor or cerebellar onset, frequent early relapses, progressive from start. PT in blocks after relapses.

Exam plan (long case)

1. Restate: age, job, course, time since last relapse, function (EDSS).
2. **ICF (5–7)** with reasoning; one-line functional diagnosis.
3. **Goals (3–5):** transfers, self-care, balance, fatigue; walking, work, lifelong exercise.
4. **Management (7–8):** phase; doses by EDSS; balance, gait; spasticity; ataxia, fine motor; fatigue and heat; bladder; work; precautions.

W17 functional diagnosis (3m): cerebellar-brainstem MS: diplopia, wide-based ataxic gait, slurred speech, intention tremor (near Charcot's triad), many MRI lesions (space); steps into house. **Draw:** three courses graph; ICF. **Underline:** demyelination, space and time, Uhthoff's, Charcot's triad, fatigue, EDSS, energy conservation, cooling, 150 minutes.

Guillain–Barré syndrome

9 long cases, often ventilated or just out of the ICU.

Full notes p. 144

Asked: a GBS long case in 9 papers (ICU with breathing difficulty S24, S25; on the ward after plasma exchange W16; tracheostomy, now standing W17; child out of ICU who must cycle to school W25; S17, S18, S20, W21): ICF with reasoning, goals, detailed PT; diagnosis with rationale (S18). **7 marks:** PT for GBS (S19), acute-stage PT (W21). **3 marks:** EMG and NCV picture (W17, within the case).

Definition and types

GBS: acute, immune-mediated **polyradiculoneuropathy**: rapidly progressive, usually symmetrical limb weakness with **absent or reduced reflexes**, worst within **4 weeks**. **LMN**: brain and cord normal. 1–2 per 100,000 a year; **two-thirds** have an infection in the 6 weeks before (diarrhoea, chest).

AIDP	Acute inflammatory demyelinating; sensory and motor; usual in Europe and Americas (~70%; Asia 30–40%)
AMAN	Acute motor axonal; pure motor; after **Campylobacter**; common in Asia
AMSAN	Motor and sensory axonal; severe, slow
Miller Fisher	Eye-muscle weakness, ataxia, areflexia (anti-GQ1b)

Hughes disability scale: 0 healthy; 1 can run; 2 walks 10 m unaided; 3 walks 10 m with help; 4 bed or chair; 5 ventilated; 6 dead.

Pathophysiology and features

- Infection (Campylobacter, cytomegalovirus, hepatitis E, Mycoplasma, Epstein–Barr, Zika) → **molecular mimicry** → attack on myelin (slows, blocks) or axon (stops).
- **Ascending** flaccid weakness, legs → arms, trunk, breathing, face; proximal as weak as distal; reflexes lost.
- Phrenic and intercostals → **up to 22% ventilated** in week 1; cranial → **bilateral facial**, swallowing; autonomic → BP and HR swings, arrhythmia, retention; sensory → tingling, lost position sense, back and leg pain; fatigue.
- Recovery: remyelination (weeks), axon regrowth (months), so axonal forms are slower. Weakest within 2 weeks; recovery can go on over 5 years.

Diagnosis: required: progressive weakness of arms and legs + reduced reflexes. Supports: progression days to 4 weeks, symmetry, mild sensory signs, bilateral facial, autonomic, pain. **Doubt:** sensory level, early bladder loss, clear asymmetry, CSF over 50 cells (cord or infection).

Investigations (NCV, EMG 3m W17)

CSF: high protein, normal cells (**albuminocytological dissociation**); normal in 30–50% in week 1. **Breathing:** VC, maximum inspiratory and expiratory pressure, single breath count, ABG. ECG; stool; MRI spine if cord possible.

	Demyelinating	Axonal
Velocity, latency	Slowed; prolonged	Near normal
F-waves	**Prolonged or absent: earliest sign** (roots first)	Reduced or absent
Amplitude	Conduction block, temporal dispersion	Low motor amplitudes
Sensory	Abnormal arms, **sural sparing**	Normal in AMAN
EMG	Reduced recruitment	Fibrillations after 2–3 weeks

Medical management

If unable to walk unaided: **IVIg 0.4 g/kg a day for 5 days** within 2 weeks, or **plasma exchange** 12–15 L over 4–5 exchanges within 4 weeks; equal; not one after the other; **no steroids**, no second IVIg. **ICU, intubate by the 20/30/40 rule:** VC under **20 mL/kg**, maximum inspiratory pressure under 30, expiratory under 40 cmH_2O; single breath count **19 or less**. DVT prophylaxis, ECG, nutrition, eye care, nerve-pain drugs.

PT assessment

Preceding infection, onset, speed, immunotherapy, ventilation, swallowing; HR, rhythm, BP lying and sitting, SpO_2; **VC, single breath count**, cough, auscultation; MMT, **MRC sum score (0–60)**, facial and bulbar; position sense, pain; ROM, skin; Hughes scale, bed mobility, sitting, transfers, 10 m walk, 6MWT; Fatigue Severity Scale.

ICF (S25: 45, man, viral fever, 10 days, breathless, ICU)

- **Structure/function: breathing first**: low VC, weak cough (phrenic, intercostals); symmetrical flaccid weakness of limbs and trunk, areflexia; facial, swallowing; tingling, pain; unstable BP, HR; chest, DVT, pressure, contracture, pressure-palsy risk; anxiety.
- **Activity:** cannot turn, sit, stand, walk; dependent for feeding, washing, toilet; cannot clear secretions.
- **Participation:** off work; family roles.
- **Environment:** ICU, immunotherapy, family (facilitators); cost, noisy ICU (barriers). **Personal:** 45, likely earner; fear of ventilator.

Goals: short-term (1–3 weeks): chest clear, VC monitored and stable; full range, no ulcers or pressure palsies; sits out with support, tolerates tilt table; feeds with set-up; short sessions without next-day exhaustion. Long-term (3–6 months): normal breathing and cough; walks indoors and outdoors (AFO if foot drop); independent ADL; endurance for work.

Acute stage: ICU (7m W21)

- **Chest:** positioning (side lying, upright), deep breathing, **incentive spirometry**, airway clearance, suctioning, assisted cough; **VC or single breath count each shift**, report VC under 20 mL/kg. Ventilated: secretion clearance, support weaning.
- **Limbs:** neutral positioning; **resting splints** (wrist, hand) and AFO or foot support; **gentle passive ROM once or twice a day**; pad elbows and fibular heads; regular turns.
- **DVT:** passive and active leg movements, stockings as prescribed. **Pain, fear:** gentle handling, explain before touching, reassure; teach the family.
- Sit up and tilt table once breathing and BP are stable; watch postural hypotension.

Recovery (7m S19)

- **Early (ward):** active-assisted → active, **gravity eliminated first**; sitting balance; rolling, bridging, sit to stand; standing in bars; short sessions with rests.
- **Rehabilitation:** progressive resistance, **few repetitions, rests, stop before fatigue**; gait: bars → walker → stick, **AFO** for foot drop; balance with vision (mirror, then eyes closed); hand function; pacing; return to school, work.
- **Fitness (Garssen 2004):** cycling **30 min at 70–90% max HR, 3 times a week for 12 weeks**, 5 min warm-up and cool-down.
- **Avoid overwork:** recovering nerves supply fewer fibres; **weaker or more tired next day → shorten or lighten**.
- **Child (W25, cycles to school):** spirometry and bubble games; strengthening through play; putty, blocks, writing; static cycling before road cycling; pacing at school; parents' home programme.

Precautions and complications

Stop and report: falling VC or breath count, rising RR, weaker cough; HR, rhythm, BP swings on turning, suctioning (arrhythmia), sitting up (tilt back). No overstretching of flaccid joints; support shoulders; upright for meals if swallowing weak; protect eyes. **Complications:** respiratory failure, aspiration pneumonia, DVT, sacral and heel ulcers, **pressure palsies** (ulnar at elbow, peroneal at fibular head), contractures (ankle PF, shoulder, fingers), pain, hallucinations, depression.

Education and prognosis

Not contagious; most recover, slowly; family: positioning, ROM, transfers; graded home programme; staged return; seek help if weakness returns (fluctuations 6–10%, recurrence 2–5%). **About 80% walk at 6 months**; death 3–10%; poorer: older, diarrhoea before, severe (modified Erasmus GBS outcome score); pain in a third and **fatigue in 60–80%** at 1 year. Worsening after 8 weeks or 3+ relapses → acute-onset CIDP (chronic inflammatory demyelinating polyneuropathy).

Exam plan (long case)

1. Restate: age, days since onset (progression, plateau, recovery), breathing and ICU status, mobility, home and role.
2. **ICF (5–7)**, breathing first; functional diagnosis.
3. **Goals (3–5)** for the present phase; walking, self-care, fatigue, role.
4. **Management (7–10):** ICU chest care, monitoring, positioning, splints, range → graded strengthening, sitting, standing, AFO gait; balance; endurance; education; precautions (autonomic, overwork).

S18 diagnosis: GBS, likely pure motor axonal: gut infection 3 weeks before, 2-day symmetrical weakness, proximal = distal, normal sensation, **delayed F-waves**; rule out myasthenia, cord lesion, low potassium, polymyositis. **Draw:** course of GBS with PT focus; ICF. **Underline:** ascending, areflexia, albuminocytological dissociation, 20/30/40, autonomic instability, avoid overwork.

Cerebellar disease and ataxia

3 long cases (Friedreich's twice, cerebellar stroke) and 3 seven-mark questions.

Full notes p. 155

Asked: long cases W24 (Friedreich's ataxia, 20-year-old woman: ICF, goals, PT), W22 (cerebellar stroke, Berg 38: ICF, PT), W18 (Friedreich's picture: structural impairments, tests, PT). **7 marks:** PT of cerebellar ataxia (S16, W16), rationale of balance training (W19). **3 marks:** rebound test (W17), assessment of sensory ataxia (S19).

Definition, types, causes

Ataxia: errors in direction, range, timing and force of movement, **not due to weakness**. **Cerebellar:** eyes open and closed, nystagmus, dysarthria, intention tremor; hemisphere → **same-side** limbs, vermis → trunk and gait. **Sensory** (posterior columns, nerves): worse eyes closed, **Romberg positive**, stamping gait. **Vestibular:** vertigo, limbs normal. **Causes:** posterior stroke, Friedreich's, MS, alcohol, phenytoin, tumour, B12 and vitamin E lack, ataxic CP.

Features, rebound (3m W17), sensory ataxia (3m S19)

Failed error correction → **dysmetria**, **decomposition**, **intention tremor**; dysdiadochokinesia, hypotonia; titubation, **wide-based staggering gait**, no tandem walk; scanning speech. Slower motor learning → more repetitions.

Rebound test: patient flexes elbow against resistance; examiner suddenly lets go, **guarding the face**; cerebellar: arm flies to the face (Holmes' rebound). **Sensory ataxia:** Romberg; position and vibration (128 Hz) lost from the toes; finger-nose, heel-shin worse eyes closed; pseudoathetosis; stamping, watches feet.

Friedreich's (W18, W24): autosomal recessive (GAA, frataxin), before 25; gait and limb ataxia, dysarthria, **lost ankle jerks + extensor plantars**, lost position sense, **pes cavus, kyphoscoliosis**, **cardiomyopathy** (main death cause), diabetes; wheelchair in 10–20 years. **W18 structural (2m):** spinocerebellar, posterior column, corticospinal degeneration; spine and foot deformity.

Assessment (W18)

Finger-nose, heel-shin, rapid alternating, rebound, circles, eyes open and closed; Romberg; tandem walk; **Berg** (below 45 = fall risk), TUG, 10 m; **SARA** (8 items, 0–40); Nine-Hole Peg; spine, feet.

ICF (W24: 20, Friedreich's 2 years, tremor, falls, poor fine motor)

- **Structure/function:** gait and limb ataxia, intention tremor; lost position sense (vision helps only partly); poor balance; dysarthria; possible scoliosis, cardiomyopathy.
- **Activity:** repeated falls, stairs, writing, buttoning, eating. **Participation:** college, travel, outings, marriage plans.
- **Environment:** family, devices (facilitators); stairs, crowded transport, stigma (barriers). **Personal:** young, educated, anxious.

Goals: short-term (4–6 weeks): knows fall triggers, aid outdoors; better Berg and SARA; writes with adapted pen; **cycles 20 min**. Long-term (3–6 months): fewer falls; walks safely at college with an aid; independent self-care and study.

W22 (65, right cerebellar stroke, diabetic, Berg 38): right limb and truncal ataxia, high fall risk, neuropathy adds sensory loss → balance over 3 h a week, right-limb coordination, stick or rollator, sugar check.

PT (7m S16, W16; rationale 7m W19)

- **Intensive coordinative training** daily for at least **4 weeks**, then home programme (Ilg 2009: gains lasted a year).
- **Proximal stability:** prone on elbows → four-point → kneeling → half-kneeling → standing; approximation; rhythmic stabilisation.
- **Balance:** reducing base, reaching, stepping, perturbation; **over 3 h a week** for falls. **Gait:** wide base then narrow, lines, obstacles, treadmill, rollator or weighted walker.
- **Arm:** slow movements to targets with feedback, elbows on table, **weighted cutlery and pens**. **Sensory part: Frenkel's**, vision first.
- **Fitness:** stationary or recumbent cycling towards 150 min a week. Friedreich's: posture, foot orthoses, seating.

Rationale (W19): the cerebellum adapts through practice and error; repeated balance challenge teaches remaining circuits to reduce sway, strengthens postural muscles, builds confidence, cuts falls.

Precautions

Gait belt, bars; Friedreich's: breathlessness, palpitations, sugar; cerebellar stroke: new headache, vomiting, drowsiness; fatigue worsens ataxia; no alcohol or sedatives.

Exam plan

Restate type, cause, falls → ICF, cerebellar vs sensory signs → tests, SARA → intensive coordination, stability, gait, arm, Frenkel's, fitness. **Draw:** ICF; wide-based gait. **Underline:** dysmetria, intention tremor, Romberg, proximal stability.

Parkinson's disease and other extrapyramidal disorders

7 long cases and 6 seven-mark questions.

Full notes p. 164

Asked: a PD long case in 7 papers (W18, S19, W19, S21, W21, W22, S22; mostly Hoehn and Yahr III): ICF with reasoning, goals, physiotherapy; home management (S19); fall prevention (S22). **7 marks:** gait abnormalities and PT (S16, W17), gait training (W16, S17), bradykinesia and rigidity (S22), PT at stage III (S26). **3 marks:** spasticity vs rigidity (W17), rigidity types (W19, S24), Hoehn and Yahr (W20), basal ganglia functions (W21), cardinal features (S25).

Definition and causes

PD: slowly progressive loss of **dopamine neurons of the substantia nigra**; basal ganglia cannot control movement. Diagnosis: **bradykinesia + rest tremor or rigidity** (MDS 2015), plus non-motor problems. About 1% over 60. Parkinsonism is the picture; PD its commonest cause.

Other parkinsonism: drug-induced (haloperidol, metoclopramide); vascular (legs, little tremor); **Parkinson-plus** (PSP: early falls, poor vertical gaze; MSA: autonomic failure, ataxia), poor levodopa response. **Risk: age**; genes 5–10%; pesticides, rural living. **Subtypes:** tremor-dominant; **PIGD**: more falls, faster progress.

Hoehn and Yahr (3m W20)

1	One side only; little or no disability (1.5: plus trunk)
2	Both sides, balance normal (2.5: recovers on pull test)
3	Both sides, mild–moderate; **some postural instability**; physically independent
4	Severe disability; can still walk or stand unaided
5	Wheelchair-bound or bedridden unless helped

Stage III = turning point: falls begin, still independent.

Basal ganglia (3m W21) and pathophysiology

- **Functions:** select and start wanted movement, suppress unwanted; **scale size, speed, force**; run learned sequences **automatically**; motor learning, habits.
- Nigral neurons die, **Lewy bodies** → ↓ striatal dopamine → thalamus and cortex over-inhibited → **slow, small movements**.
- Automatic control lost → walking needs attention; worse with a second task; freezing. **External cues** supply the timing.
- Non-motor signs may come years before motor signs.

Cardinal features (3m S25), rigidity (3m W19, S24)

TRAP

Tremor at rest, 4–6 per second, pill-rolling, absent in sleep. **Rigidity** whole range, flexors and extensors. **Akinesia/bradykinesia:** slow, hard to start, movements shrink; mask face, soft voice, small handwriting. **Postural instability:** later; falls back on pull test.

Lead-pipe: smooth, even, any speed. **Cogwheel:** ratchet (tremor on lead-pipe). **Decorticate:** arms flexed, legs extended; above red nucleus. **Decerebrate:** all extended, arms IR; brainstem (worse).

(3m W17)	Spasticity	Rigidity
Lesion	UMN (pyramidal)	Extrapyramidal
Speed	**Velocity-dependent**	Same at any speed
Pattern	Clasp-knife; antigravity	Lead-pipe, cogwheel; both groups
Reflexes	Brisk, clonus, Babinski	Normal

Gait (7m S16, W17) and other features

Flexed posture (line of gravity in front of feet); short **shuffling** steps, no heel strike; **festination** (shorter, faster steps chasing the centre of gravity; propulsion, retropulsion); ↓ arm swing and trunk rotation; **en bloc turns**; start hesitation and **freezing** (doorways, turns, talking); worse with dual task and "off" period.

Non-motor: loss of smell, constipation, acting out dreams, depression, dizziness on standing, urgency, later dementia; dysphagia.

Investigations and medicines

Clinical diagnosis; levodopa response supports; DaTscan if uncertain; MRI excludes vascular, NPH. **Levodopa + carbidopa** (most effective; later "off" periods, dyskinesia); agonists, MAO-B, COMT, amantadine; anticholinergics avoided in elderly; **DBS**. Physio: **train in the "on" period**; note last dose time.

PT assessment

Medicine timing, falls, freezing, ADL, home. **H&Y; MDS-UPDRS III**. Kyphosis, contractures, chest expansion, Five Times Sit to Stand. **Mini-BESTest** or Berg; pull test. 10 m walk, **TUG ± dual task**, 6MWT, freezing questionnaire. Lying and standing BP; PDQ-39.

ICF (W21: 65, retired teacher, akinesia, tremor, slow gait)

- **Structure/function:** PD (H&Y II–III); akinesia, bradykinesia, rigidity, rest tremor, flexed posture, ↓ trunk rotation, balance (stage III), ↓ fitness.
- **Activity:** slow short-stepped gait; chair rise, turning in bed, dressing, buttoning, writing, eating.
- **Participation:** social, religious, community outings; tutoring.
- **Environment:** family, medicines, exercise groups (facilitators); rugs, poor light, low seats, Indian toilet (barriers). **Personal:** educated, follows strategies.

Goals: short-term (4–6 weeks): chair rise in one attempt; longer steps with cue; more trunk rotation; better Mini-BESTest; exercises 3 days a week. Long-term (3–6 months): daily tasks without help; outdoors without cues; **no falls**; community class, keeps roles.

Principles and stages (stage III: 7m S26)

Exercise early and for life; **big, fast, purposeful**; cues and strategies bypass lost automatic control; real tasks; prevent falls; **"on" period**. **APTA 2022 strong:** aerobic, resistance, balance, gait, cueing, community exercise; weak: flexibility. **Stages 1–2:** fitness, community classes; **2–4:** add cued gait, strategies, falls, posture, home; **5:** positioning, pressure care, breathing, carer.

Aerobic	**≥3 days a week, 30–40 min, 60–75% to 75–85% max HR**; use RPE

Resistance	1–2 days a week; **80% 1RM** strength, **40% 1RM fast** for power
Balance	Reactive steps, turning, dual task: **2–3 times a week, 16–30 h over 5–10 weeks**
Posture	Trunk extension and rotation, chest expansion; prone lying

Classic vs current: not gentle ROM only; moderate–high intensity is safe (SPARX 2018).

Gait training (7m W16, S17), bradykinesia and rigidity (7m S22)

- **Cues:** floor stripes at step length; metronome, music; "big step". **Attention** to long steps, heel strike.
- Arm swing; treadmill; Nordic poles; **wide-arc turns**; doorways; stops, starts. **Festination:** stop, stand tall, long step.
- **Dose: 20–60 min, 3–5 days a week, 4–12 weeks**; cueing 2–5 times a week.
- **Freezing:** stop, stand tall, shift weight; step over a line; count or march; target beyond doorway.
- **Cognitive strategies:** chair rise in steps: front of seat, feet back, **"nose over toes"**, push, stand tall.
- **Bradykinesia: LSVT BIG** (16 one-hour sessions, 4 days a week, 4 weeks); speed, power.
- **Rigidity:** slow rhythmic rotation; **rhythmic initiation** (passive → active); stretch pectorals, hip flexors, hamstrings; relaxation.

Precautions and fall prevention (4m S22)

Complications: falls (~**60%** yearly, **39%** repeatedly), kyphosis, contractures, deconditioning, orthostatic hypotension, aspiration pneumonia, pressure sores (stage 5). **Take care:** train in "on" period; gait belt; lying and standing BP; **RPE as HR may be blunted**; dual tasks only when safe; no reclined exercise after meals. **Falls:** find risks; balance + leg strength; wide turns, freezing cues; home safety, footwear; stand slowly; teach floor rise.

Home management (5m S19) and prognosis

Remove rugs and clutter; lighting, switch at bed; grab bars; raised western toilet or commode; firm chairs with armrests; wide turning paths; **floor stripes at doorways**. Demanding tasks in "on" period; sit to dress; Velcro, elastic laces; weighted cutlery, non-slip mat; bed rope or rail. Daily programme + community class; medicines on time; fluids, fibre. **Prognosis:** slow; PIGD faster; falls, dysphagia, dementia mark late stages (death often pneumonia). PT in blocks with review.

Exam plan (long case)

1. Restate: age, years, H&Y stage, medicines, falls, freezing, roles.
2. **ICF (5–7)** with reasoning; functional diagnosis.
3. **Goals (3–5)**: gait, transfers, posture, balance; no falls, lifelong exercise.
4. **Management (7–10)**: principles; doses; cued gait; freezing; strategies; rigidity; falls, home; "on" timing.

Draw: parkinsonian posture with line of gravity; ICF.
Underline: dopamine, bradykinesia, lead-pipe, cogwheel, festination, freezing, external cues, "on" period.

Anterior horn cell diseases: MND, SMA and poliomyelitis

ALS long cases (S16, W16); MND types and terminal care.

Full notes p. 177

Asked: ALS long cases (W16: 29-year-old teacher; S16: 36-year-old dentist): impairments or ICF with reasoning, goals, PT. **7 marks:** classify MND with features (S23), PT for a terminally ill patient (W25). **3 marks:** diagnostic criteria (S18), pseudobulbar palsy (S19), types of MND (S21, W24).

Definition and types (3m S21, W24; 7m S23)

MND: progressive degeneration of **UMN and LMN** (cortex, brainstem, anterior horns): weakness, wasting, spasticity, then breathing failure; **sensation spared**. 1–3 per 100,000 a year; after 50; ~10% familial (SOD1, C9orf72).

ALS (commonest)	UMN + LMN: wasting, fasciculations with spasticity, brisk reflexes, Babinski; one hand or foot first
Progressive bulbar palsy	Brainstem nuclei: slurred speech, dysphagia, wasted fasciculating tongue; worse outlook
Progressive muscular atrophy	LMN only: wasting, fasciculations, lost reflexes
Primary lateral sclerosis	UMN only: slow spasticity; best outlook

Criteria (3m S18): Gold Coast 2020: progressive motor loss after normal function; **UMN + LMN in 1 region or LMN in 2**; tests exclude others.

Features and pseudobulbar palsy (3m S19)

LMN loss → wasting, **fasciculations**, cramps; UMN loss → spasticity, brisk reflexes even in wasted limbs. **Split hand**; foot drop; head drop; night hypoventilation; eyes, sensation, bladder spared.

	Pseudobulbar (UMN)	Bulbar (LMN)
Lesion	Both corticobulbar tracts (strokes, MND, MS)	Nuclei IX–XII (MND, GBS, polio)
Tongue	Small, stiff, no wasting	Wasted, fasciculating
Signs	Spastic speech, **brisk jaw jerk**, **emotional lability**	Nasal speech, weak gag

Investigations, medical care, assessment

EMG (denervation, normal sensory); MRI excludes myelopathy; **FVC**, night SpO_2. **Riluzole** (+2–3 months); **NIV**; gastrostomy; palliative care. **PT:** MRC, **ALSFRS-R (0–48)**, grip, timed walk, Berg, FVC sitting and lying, cough peak flow.

ICF (W16: 29, teacher, sole earner, ALS 4 months)

- **Structure/function:** shoulder, hand and leg weakness; **fatigue** (fewer motor units overworked); cramps, spasticity; check breathing.
- **Activity:** writing on the board, standing through a lesson, long walks, buttoning.
- **Participation:** full teaching days; family income at risk.
- **Environment:** family, school may adapt (facilitators); blackboard, stairs, cost (barriers). **Personal:** sole earner; coping with diagnosis.

Goals: short-term (4–6 weeks): energy conservation, rests between lessons; lower whiteboard or projector, sits to teach; no falls. Long-term (3–6 months): keeps teaching with adaptations; self-care as long as possible; NIV and equipment in place before urgent. **S16 dentist (split hand):** thumb and wrist splints, thicker instrument handles.

PT by stage

- **Early:** moderate, **non-fatiguing** strengthening of muscles **grade 3 or more** (few reps, rests); walking, cycling, swimming; daily ROM; pacing; work ergonomics.
- **Middle: AFO**, wrist and thumb splints, neck collar; aids, then wheelchair; transfers; breathing, airway clearance; stretch for cramps.
- **Late:** positioning, pressure care, passive ROM, assisted cough, NIV support, hoist, tilt-in-space chair, carer training.

Terminally ill (7m W25): comfort positions, profiling bed, turning; gentle passive ROM (shoulders); breathlessness: forward lean, **hand-held fan**, breathing control, NIV mask comfort; secretions: assisted cough, cough machine, suction; hoist, head support; respect wishes and advance care plan; palliative team.

Precautions and prognosis

Never to exhaustion (Cochrane 2013: moderate exercise no harm); no heavy eccentric work or strengthening below grade 3; stop if cramps or weakness rise next day; refer orthopnoea, morning headache; upright for meals + 30 min. Median survival ~**3 years**; bulbar onset faster, PLS slowest.

Exam plan

Restate → **ICF (3–7)** with reasons → **goals (4–5)** → **management (4–7)** by stage, precautions. **Draw:** motor pathway, UMN and LMN levels. **Underline:** UMN and LMN together, sensation spared, non-fatiguing, NIV.

Myopathies and myasthenia gravis

Myasthenia gravis as 7 marks (W19) and 3 marks (W24).

Full notes p. 186

Asked: myasthenia gravis only. **7 marks:** PT management of MG (W19). **3 marks:** three features of MG (W24). Expect also: myasthenic vs cholinergic crisis, Lambert–Eaton, myotonia.

Myasthenia gravis: definition and types

MG: autoimmune disease of the **neuromuscular junction**: antibodies block and destroy **ACh receptors**, so muscles weaken with use and recover with rest. AChR antibodies in ~85% of generalised MG; MuSK in some. Ocular or generalised (ocular usually generalises within 2 years). Peaks in younger women, older men; **10–15% thymoma**.

MGFA: I eyes only; II mild; III moderate; IV severe (IVb feeding tube); V intubated. II–IV: a limb and trunk, b bulbar and breathing.

Features (3m W24)

- **Fatigable weakness:** worse with repetition and by evening, better with rest.
- **Ptosis and diplopia** (commonest first symptom).
- **Bulbar:** nasal speech, chewing and swallowing difficulty, "snarling" smile.
- Neck (head drop), proximal limbs; breathlessness lying flat; may fail suddenly (crisis).
- **Reflexes, sensation, bladder normal**; wasting unusual. Triggers: infection, overexertion, **heat**, stress, surgery, some drugs.

Diagnosis, medicines, crisis

Antibodies; repetitive nerve stimulation (falling responses), single-fibre EMG; ice-pack test; chest CT. **Pyridostigmine** peaks **1–2 h after a dose**; steroids, immunosuppressants; thymectomy (sternal precautions); IVIG or plasma exchange in crisis.

	Myasthenic	Cholinergic
Cause	Flare: infection, missed dose, drug	Too much anticholinesterase
Signs	Rapid weakness, breathing and swallowing fail	Weakness + sweating, saliva, cramps, diarrhoea, small pupils, twitching

Either: stop, sit upright, call the team, support airway.

PT for MG (7m W19)

Assess strength before and after repetition, VC, cough, swallowing. **ICF:** fatigable weakness, ptosis → stairs, chair rise, reading → full work day; heat a barrier.

- **Exercise:** moderate resistance and aerobic in stable MG, start low, short sets with rests; aim **150 min a week** (Gilhus 2021).
- **Timing: 1–2 h after pyridostigmine**, earlier in day, cool room.
- **IMT** + breathing retraining: **3 times a week, 8 weeks, 20% → 60% MIP** (Fregonezi 2005).
- Airway clearance: huffing, assisted cough, positioning.
- **Energy conservation:** plan, prioritise, pace, sit; rest before tired.
- Soft collar; eyelid support or alternating eye patch; fall-proof home. After thymectomy: breathing, early walking, sternal precautions.

Classic vs current: not "rest and avoid exercise"; graded training is safe when stable.

Precautions (MG)

No exercise in flare, infection or crisis; stop if breathing, speech or swallowing worsen; **avoid heat** (hot packs, pools) and exhaustion; watch orthopnoea, weak cough, choking, falling VC; tell team before botulinum toxin.

Other muscle diseases

- **Lambert–Eaton:** calcium-channel antibodies, ↓ ACh release; half small-cell lung cancer; proximal leg weakness, dry mouth; **strength and reflexes improve briefly with effort**.
- **Myositis:** dermatomyositis (rash, screen for cancer); polymyositis (high CK); inclusion body (over 50, quads, finger flexors, falls). Active: ROM, isometrics, breathing; stable: progressive resistance, aerobic, functional.
- **Myotonic dystrophy 1:** distal weakness, ptosis, **myotonia** (slow grip release; warm-up eases), cataracts, heart conduction: ECG follow-up, AFO.
- **FSHD:** face (cannot whistle), winged scapulae. **Limb-girdle:** hip and shoulder girdles. **Becker:** milder dystrophin, calf enlargement, cardiac (DMD (p. 55)).

Rules: moderate aerobic and strength training safe (Cochrane 2019, low quality); **no eccentric, high-intensity or exhausting work**; review if pain, cramps or weakness last to next day or **dark urine**; screen heart, VC sitting and lying; orthoses early.

Exam plan (MG, 7 marks)

Intro (1): NMJ, fatigable weakness → assessment and aims (1) → treatment (4): graded training after pyridostigmine, IMT, clearance, energy conservation, aids → precautions (1): heat, crisis, triggers. **Draw:** NMJ with antibodies blocking receptors. **Underline:** fatigable weakness, ptosis and diplopia, pyridostigmine timing, IMT, myasthenic crisis, no exhaustion.

Disorders of the autonomic nervous system

Horner's syndrome; blood-pressure disorders; autonomic dysreflexia.

Full notes p. 193

Asked: 3 marks: clinical features of Horner's syndrome (S23). Autonomic dysreflexia comes in most papers inside SCI (full answer on spinal cord (p. 15)). Expect orthostatic hypotension and autonomic care in tetraplegia.

Autonomic nervous system

	Sympathetic	Parasympathetic
Outflow	**Thoracolumbar (T1–L2)**	**Craniosacral** (III, VII, IX, X; S2–S4)
Actions	"Fight or flight": ↑ HR and BP, sweating, wide pupils	"Rest and digest": slows heart, gut, bladder emptying
Physio	Lost below cord lesion **at or above T6**: low BP, dysreflexia, no sweating	Vagus intact: **slow HR on suction or turning** in acute tetraplegia

Horner's syndrome (3m S23)

Definition: loss of sympathetic supply to one side of the face and eye. **Same side:** mild **ptosis**, **miosis** (clearest in dim light; still reacts to light), **anhidrosis** of the face, apparent **enophthalmos**.

Neuron	Course; causes
1st (central)	Hypothalamus → brainstem → cord to T1; lateral medullary stroke, syringomyelia, cervical SCI, MS
2nd (preganglionic)	T1 root over lung apex → superior cervical ganglion; **T1 avulsion (Klumpke's)**, Pancoast tumour, neck surgery
3rd (postganglionic)	Along internal carotid to eye; carotid dissection, cluster headache, cavernous sinus

Physio link: Horner's in a plexus injury = T1 root torn from the cord → **poor recovery of the hand** (brachial plexus (p. 52)).

Orthostatic hypotension

Definition (2011): sustained fall of **≥20 mmHg systolic or ≥10 mmHg diastolic within 3 minutes** of standing or head-up tilt. **Causes:** SCI at or above T6, PD and MSA, diabetic and autonomic neuropathies, GBS, bed rest, drugs (BP tablets, levodopa, diuretics). **Symptoms:** dizziness, greying vision, yawning, nausea, fainting; tetraplegia: few symptoms despite big falls.

PT for orthostatic hypotension

- **Measure BP lying and after sitting or standing**; record symptoms.
- **Gradual uprighting:** head of bed up → **tilt table** or sitting, raise angle in small steps as BP allows.
- Reduce pooling: **abdominal binder**, compression stockings, ankle pumps before rising.
- Slow changes; sit on bed edge first; fluids as advised; no heat or big meals before exercise.
- **Counter-manoeuvres:** cross legs, tense leg and buttock muscles, squat.
- Symptoms: **stop, recline or lower tilt, raise legs, check BP**.

Dysreflexia and other problems

Autonomic dysreflexia: SCI at or above T6; stimulus below lesion (most often **full bladder**) → sudden high BP, pounding headache, flushing and sweating above lesion, slow pulse. **Sit up, loosen clothing, check bladder and bowel, monitor BP.**

- **Poikilothermia** (high SCI): no sweating or shivering below lesion; cool room, avoid heat and cold.
- GBS dysautonomia: rhythm and BP swings. Brain injury: **paroxysmal sympathetic hyperactivity** (fast HR, high BP, fever, sweating, posturing): calm handling.
- Bladder and bowel: bladder and bowel (p. 41); CRPS: neuropathies (p. 19).

C6 tetraplegia (7m practice): binder, stockings, gradual tilt; dysreflexia steps, no tight straps, check catheter; cool room; slow HR with suction; bladder and bowel routine first; teach carers.

Exam plan (Horner's, 3 marks)

Definition (sympathetic loss, one side) → triad **ptosis, miosis, anhidrosis** + enophthalmos → causes by neuron level with the T1 avulsion link. **Draw:** three-neuron pathway: hypothalamus → T1 → superior cervical ganglion → carotid → eye. **Underline:** ptosis, miosis, anhidrosis, T1 root avulsion, 20/10 in 3 minutes, tilt table, binder, counter-manoeuvres, poikilothermia.

Psychosomatic pain and paralysis

In the syllabus; not asked on its own in 2016–2026.

Full notes p. 198

Asked: not asked on its own. Expect a 3-mark question on Hoover's sign, features of functional paralysis or functional weakness vs malingering, or a 7-mark question on physiotherapy for functional paralysis.

Definition and types

Functional neurological disorder (FND) (older: conversion disorder, hysteria, psychosomatic paralysis): a problem of how the nervous system **works, not its structure**; **real, involuntary** weakness, abnormal movements, seizures or numbness without structural damage. Common; often after a trigger (injury, illness, pain, panic, stress). Can coexist with other disease (functional seizures with epilepsy).

Not malingering (deliberate faking for gain: money, leave) nor factitious disorder (producing symptoms to be a patient). People with FND are not pretending.

Types: functional weakness (one limb, one side, both legs); movement disorders (tremor, fixed dystonia: clenched fist, inverted ankle; jerks; gait); dissociative seizures; sensory, dizziness, cognitive; with fatigue and pain.

Positive signs (diagnosis is positive, not only exclusion)

Hoover's sign	Hip extension of the weak leg weak on direct test, **returns to normal when the other hip flexes against resistance** (automatic); heel presses into examiner's hand
Hip abductor sign	"Weak" leg abducts strongly when the other leg abducts
Give-way weakness	Normal power, then sudden collapse
Drift without pronation	Arm drifts down without pronation (pyramidal drift pronates)
Dragging gait	Leg dragged behind, hip turned in or out; not circumducted
Entrainment (tremor)	Tremor takes up the rhythm copied by the other hand, pauses or turns irregular
Distractibility	Less when attention is elsewhere

Reflexes, tone and plantars normal; no early wasting.

PT for functional motor disorders (Nielsen 2015)

- **Explain positively:** symptoms real, nervous system not damaged, can relearn; **show the positive sign** (normal movement in Hoover's).
- Set expectations; agree goals.
- **Retrain with attention directed away** from the limb: automatic movements, rhythm, dual tasks, music, imagery; start with what works (walking backwards, sideways, treadmill) → normal patterns.
- **Minimal aids**, braces and hands-on support; withdraw gradually.
- **Self-management:** daily practice, pacing, setback plan, return to work.
- Team: psychology (CBT), psychiatry for anxiety, depression, trauma.

Classic vs current: not "nothing is wrong, refer to psychiatry"; physio is core. Physio4FMD 2024: no better 12-month function score than community physio, but more felt improved; likely cost-effective.

Practice case (28, clerk, right leg after a fall, stick): short-term walks indoors without stick; long-term normal outdoor walking, back to work; treadmill, rhythm, backwards, dual tasks; stick withdrawn; graded return to work.

Chronic pain with psychological factors

Central sensitisation; worse with fear, low mood, poor sleep, inactivity; pain is real. Pain education, **graded activity and graded exposure**, exercise, pacing, sleep and stress strategies, cognitive-behavioural approach; fewer passive treatments. Screen red flags first; refer if distress high.

Exam plan (7 marks)

Define FND vs malingering (1) → positive signs (1–2) → explanation, goals, retraining with redirected attention, minimal aids, graded progression, self-management, team (4). **Draw:** Hoover's test, hand under the heel of the "weak" leg while the other leg lifts. **Underline:** real and involuntary, not malingering, positive signs, Hoover's, entrainment, redirected attention, self-management.

Electrotherapy in neurological rehabilitation

Pain relief and functional re-education with electrical stimulation.

Full notes p. 204

Asked: not asked on its own. Expect 3 marks on FES, TENS types or the strength-duration curve, or 7 marks on electrotherapy in hemiplegia or nerve injury.

Uses in neurology

Pain: TENS. **Muscle activation:** NMES, FES. **Feedback:** EMG biofeedback. **Denervated muscle:** long-pulse stimulation (limited evidence). **Electrodiagnosis:** strength-duration curve (largely replaced by EMG and nerve conduction).

TENS for pain

Type	Settings; mechanism
Conventional	**80–100 Hz**, short pulses, strong comfortable tingling, no twitch; **gate control**, works while on
Acupuncture-like	**2–4 Hz**, longer pulses, visible twitches; **endorphins**, may outlast session
Burst, brief intense	Bursts of high-frequency pulses; brief intense for procedures

Neuro uses: painful hemiplegic shoulder (with positioning and movement), postural musculoskeletal pain, some neuropathic pain. Electrodes over the area or along the nerve; patient must feel the tingling.

NMES and FES

NMES: stimulates an innervated nerve for strength or re-education. **FES:** stimulation **timed to a functional task** (dorsiflexors in swing; wrist and finger extensors in grasp and release). **Settings: 30–50 Hz**; **200–300 µs** (longer for big muscles); ramp up and down; on–off **1:3 to 1:5** at first; strong comfortable contraction; **20–30 min**. Always with active effort and task practice.

Shoulder subluxation	**Supraspinatus + posterior deltoid**, early: ↓ subluxation, ↑ pain-free lateral rotation, no clear effect on pain (Cochrane); UK 2023: consider within 6 months
Foot drop (stroke, MS)	Peroneal stimulation by heel switch or tilt sensor: **as good as an AFO** (Bethoux 2014)
Arm and hand	Wrist and finger extensors with task practice; add-on, some movement needed
SCI	FES cycling, standing, grasp: muscle and fitness
Disuse	Quadriceps NMES in bed-bound or ICU patients

EMG biofeedback

Surface electrodes show muscle activity as sound or light: increase activity in a weak muscle or reduce it in an overactive one. Early re-education after nerve injury or stroke (flicker before visible movement), relaxation, after tendon transfer. Add-on to exercise, not a replacement.

Denervated muscle and strength-duration curve

Denervated muscle responds only to **long pulses** (tens to hundreds of ms; interrupted direct or triangular); limited evidence; **no benefit in Bell's palsy**.

- **SD curve:** current for a minimal contraction plotted against pulse duration.
- **Rheobase:** smallest current that causes a contraction with a long pulse.
- **Chronaxie:** pulse duration at **twice the rheobase**; **under about 1 ms** normally, much longer when denervated.
- Denervation shifts the curve **up and to the right**; a **kink** = partial denervation; serial curves show reinnervation.

Precautions and contraindications

SCI at or above T6: stimulation below lesion can trigger **autonomic dysreflexia**: monitor BP, stop at once. Pacemakers, defibrillators (avoid chest); carotid sinus, front of neck, eyes; malignancy, infection, **DVT**, broken skin; abdomen or low back in pregnancy; head and neck in uncontrolled epilepsy. Numb skin or poor understanding: low intensity, frequent checks. **Check skin before and after**; explain the sensation.

Exam plan

3m FES: definition (timed to task) → settings + peroneal stimulator example → evidence (as good as AFO) or one precaution. **7m hemiplegia:** with task practice; NMES shoulder; FES grasp; peroneal FES; biofeedback; TENS; settings; precautions. **Draw:** SD curve, normal and denervated, rheobase and chronaxie marked. **Underline:** gate control, 30–50 Hz, timed to the task, shoulder subluxation, chronaxie, autonomic dysreflexia.

Neurotherapeutic approaches: Brunnstrom, Rood, Bobath, MRP, PNF and CIMT

Rood, Bobath, MRP and PNF in 8 papers.

Full notes p. 209

Asked: no long case. **7 marks:** principles of PNF (W25), MRP for the hemiplegic upper limb (W18), principles and techniques of Rood's approach (S17). **3 marks:** three Rood techniques for spasticity (S16), three Bobath principles (W16), three MRP principles (S18), reflex-inhibiting patterns with an example (W19), three Rood principles (S22).

Approaches and evidence

- **Neurophysiological** (1940s–1970s): Rood, Bobath, Brunnstrom, PNF: reflexes, sensory stimulation, handling to change tone and bring out movement.
- **Motor learning:** MRP, task-oriented training, CIMT: patient as an **active learner** practising real tasks.
- **Cochrane 2014 (Pollock):** rehabilitation combining components works after stroke; **no one named approach is more effective**; do not use compartmentalised approaches.

Classic vs current: not one approach for the whole programme; evidence-based components centred on **active, repetitive, task-specific practice**; passive handling alone does not improve function. End every answer: "used within task-oriented practice; no single approach superior (Cochrane 2014)".

Rood: principles (3m S22; 7m S17)

- Normalise tone and evoke the response with the **right sensory stimulus**: some facilitate, some inhibit.
- Sensorimotor control develops in sequence (**ontogenetic sequence**): treatment follows developmental patterns.
- Movement is **purposeful**: activities call out the response automatically.
- **Repetition** of sensorimotor responses is needed for learning.

Ontogenetic patterns: supine withdrawal → roll over → pivot prone → neck co-contraction → prone on elbows → all fours → standing → walking. **Levels of control:** mobility → stability → controlled mobility → skill.

Rood: techniques (spasticity 3m S16)

Stimulus	Facilitatory (hypotonia)	Inhibitory (spasticity)
Touch	Fast brushing over the muscle	**Slow stroking** down the back, **3–5 min**
Temperature	Quick icing	**Neutral warmth:** blanket wrap **10–20 min**; prolonged cold
Stretch, pressure	Quick stretch; tapping belly or tendon; high-frequency vibration	**Prolonged stretch**; maintained pressure over tendon insertion
Joint	Heavy compression; resistance	Light compression
Vestibular	Fast rocking, spinning	Slow rocking or rolling

Three for spasticity: slow stroking, neutral warmth, prolonged stretch. Effects short-lived, limited research: use to prepare for active practice, not instead.

Bobath / NDT (3m W16, W19)

- Abnormal tone and postural reflexes block normal movement: **inhibit first**.
- **Facilitate** normal automatic postural reactions (righting, equilibrium, protective) and normal patterns.
- Handling through **key points of control**: proximal (head, shoulder girdle, pelvis); distal (hands, feet).
- Treat the whole person; include the affected side in all activity (bilateral, symmetrical).

Reflex-inhibiting pattern (RIP): position or movement **opposite to the spastic pattern** to reduce tone before facilitating. **Arm:** shoulder forward and ER, elbow extended, forearm supinated, wrist and fingers extended, thumb abducted; e.g. **sitting with weight on the extended arm** beside or behind the body. **Leg:** hip and knee flexed, hip abducted and ER, ankle dorsiflexed. **Child with extensor thrust:** flexed side lying or carried in flexion.

Now: contemporary Bobath = problem-solving for postural control and function; not superior after stroke (Pollock 2014); passive NDT ineffective for motor goals in CP (Novak 2020).

Brunnstrom

Based on stages of recovery (flaccid → synergies with spasticity → voluntary synergies → out of synergy → isolated; stroke (p. 11)). Early: use associated reactions, tonic reflexes, synergies to bring out movement; then break out of synergy. **Phenomena:** associated reactions; **Souques'** (arm above ~150° releases finger extension); **Raimiste's** (resisted hip abduction or adduction of strong leg → same in weak leg); homolateral limb synkinesis. **Today:** stages for assessment only; do not train synergies.

Motor Relearning Programme (3m S18; 7m W18)

- Recovery = **relearning real-life tasks**: task-specific practice in context.
- Patient is an **active learner**; instruction, demonstration, feedback.
- **Eliminate unnecessary muscle activity** and compensation; practise missing components.
- **Practice and carry-over**: many repetitions on ward and at home; biomechanics and motor learning.

Four steps: analyse task → practise **missing components** → practise whole task → **transfer** to daily life. **Seven sections:** upper limb; oro-facial; sitting up over the side of the bed; balanced sitting; standing up and sitting down; balanced standing; walking.

Upper limb (W18): analyse reach, grasp and release, manipulation → in lying, reach to ceiling and control arm at points of range; in sitting, reach forward and sideways; wrist extension while grasping a cup; release; thumb opposition to each finger; turning hand over → whole tasks: drink from cup, place objects, comb, write → avoid shoulder hitching, trunk leaning; protect shoulder, prevent shortening → practice sheets for ward and home with family.

PNF principles (7m W25)

Philosophy: positive, builds on what the person can do, functional, strong parts help weak, whole person, motor learning.

- **Resistance:** optimal, to facilitate contraction and learning.
- **Irradiation and reinforcement:** strong muscles spread activity to weak in the pattern.
- **Manual contact:** lumbrical grip guides direction. **Body position** and mechanics in line with movement.
- **Verbal commands** short, clear, timed; **vision**: patient watches.
- **Traction** facilitates movement; **approximation** stability and weight-bearing.
- **Stretch** stimulus and reflex; **timing**: normal distal to proximal, timing for emphasis.
- **Patterns:** diagonal, spiral mass movements.

Sherrington basis: after-discharge, temporal and spatial summation, irradiation, successive induction, reciprocal innervation.

	Upper limb (shoulder)	Lower limb (hip)
D1 flexion	Flex, add, ER; wrist, fingers flex (hand to opposite ear)	Flex, add, ER; DF + inversion
D1 extension	Ext, abd, IR; wrist, fingers extend	Ext, abd, IR; PF + eversion
D2 flexion	Flex, abd, ER; wrist, fingers extend (drawing sword)	Flex, abd, IR; DF + eversion
D2 extension	Ext, add, IR; wrist, fingers flex (sheathing)	Ext, add, ER; PF + inversion

Techniques: agonist: **rhythmic initiation** (passive → assisted → active → resisted; starting, rigidity), combination of isotonics (control), repeated stretch; reversals: dynamic reversals (coordination), stabilising reversals and **rhythmic stabilisation** (stability); relaxation: **contract-relax, hold-relax** (range; hold-relax for pain).

Imagery, CIMT, transfer

Mental imagery added to practice improves arm function after stroke (low-quality, Cochrane 2020): short guided sessions, own view, of practised tasks. **CIMT:** restrain strong hand + intensive shaped practice; needs some wrist and finger extension; effective in children (Novak 2020).
Transfer: context-specific practice; part → whole; bilateral transfer; varied practice.

Exam plan (7 marks, any approach)

1. **Background (1):** who developed it, the idea.
2. **Principles (3):** four to six, one line each.
3. **Techniques (2–3):** three or four with use, one patient example.
4. **Evidence (0–1):** within task-oriented practice; no approach superior.

Draw: D1 and D2 arm diagonals as arrows across the body; RIP for hemiplegic arm. **Underline:** sensory stimulation, ontogenetic sequence, key points of control, reflex-inhibiting pattern, missing components, irradiation, diagonal patterns, rhythmic initiation, no approach superior.

Managing tone: spasticity, rigidity and hypotonia

Tone techniques in 9 papers.

Full notes p. 216

Asked: 7 marks: tone-inhibiting techniques (S19), PT of spasticity (W20), tone + PT of hypotonia (W17), facilitatory techniques (W24). **3 marks:** three facilitatory techniques (W16), slow icing (W18), Modified Ashworth (S19, W21), define and classify tone (S20), three methods to reduce tone (S23).

Tone: definition and types (3m S20)

Tone: resistance of a relaxed muscle to passive stretch; neural (stretch reflex, descending control) + non-neural (muscle, connective tissue stiffness). Normal: holds posture, allows selective movement.

Spasticity	**Velocity-dependent**, clasp-knife, brisk reflexes, clonus; UMN (stroke, SCI, MS, CP)
Rigidity	Same at any speed, lead-pipe or cogwheel; basal ganglia (PD)
Dystonia	Contractions, twisting postures; dyskinetic CP
Hypotonia	Floppy, hypermobile, ↓ reflexes; LMN, cerebellum, early stroke, spinal shock, Down's
Fluctuating	Low to high; athetoid CP

Spasticity (Lance 1980): velocity-dependent ↑ tonic stretch reflexes with exaggerated jerks, part of the UMN syndrome; negative features (weakness, poor dexterity) limit function more.

Modified Ashworth (3m S19, W21) and Tardieu

0 no increase; **1** catch and release or minimal resistance at end of range; **1+** catch then minimal resistance through **less than half** the remaining range; **2** marked through most of range, moves easily; **3** considerable, passive movement difficult; **4** rigid. Test relaxed, full range in ~**1 second**, max 3 times, grade the first. **Tardieu:** R2 slow full range, R1 fast catch; **large gap** = dynamic (stretch, cast, toxin); small gap = contracture.

Reducing tone (7m S19, W20; 3m S23, W18)

Treat when a problem (pain, hygiene, contracture, seating); extensor tone may help standing. **Triggers first:** pain, full bladder, UTI, constipation, sores, ingrown toenails, tight clothes.

- **Positioning** opposite the spastic pattern.
- **Prolonged stretch** held minutes: positioning, splint, standing frame, cast.
- **Weight-bearing** through extended arm; plantigrade standing (stretch + compression).
- **Slow icing (W18):** wrapped ice pack or immersion **15–20 min**; slows conduction, calms spindle; use window to stretch and practise (quick icing facilitates).
- Neutral warmth, slow stroking; **rhythmic rotation**; serial casts, resting splint, AFO.
- Electrical stimulation of antagonist (reciprocal inhibition); **active task practice**, strengthen antagonists.

Three (S23): prolonged stretch, slow icing, weight-bearing. **Team:** baclofen, tizanidine, dantrolene; **botulinum toxin + stretch, splint, task practice** (AHA/ASA 2026); intrathecal baclofen. **Classic vs current:** stretch under 7 months: no important effect on mobility, little on spasticity (Cochrane).

Facilitation and hypotonia (3m W16; 7m W17, W24)

- **Quick stretch, tapping** before movement (spindle).
- **Approximation** in weight-bearing (co-contraction); light graded **resistance** (irradiation).
- Fast brushing, quick icing (Rood); high-frequency **vibration** (tonic vibration reflex).
- Weight-bearing, bouncing on a ball; fast vestibular input; PNF repeated stretch, rhythmic stabilisation; electrical stimulation.

Hypotonia PT: protect hypermobile joints (shoulder support, no overstretch); supported anti-gravity positions (prone over wedge, sitting, standing frame); facilitation just before active movement; co-contraction in kneeling and standing; head and trunk control, sit-to-stand, walking; carer handling.

Precautions

Cold: check sensation and circulation; avoid in Raynaud's. **Never force a spastic joint.** Facilitation can raise tone elsewhere: stop. Fast vestibular: not in seizures, raised ICP, nausea. Splints and casts: check skin daily.

Exam plan (spasticity, 7 marks)

Define (Lance) + MAS, Tardieu, function, triggers (1–2) → techniques with reason (4) → team: toxin with therapy, drugs, intrathecal baclofen (1). **Draw:** stretch reflex arc (spindle, Ia, alpha motor neuron) with lost descending inhibition. **Underline:** velocity-dependent, triggers, prolonged stretch, slow icing, weight-bearing, approximation, task practice.

Balance and coordination training

Balance strategies and training in 4 papers.

Full notes p. 223

Asked: 7 marks: training static and dynamic standing balance (S16), evaluating sensory and motor strategies of balance (W18). **3 marks:** three strategies to improve dynamic balance (S17), three motor strategies of balance (S18).

What balance needs

Balance: keeping the **centre of mass over the base of support**: static, dynamic, reactive.

- **Sensory:** vision, vestibular, somatosensation; brain **reweights** them (on foam: more vestibular and vision).
- **Central:** cerebellum, basal ganglia, brainstem, cortex.
- **Motor:** strength, range (ankle), postural strategies; **anticipatory** (before movement) and **reactive** (after disturbance).
- **Cognition:** worse with dual tasks.

Motor strategies (3m S18)

Ankle	Small, slow disturbance, firm wide surface; sways as one piece about ankles; **distal → proximal**
Hip	Larger, faster, or narrow or soft surface; quick hip flexion or extension; **proximal → distal**
Stepping (reaching)	Centre of mass outside base: rapid step or grab makes a new base

Sometimes added: suspensory (bend knees, lower centre of mass).

Evaluating strategies (7m W18)

Sensory: mCTSIB: 30 s each in four conditions: firm eyes open, firm eyes closed, foam eyes open, foam eyes closed. Falls firm eyes closed = **somatosensory loss**; foam eyes closed = **vestibular loss** or poor reweighting. Romberg, sharpened Romberg (sensory ataxia); Sensory Organisation Test (six conditions).

Motor: push and release or nudge each direction: ankle, hip or step, quick and effective?; pull test (PD). **Functional reach** along a ruler at shoulder height (**less than ~15 cm** = high fall risk); multidirectional reach; rising on toes. **BESTest / Mini-BESTest** (biomechanics, stability limits, anticipatory, reactive, sensory orientation, gait). **Berg (0–56)**, TUG, DGI or FGA, dual-task walking.

Static and dynamic training (7m S16)

Change	Easier → harder
Base	Feet apart → together, semi-tandem, tandem, single leg
Surface	Firm → foam, wobble board, uneven
Vision	Eyes open → closed, moving scenes
Support	Hands on bars → no hands → holding an object
Head; attention	Still → turning, nodding; single → talking, counting back, tray

Static: hold (e.g. **30 s**) before progressing; even weight with mirror or scale (stroke).

Dynamic (3m S17):

- **Reaching** beyond arm's length, to floor and overhead.
- **Weight shifts and stepping** all directions, onto a step, over obstacles; turning.
- **Perturbation** training: expected, then unexpected pushes (ankle, hip, step).
- Walking: speed, direction, narrow paths, head turns, carrying, dual task; tai chi, dance, ball games.

Dose: challenging balance **more than 3 hours a week** = largest effect, **39% fewer falls** (Sherrington 2017); near limit of ability, safely.

Coordination

Tests (eyes open and closed: cerebellar vs sensory): finger-nose, finger-finger, heel-shin, **dysdiadochokinesia**, rebound, foot circle, tapping. **Frenkel's:** slow, precise, repeated movements of rising difficulty, lying → sitting → standing → walking, watching the limb, to counts or floor marks; vision replaces position sense: best for **sensory ataxia**. Also PNF rhythmic stabilisation, stabilising and slow reversals, approximation, light wrist or ankle weights, task practice (ataxia (p. 27)).

Precautions

Gait belt, bars or harness; stand on weaker side; clear obstacles; BP before standing in orthostatic hypotension; **progress one variable at a time**; challenging but safe.

Exam plan (S16, 7 marks)

Assessment: systems, Berg, mCTSIB (1) → static progression: base, surface, vision, support, head, attention (2) → dynamic: reaching, stepping, perturbation, walking variations, functional (3) → dose over 3 h a week, guarding (1). **Draw:** stick figures of ankle, hip, stepping strategies. **Underline:** centre of mass, base of support, reweighting, anticipatory and reactive, mCTSIB, functional reach, perturbation, dual task.

Tools and adaptive equipment for neuro-rehabilitation

Balls, tilt boards, bolsters, wedges, benches and mats.

Full notes p. 228

Asked: not asked on its own in 2016–2026. Expect a **practical demonstration** or a 3-mark note on the uses of one piece of equipment (ball, wedge); a 7-mark "equipment to train balance" is possible.

The named equipment: what each trains

Therapy (vestibular, Swiss) ball	**Postural, equilibrium and protective reactions**, trunk control, vestibular input; relaxation by slow rocking
Tilt and wobble boards	Equilibrium (tilting) reactions, **ankle and hip strategies**, sitting and standing balance
Bolsters and rolls	Arm weight bearing, hip abduction (straddle sitting), positioning, trunk control
Wedges	Head and upper-trunk control in prone; positioning
Graded benches and steps	Sit to stand, step-ups, stairs, seated balance at different heights
Therapeutic mats	Bed and floor mobility: rolling, crawling, kneeling, getting up from the floor

Activities and progression

- **Ball:** child prone over the ball (head and arm weight bearing); sitting with weight shifts; tip slowly → **protective arm extension**. Progress: larger, faster tilts, less support; **peanut ball** for more stability.
- **Boards:** rocker tilts one direction, **wobble all directions**. Sit, kneel or stand on it. Progress: sitting → standing, eyes open → closed, holding on → free hands, add reaching or catching.
- **Bolster:** prone over a roll for elbow and hand weight bearing; **straddle sitting reduces adductor tone** (spastic diplegia) and trains sitting balance; side-to-side rolling for reactions.
- **Wedge:** prone with forearm support, toys in front (head lifting); supported half-lying for feeding or chest care. Progress: lower wedge → flat prone.
- **Bench:** sit to stand from a higher bench, lowered stepwise; step-ups on graded steps; stable seat for reaching.
- **Mat programme:** rolling → prone on elbows → four-point kneeling → kneeling → half-kneeling → floor to stand; **push-ups in long sitting** for SCI.

Other common equipment

Parallel bars: early standing, walking, weight shift; move to a walker or stick as soon as possible. **Tilt table:** gradual uprighting for orthostatic hypotension, early weight bearing, calf stretch. **Standing frame, prone stander:** supported standing. **Treadmill** (± body-weight support): repetitive walking, endurance. **Mirror:** posture and symmetry feedback; mirror therapy after stroke. **Foam pads, beams, cones, obstacles:** unstable, narrow-base and obstacle walking. Weights, bands, pulleys, static cycles: strength and fitness. Hand boards, pegboards, putty: grasp, release, dexterity.

Safety and good practice

- **Guard** on unstable equipment; **gait belt**; clear space; **mats under balls and boards**.
- Ball properly inflated and right size: **hips and knees about 90°** when sitting.
- No fast spinning or rocking if prone to seizures or autonomic problems; watch dizziness and BP.
- Down's syndrome with **atlantoaxial instability**: no forced neck flexion or trampoline-type bouncing.
- Clean between patients; **latex-free** balls and bands where latex allergy is possible (spina bifida).
- Equipment supports the task; the goal is a real-life skill **without it**.

Exam plan

3 marks (uses of one piece): description → what it trains with two activities and a progression → one precaution.

7 marks (balance equipment): balls (sitting reactions); rocker and wobble boards; foam and beams; parallel bars; steps and benches; cones; mirror → progress support, base, eyes, surface, speed, dual task → safety: guarding, gait belt, mats.

Practice LAQ (diplegic child, 4 years, sits but cannot stand): roll with forearm weight bearing 5 min → bolster straddle 10 → ball tilts 10 → mat kneeling and half-kneeling 10 → graded bench sit to stand 10 → standing frame, then rocker board with hands held 10 → stretches and parent teaching 5.

Spots: identify and demonstrate ball, peanut ball, rocker and wobble boards, bolster, wedge, steps, mat, standing frame, tilt table. **Underline:** equilibrium reactions, protective reactions, weight bearing, graded progression, mat programme, guarding, task transfer.

Transfers, functional re-education, posture and gait training

Gait training appears inside most stroke and Parkinson's answers. Full notes p. 232

Asked: not on its own in 2016–2026; transfers and gait training fill **4–7 management marks** in most long cases ("trunk control and ambulation"). Named gaits are asked on condition pages: hemiplegic, parkinsonian, ataxic, DMD, L3 fracture ambulation.

Bed mobility, transfers, posture

- **Rolling:** head and arm lead, knees bent; hemiplegia: both ways (to the strong side trains the weak side). **Supine to sit:** side lying, legs over the edge, push up.
- **Sit to stand:** slide forward, feet back, trunk forward (**"nose over toes"**), push through both legs; sit down with control. **Raise the seat to ease, lower to progress.**
- **Transfers:** stand-pivot (hemiplegia: usually **towards the stronger side**, guard the weak knee); sliding board or depression lift (paraplegia); car, toilet, bath; up from the floor after a fall. Teach the family safe help and equipment.
- **Re-education:** whole tasks in real settings; **part practice** only for missing parts; many repetitions, progressively harder.
- **Posture:** pelvis upright, weight even; correct flexed (PD) or asymmetrical (stroke) posture with mirror, verbal cues, extensor and trunk strengthening.

Normal gait cycle and measures

Stance about 60%: loading response 0–10%, midstance 10–30%, terminal stance 30–50%, pre-swing 50–60%. **Swing about 40%:** initial 60–73%, mid 73–87%, terminal 87–100%. **Double support** 0–10% and 50–60% (about 10% each); slower walking lengthens stance and double support.

Measures: 10 m walk (speed), 6MWT (endurance), step and stride length, cadence, symmetry. **Stroke speed (Perry 1995):** below ~**0.4 m/s** household; 0.4–0.8 limited community; above ~**0.8 m/s** community walker.

Pathological neurological gaits

Hemiplegic	Short weak-side stance, stiff knee, foot drop, **circumduction** or hip hiking (stroke)
Scissoring, crouch	Thighs cross; hips and knees bent, toe walking (diplegic CP)
Festinating	Flexed, short steps that speed up, little arm swing, freezing, turns en bloc (PD)
Cerebellar ataxic	Wide-based, staggering; no better with eyes open (cerebellum, MS)
Sensory ataxic	High steps, heels stamp, eyes on feet; worse in the dark; Romberg + (neuropathy, dorsal column)
High-stepping	Hip and knee high to clear a dropped foot; foot slaps (peroneal palsy, CMT, polio)
Waddling	Trunk sways, wide base, lordosis (DMD, myopathies)
Trendelenburg	Pelvis drops on swing side (weak abductors); hand-on-thigh: weak quads (polio)

Principles of gait training

- **Walking practice improves walking**: moderate to high intensity in chronic stroke, incomplete SCI, brain injury (Hornby 2020).
- Overground with variety (turns, slopes, obstacles); **treadmill ± body-weight support**; dual task; cueing in PD.
- **Prerequisites:** standing balance, weight shift, strength (hip extensors, quads, plantar- and dorsiflexors), range (hip extension, DF), endurance.
- AFO for foot drop (orthoses (p. 43)); **the least aid that keeps walking safe**.
- **Progression:** parallel bars → walker → crutches or stick → no aid; level → slopes, stairs, outdoors; short → long; supervised → independent; re-measure speed and distance.

Aids, crutch gaits, stairs

Height: handgrip at the wrist crease (or greater trochanter), elbow **20–30°** bent; stick in the hand **opposite the weak leg**. **Four-point:** crutch, opposite foot, other crutch, other foot (most stable). **Two-point:** crutch and opposite foot together (faster). **Three-point:** both crutches with the affected leg, then the good leg (non- or partial weight bearing). **Swing-to, swing-through:** paraplegia with KAFOs. **Stairs: "up with the good, down with the bad"** (stick leads down).

Exam plan (gait training in a long case)

1. Gait analysis: deviations in each phase and their causes.
2. Prerequisites: balance, weight shift, strength, range.
3. Walking practice with intensity and variety; treadmill; orthosis and aid.
4. Progression (aids, surfaces, distance, stairs) and outcome measures (speed, distance).

Draw: gait cycle bar with phases and %. **Underline:** task-specific, repetition and intensity, stand-pivot, stance 60%, walking speed, treadmill, progression of aids, up with the good.

Revised ☐☐☐

Bladder and bowel training

Types of neurogenic bladder in 9 papers.

Full notes p. 238

Asked: 9 of 20 papers. **7 marks:** types of bladder and PT (W25, S26), classify and manage one (S21), spastic vs flaccid with PT (W16, W17). **3 marks:** three types (S17), pathological bladders (W20, S23), reflex bladder (W22). **Long case:** bladder type after T9–T10 fracture (S26).

Definition and control

Neurogenic bladder: abnormal storage or emptying from a brain, cord or nerve lesion (SCI, MS, stroke, PD, spina bifida, cauda equina, diabetes). Risks: infection, stones, dysreflexia, **kidney damage from high pressure**. **Sympathetic** (T10–L2) stores; **parasympathetic** (pelvic, **S2–S4**) contracts the detrusor; **pudendal** (S2–S4) holds the external sphincter; **pontine centre** co-ordinates emptying.

Types (3m S17, W20, W22, S23)

Lesion level (EAU 2026): brain → overactivity, little residual; cord → overactivity + **detrusor-sphincter dyssynergia** (DSD); sacral and below → weak detrusor, retention.

Uninhibited	Brain: sensation present; small; urgency, urge leaks; empties fully
Reflex	Complete cord lesion above S2: no sensation; reflex emptying; DSD
Autonomous	Sacral cord, cauda equina, spinal shock: large; overflow
Sensory paralytic	Diabetes, tabes, B12: no fullness sense; large residual
Motor paralytic	Polio, root injury: feels full, cannot empty

Reflex bladder (W22): sacral arc intact, cut off from the brain → reflex contraction against a tight sphincter → high pressure; CIC + detrusor-calming drugs. **S26 case:** reflex (UMN) bladder; fullness sense = spared sensory fibres.

Spastic vs flaccid (7m W16, W17)

	Spastic (UMN)	Flaccid (LMN)
Lesion	Above S2–S4 (fracture above ~T12)	Sacral cord, cauda equina; spinal shock
Bladder	Arc intact; **small**, overactive; DSD; reflex leaks	Arc broken; **large**; overflow dribbling
Danger	Kidney damage; dysreflexia (T6 and above)	Overstretch, infection, stones
Emptying	CIC + drugs	CIC

Assessment and PT (7m S21, W25, S26)

- **Assess:** bladder diary ≥ 3 days; residual urine; urodynamics; hand function and sitting balance for self-catheterisation; toilet transfers; home (Indian squat toilet not possible for most paraplegics).
- **CIC** (preferred after SCI): **4–6 times a day**, volume **under 400–500 mL**; plan fluids; 12–16 Fr, hydrophilic. Teach hygiene, long sitting or wheelchair position, balance with both hands free, recording; C6–C7: adapted aids.
- **Reflex:** CIC + antimuscarinics (botulinum toxin if needed); sheath only if pressure is safe. **Areflexic:** CIC; indwelling or suprapubic only if CIC impossible. **Sensory paralytic:** void by the clock.
- **Uninhibited: timed voiding** by the clock; **urge suppression** (still, quick pelvic floor squeezes, walk calmly); **PFMT supervised ≥ 3 months** (MS, stroke, incomplete SCI); tibial nerve stimulation may help; faster transfers, bedside commode, easy clothing; less tea and coffee late.
- **Current:** tapping, Credé, Valsalva raise pressure: only if urodynamics show it is safe.

Bowel programme

Reflexic (UMN, above conus): tight sphincter, constipation → daily or ≥ 3 times a week; suppository, **digital rectal stimulation**, upright on a commode. **Areflexic (LMN):** slack sphincter, leaks → daily; **manual removal**, firmer stool. Same time daily after a meal (**gastrocolic reflex**); Bristol 3–4; under 30 min, never over 1 h. Physio: sitting balance, commode transfers, aids for weak hands. **Current:** no abdominal massage or Valsalva (2020).

Precautions

SCI **T6 and above**: full bladder or bowel → **autonomic dysreflexia**; empty before exercise. Infection: fever, cloudy or smelly urine, more spasms or leaks. Keep skin dry; never clamp or block tubing on the tilt table or in transfers.

Exam plan (7 marks)

Control (1) with lesion figure → **types (2–3)** and spastic vs flaccid table → **PT (3–4)**: diary, CIC, timed voiding, PFMT, toilet access, dysreflexia. **Draw:** lesion-level figure. **Underline:** level of lesion, DSD, residual urine, CIC, 400–500 mL, timed voiding, autonomic dysreflexia.

A philosophy of caring

Patient- and family-centred care.

Full notes p. 246

Asked: not on its own in 2016–2026. Shows as patient and parent education in long cases and as the **affective domain** in the practical. Expect 3 marks on principles of family-centred care, or 7 on a philosophy of caring.

Person-centred care

- Start from the person's **life, values and goals** (work, family roles, faith, hobbies), not only the diagnosis.
- **Set goals together**, written in the person's words (ICF and goals (p. 3)).
- **Autonomy:** explain options, risks, benefits; accept informed choices, including no.
- Build **self-management and self-efficacy** so progress continues after discharge.

Family-centred care in children (likely 3m)

- **Parents are partners** and main therapists: they decide priorities with you.
- Clear, honest information in plain language; teach handling, positioning and a **short home programme that fits the family's day**.
- Build on family strengths; respect culture, beliefs, resources; support siblings and grandparents in joint families.
- Co-ordinate with school and services; plan transitions (preschool, school, adulthood).

Communication and the relationship

- Introduce yourself, explain what and why, **consent every session** before touching.
- Listen actively; **teach-back** (ask them to explain or show it back); their language, no jargon.
- **Aphasia, hearing loss, neglect, cognition:** short sentences, gestures, pictures, demonstration, quiet room.
- **Empathy and realistic hope**; praise effort and progress.
- **Dignity and privacy:** drape, curtains, chaperone where appropriate, respect modesty and cultural norms.

Ethics in everyday practice

Autonomy	Informed consent; their choices; involve them even with reduced capacity
Beneficence	Evidence-based treatment aimed at their goals
Non-maleficence	Safe handling, precautions, no unnecessary or ineffective treatment
Justice	Fair care regardless of income, caste, religion, gender, disability; help with schemes
Confidentiality	Share only with consent and the care team

Carers, teamwork, difficult situations

- **Carer burden:** ask how they cope; safe handling; respite, support groups, schemes.
- **Teamwork:** doctors, nurses, occupational and speech therapists, psychologist, social worker, orthotist, teachers; **one shared plan**.
- **Progressive and terminal disease** (MND, advanced PD, DMD): honest, compassionate talks; plan ahead; comfort, function, quality of life; respect advance wishes.
- **Reflective practice:** what went well, what to change; seek supervision.

Exam plan

3 marks (family-centred care): parents as partners setting priorities; honest information; handling and a home programme fitting the routine; family strengths, culture, resources; co-ordinate with school.

7 marks (philosophy of caring): person- and family-centred goals → autonomy and consent → adapted communication → empathy and realistic hope → dignity, privacy, culture → ethics table → carers → teamwork → honesty in progressive disease → reflection.

Practical: introduce, consent, drape, guard, explain each step: affective-domain marks (practical exam (p. 60)).
Underline: person-centred, family-centred, shared decision-making, informed consent, dignity, realistic hope, carer burden, teamwork.

Orthoses and splints

AFO, cock-up splints and CP foot orthoses; a practical spot.

Full notes p. 249

Asked: 5 of 20 papers. **7 marks:** foot orthoses for CP children (S22). **3 marks:** AFO (W21), orthosis for radial nerve injury (W17), principle and uses of dynamic cock-up (S25), cock-up splint (W25).

Definition, aims, principles

Orthosis: external device that supports, aligns, protects or improves function of a body part. **Static** holds still; **dynamic** allows or assists. **Aims:** function, stabilise, prevent deformity, protect, position.

- **Three-point force system:** two forces one side, one opposite between. Foot-drop AFO: footplate lifts forefoot, strap at front of ankle, calf shell at back of upper calf.
- **Pressure = force ÷ area**: wide contact protects numb skin. **Ground reaction force** vs knee decides bend or straighten.

Prescription and check-out

Assess ROM (fixed or flexible), tone, strength, sensation, gait → aim in ICF terms → design with the orthotist → check fit, pressure, shoe → train donning, **gradual wearing schedule** → review skin, growth; measure gait speed, falls.

AFO (3m W21) and choosing one

Parts: calf section and strap, ankle (solid or hinged), footplate, in a shoe. **Uses:** foot drop (stroke, peroneal injury, MS, CMT), spastic equinus, ankle instability, crouch, knee hyperextension. AFOs raise stroke walking speed; **peroneal FES as good as an AFO** (Bethoux 2014).

Swing foot drop, little spasticity	Posterior leaf spring (or peroneal FES)
Spasticity or wobbly ankle	Hinged with **plantarflexion stop**
No ankle control, non-walker, night stretch	Solid, ankle at 90°
Crouch, full passive knee range	**Ground-reaction** (anterior shell)
Weak quads, knee buckles	Ground-reaction if knee straightens fully; else KAFO
Flat valgus foot	Supramalleolar orthosis

KAFO: weak quads, unstable knee (polio, SCI); **drop lock**, offset, Swiss lock. **RGO:** paraplegia; one hip extends, other leg swings.

Foot orthoses in CP (7m S22)

Solid AFO (poor control, non-walkers, night); hinged with plantarflexion stop (walkers, enough calf length); ground-reaction (crouch, good passive range); leaf spring (mild foot drop); supramalleolar (flat valgus); dynamic thin AFO (mild tone). Aims: stable base, control equinus, gait, range. **Review with growth**; add stretching, strengthening. See CP (p. 47).

Upper limb (3m W17, S25, W25), shoulder, spine

Static cock-up (W25)	Wrist **20–30° extension** (neutral for carpal tunnel), to the palmar crease, fingers free; radial palsy rest and night
Dynamic (lively) cock-up (S25, W17)	Wrist support + **outrigger with elastic slings** re-extend fingers after each grasp; radial palsy by day
Knuckle-bender	Claw hand: blocks knuckle hyperextension
Opponens, web spacer	Median palsy: thumb opposed, web open
Resting hand splint	Very weak or spastic hand; little effect on contracture
Shoulder, collar	Arm trough or lap tray (sling briefly); collar for head drop

Making a temporary splint

Position → paper pattern (cock-up: two-thirds of forearm to palmar crease) → cut low-temperature thermoplastic → heat in water at **65–70 °C** (1–2 min) → test on own skin, mould over stockinette till set → trim, flare, pad, straps → check red areas; spot-heat → teach schedule, keep from heat. **Serial casting:** changed every few days.

Precautions and current evidence

Skin check after every use (lasting redness, blisters, numbness → adjust); allow for fixed deformity; lightest that works; review children every few months. **Current:** sustained stretch by splints or casts has no important effect on joint mobility (Cochrane 2017): prescribe for a **stated functional aim**; active use and positioning prevent contracture.

Exam plan

3 marks (any orthosis): definition, parts, sketch → principle (three-point force or assist) → uses and one precaution. **7 marks (CP):** aims → each AFO matched to the gait problem → check-out, growth, exercise.

Draw: AFO selection chart; static and dynamic cock-up with outrigger. **Underline:** three-point force system, check-out, plantarflexion stop, ground-reaction AFO, drop lock, outrigger, skin check.

Lifting, wheelchairs and adaptive devices

Wheelchair prescription in DMD (S26). Full notes p. 257

Asked: 1 of 20 papers: **long case S26**, 8-year-old rural boy with DMD walking only with support: steps of wheelchair prescription, short- and long-term goals (full case on DMD (p. 55)). Also used in most severe-case management answers.

Safe lifting and handling

- **Assess** task, person, environment, equipment → **plan and communicate**; patient does as much as possible.
- **Equipment instead of lifting:** slide sheets, transfer boards, stand-aids, **hoists** for dependent patients.
- Wide base, bend knees, load close, back upright, turn with the feet.
- **Never the drag lift** (under the arms): injures the hemiplegic shoulder and the carer's back.
- **Current:** technique training alone did not prevent back pain (Cochrane 2011): use equipment.
- **Transfers:** independent (pivot, sliding board) → one helper (pivot to stronger side, block weak knee) → two helpers, stand-aid → hoist.

Wheelchair parts and types

Parts: frame; seat, back, cushion; rear wheels, push rims; castors; brakes; armrests; footrests or legrests; push handles; anti-tip bars; lap belt.

Lightweight rigid	Active users with good arms (paraplegia)
Hemi-height, one-arm drive	Hemiplegia: propel with good foot, double rim or lever
Tilt-in-space, reclining	Poor trunk and head control, pressure relief (MND, high tetraplegia, severe CP)
Powered	Weak arms, long distance (DMD, MND); joystick, chin, breath
Hand-propelled tricycle	Rural, outdoor India

Prescription (LAQ S26)

WHO 8 steps: referral → assessment → prescription → funding and ordering → product preparation → fitting → **user training** → follow-up and repairs.

Assess: stable or progressive, trunk and head control, arm strength, deformity (pelvic obliquity, scoliosis), skin, breathing, growth; home, school, **rough rural roads**, carer, budget, local repair.

Measurements (seated on the cushion)

Seat width	**Hip width + 5 cm**
Seat depth	**Buttock to back of knee − 5 cm** (too deep: slumps)
Footplate	Thighs level; **5 cm** off the ground
Back height	Below scapulae (active); higher + head support for poor trunk
Armrest	**Elbow height + 2.5 cm** (elbow 90°)

Seating: solid seat and back, laterals, hip guides, pommel, headrest, lap belt; pressure cushion.

Modifications by condition

- **Hemiplegia:** hemi-height seat, arm trough or lap tray, brake extension on the weak side.
- **Paraplegia:** light rigid frame, low back, good cushion, removable armrests for board transfers.
- **C5–C6:** rim projections or coated rims, gloves, higher back; power for distance. **C4 and above:** power with chin or breath control and tilt.
- **CP:** custom seating (pelvic belt, pommel, laterals, headrest), tilt.
- **DMD:** manual for distance early → **power with tilt, recline, elevating legrests, spinal supports**. **MND:** tilt, head support, plan power early.

Pressure care and skills

Cushion (foam, gel, air) not bottomed out; **weight shifts every 15–30 min, held ≥ 1 min** (push-up, forward or side lean, tilt); skin checks twice daily with a mirror. **Teach:** brakes, long strokes, turns, ramps, kerbs (back-wheel balance), transfers, getting back in after a fall, maintenance.

Adaptive devices and home

Aids: stick, tripod, quadripod, crutches, walker, rollator; handgrip at wrist crease, elbow **20–30°**; axillary top 5 cm below armpit; stick opposite the weak leg. **ADL:** built-up cutlery, button hook, reacher, raised toilet seat, grab rails, shower chair. **Ramp:** no steeper than **1 in 12**, ≥ 1200 mm wide, rails both sides at 760–900 mm (India 2021). **ADIP scheme:** free or subsidised aids via ALIMCO with disability certificate ≥ 40%.

Exam plan (prescription in a long case)

Assess person and setting → type with reasons → measurements, seating, cushion → fit, train user and family → review with growth or progression.

Draw: side view of a wheelchair, parts labelled. **Underline:** risk assessment, no drag lift, WHO eight steps, hip + 5 cm, tilt-in-space, weight shifts, user training, ramp 1:12.

Ergonomic advice and caregiver education

Home-care advice to parents and caregivers in several long cases.

Full notes p. 263

Asked: not on its own in 2016–2026; inside long cases as **home-care advice**: 2-year-old with spastic diplegia (5 marks, W17; see CP (p. 47)), 5-year-old with DMD (W16; see DMD (p. 55)). Possible 3 marks: energy conservation.

Ergonomic advice for patients

- **Posture:** pelvis upright, feet and back supported; change position often; no long sustained postures.
- **Workstation: screen at eye level**, **elbows about 90°**, forearms supported, chair and desk heights adjusted; wheelchair users: knee clearance, items within reach.
- **Energy conservation** (MS, MND, myasthenia, post-polio, PD): **plan, prioritise, pace**, sit to work, rest before fatigue, heavy tasks when strongest; items within reach, trolleys; avoid heat (MS).
- **Joint protection:** large joints and both hands, no sustained grips, built-up handles, trolleys; protect the hemiplegic shoulder.
- **Home layout:** frequent items between hip and shoulder height; clear walkways, good light, rails, non-slip surfaces (adaptive devices (p. 44)).
- **Return to work:** graded hours, task changes, rest breaks, travel plan, workplace visit.

Educating parents and caregivers

- **Carrying matched to tone:** spastic: break the stiff pattern, **hips bent and apart**; athetoid: limbs together and bent, close to the carer.
- **Positioning through the day:** side lying, prone on a wedge, supported sitting; **no W-sitting**, no long periods supine.
- **Feeding:** upright, **head slightly forward**, trunk supported; coughing or chest infection → swallow assessment.
- **Dressing:** stiffer or weaker limb **first in, last out**.
- **Adults:** slide sheets, boards, hoists; **never lift under the arms**.
- **Home programme:** short, written, illustrated, built into routines (bath, play, meals); demonstrate, then **teach-back** (carer shows it back).
- **Warning signs:** shunt problems, autonomic dysreflexia, breathing trouble, skin redness.

Protecting the carer

Equipment rather than lifting; bed and chairs at working height; load close; no twisting. Lifting-technique training alone does not prevent back pain (Cochrane 2011): equipment and planning matter more. **Carer strain:** tiredness, low mood, back pain → involve family, respite, support groups.

Exam plan

Home-care advice (5 marks):

1. Positioning and handling, day and night.
2. Daily activities: feeding, dressing, bathing, toileting.
3. Home programme built into routines; play.
4. Equipment and home changes.
5. Warning signs, follow-up; carer's own health.

CP parents (7m practice): side lying, wedge, supported sitting, no W-sitting; carrying by tone; feeding upright; stiffer limb first; play for head control, sitting, reaching; standing frame time; splint and skin care; **no baby walkers**; written programme shown back; warning signs.

Underline: posture, workstation, energy conservation, joint protection, handling and carrying, positioning, home programme, teach-back, carer strain.

Developmental neurology: milestones, reflexes and reactions

Milestones as 7 marks in 3 papers; ATNR, parachute and other reflexes.

Full notes p. 267

Asked: 6 of 20 papers. **7 marks:** gross and fine motor milestones to 5 years (W19), to 3 years (S21); gross motor to 5 years (S23). **3 marks:** ATNR (S20, S21), any one primitive reflex (W20), cortical reflexes (W20), parachute reactions (W25).

Principles

Sequence same in all children, **rate varies**; continuous; **cephalocaudal** (head → sitting → standing); **proximodistal** (trunk → fingers); general to specific; primitive reflexes fade so milestones appear. Born before 37 weeks: **corrected age** (age minus weeks early) until about 2 years.

Milestones (7m W19, S21, S23)

Age	Gross motor	Fine motor
3 m	**Neck holding**; chest up on forearms in prone	Watches hands; hands to midline (3–4 m)
4–5 m	Rolls over (5 m)	Bidextrous reach (4 m)
6 m	Sits with support; prone on straight arms	Unidextrous reach; palmar grasp; transfers by 6–7 m
8–9 m	**Sits alone** (8); stands holding on, crawls (9)	Immature pincer (9)
12 m	Stands alone; walks with one hand held	**Mature pincer**
15 m	**Walks alone**; creeps upstairs	Imitates scribble; tower of 2
18 m	Runs stiffly; climbs onto furniture	Scribbles; tower of 3; spoon
2 y	Runs well; upstairs two feet per step; jumps; kicks a ball	Tower of 6; vertical and circular strokes
3 y	Upstairs alternate feet; tricycle	Tower of 9; **copies circle**; dresses with help
4 y	Hops on one foot; downstairs alternate feet	Copies cross; bridge of blocks
5 y	Skips; hops well on either foot	Copies triangle; ties laces

WHO (2006): walking alone is normal anywhere from about **8 to 18 months** (median 12).

Primitive reflexes (3m W20: any one; ATNR 3m S20, S21)

Brainstem and spinal; integrated mostly by **4–6 m**. Absent, asymmetrical, obligatory or persisting = brain damage (CP). Rooting, sucking to 3–4 m; palmar grasp ~6 m; plantar grasp ~15 m; stepping 2–3 m; placing ~1 y; Galant 2–4 m; STNR gone 9–12 m; Babinski normal to 1–2 y.

Moro: sudden drop of supported head → arms abduct, extend, hands open, then adduct and flex, often a cry; to 3–4 m (normal to 6). Absent: brain depression, hypotonia; **asymmetrical: brachial plexus palsy, fractured clavicle or humerus, hemiplegia**; after 6 m: CP.

ATNR: supine, head midline, turn head → **face-side arm and leg extend, skull-side flex ("fencing")**; integrated by **~6 m**. **Obligatory** at any age or after 6 m = CP. Persistence: no hands to midline or mouth, cannot roll or sit symmetrically, poor hand–eye, scoliosis and hip risk, cannot cross midline. PT: midline positioning, toys in front.

Reactions: parachute (3m W25), cortical (3m W20)

Righting: head upright, body aligned (neck, body, labyrinthine, optical); Landau 3 m–2 y. **Parachute:** held prone, moved suddenly head-first → **both arms extend and abduct, hands open**; forward **6–9 m, for life**, then sideways, backward. Absent or asymmetrical: hemiplegic CP, brachial plexus palsy, delay. Train on ball, roll, tilt board.

Levels (Fiorentino): spinal (withdrawal, extensor thrust, crossed extension) → brainstem (ATNR, STNR, tonic labyrinthine, supporting reactions) → midbrain (righting, protective) → **cortical: equilibrium reactions** in prone, supine, four-point, sitting, kneel-standing, standing + hopping, dorsiflexion; for life. **Current:** a teaching model; train skills directly.

Red flags and screening

Not by **6 m**: rolls, pushes up on straight arms; **9 m**: sits alone; **12 m**: pulls to stand, pincer; **18 m**: walks alone; **2 y**: runs, kicks. **Loss of skills**; asymmetry; hand preference before 12–18 m; very low or high tone. Tools: Trivandrum chart, Denver II, AIMS (0–18 m), Bayley; Hammersmith, General Movements (early CP).

Exam plan

7 marks: principles (1) → gross motor table to the age asked (3) → fine motor (3) + one line of red flags. **3 marks:** elicit, response, timing, abnormal. **Draw:** WHO windows chart; ATNR "fencing" figure. **Underline:** cephalocaudal, sequence constant, corrected age, neck holding 3 months, sits alone 8 months, walks alone 12–15 months, pincer, obligatory ATNR, equilibrium reactions.

Cerebral palsy

13 long cases in 20 papers, the most of any topic.

Full notes p. 275

Asked: a CP long case in 13 papers (athetoid girl W19, S21, W22, W24; diplegia S17, W17, W18, S18, S20, S26; hypotonic S22; hemiplegia S25; dyskinetic W25). **7 marks:** carrying positions (S18), PT for spastic CP (S19), foot orthoses (S22), classification and causes (S24), athetoid and diplegic posture and movement (S26). **3 marks:** causes (W16, W17, S18, W19, W22), spinal deviations (W25), hypotonic impairments, classification (S26).

Definition and classification (7m S24; 3m S26)

CP: permanent disorders of movement and posture, causing activity limitation, from a **non-progressive** disturbance in the **developing** brain; with sensory, cognitive, communication problems, epilepsy, secondary musculoskeletal problems (Rosenbaum 2007). Lesion static, picture changes with growth. India about **3 per 1000**.

Spastic (commonest)	**Hemiplegia** (arm more than leg; stroke or bleed); **diplegia** (legs much more; periventricular leukomalacia, **PVL**, in preterm); **quadriplegia** (four limbs and trunk)
Dyskinetic	Athetosis, chorea, dystonia; fluctuating tone; basal ganglia (asphyxia at term, kernicterus)
Ataxic	Low tone, wide base, intention tremor; cerebellum
Hypotonic; mixed	Floppy, often later dyskinetic or ataxic; mixed usually spastic + dyskinetic

GMFCS: I walks without limits; II with limits, rail on stairs; III hand-held device; IV powered mobility; V transported. **MACS** hand, **CFCS** communication.

Causes (3m W16, W17, S18, W19, W22)

Prenatal: maternal infection (rubella, cytomegalovirus, toxoplasmosis, Zika); multiple pregnancy; placental insufficiency; pre-eclampsia, thyroid; malformations; Rh incompatibility. **Perinatal: prematurity** (before 32 weeks), low birth weight; **birth asphyxia** ("delayed cry"); prolonged labour; **kernicterus**; neonatal infection. **Postnatal (to 2 years):** meningitis, encephalitis, head injury, near drowning, hypoglycaemia.

Pathophysiology to impairments

Primary: abnormal tone, weakness, poor selective control and balance, sensory loss. **Secondary:** muscle lags bone growth → **contractures** (calf, hamstrings, hip adductors, flexors); bony torsion; hip migration; scoliosis. Muscle grows only under stretch in daily use, so active use, standing, casting, orthoses through childhood. **Tertiary:** toe walking, crouch, W-sitting. **Associated:** epilepsy, intellectual disability, vision, hearing, swallowing, drooling, low bone density.

Features by type (7m S26; 3m W25, S26)

- **Diplegia:** hips flexed, adducted, IR; knees flexed; **equinus**; sacral or W-sitting; **scissoring**, toe walking, crouch, in-toeing, arms in high guard; intelligence often normal.
- **Athetoid:** writhing movements ↑ with effort and excitement; **fluctuating tone**; poor head and midline control, persistent ATNR; **poor proximal stability**, cannot hold or grade; extensor thrusts; drooling, unclear speech; hearing loss; late wide-based walking.
- **Hemiplegia:** fisted hand, thumb in palm; equinovarus; hand preference before 12 months.
- **Spinal deviations, quadriplegia (W25):** neuromuscular **C-curve** scoliosis, pelvic obliquity, kyphosis, ↑ lordosis with hip flexion contracture, **windswept hips**.
- **Hypotonic (S26):** weak antigravity muscles; poor head and trunk control; hypermobile joints; **frog-leg** posture; very delayed milestones; weak cough; feeding difficulty.

Investigations and PT assessment

Diagnosis before 6 months: MRI + **General Movements** (absent fidgety) under 5 months; MRI + **Hammersmith examination** after. Hip X-ray **migration percentage**; EEG, hearing, vision; exclude progressive disease (loss of skills). **Assess:** birth history, milestones, family goals; GMFCS, **GMFM**; MAS, **Modified Tardieu**; reflexes; Thomas test, **popliteal angle**, DF knee straight and bent; gait; MACS.

Medical and surgical

Botulinum toxin + stretch, casting, orthoses, practice; baclofen (oral, intrathecal); selective dorsal rhizotomy (diplegia). Tendon lengthening, multilevel surgery, hip surgery, spinal fusion.

ICF (W22: 5-year-old athetoid girl, sits alone, cannot stand, at school)

- **Structure/function:** basal ganglia lesion; involuntary movement; fluctuating tone; poor co-contraction; poor head and midline control, ATNR; poor standing balance; drooling; hip and spine risk.
- **Activity:** cannot stand or walk; transfers, reaching, writing, feeding, toileting.
- **Participation:** play with children, classroom and playground, outings.
- **Environment:** motivated, well-off parents, school (facilitators); stairs, toilets, no adapted seat (barriers). **Personal:** 5, likely normal intelligence. Near **GMFCS III–IV**.

Goals: short-term (6–8 weeks): head midline while reaching; standing frame 60 min a day; sit to stand with help; grasp-release at midline. Long-term (6–12 months): bench sitting hands free 15 min; stands holding 1 min; walker in class; adapted spoon; no hip displacement.

Management (7m S19)

Family-centred, goal-directed, active task practice through play; home programme; prevent deformity. **Effective (Novak 2019):** CIMT, bimanual, goal-directed, treadmill, strength, fitness, casting, hippotherapy, home programmes, botulinum + therapy, hip surveillance. **Not effective: passive NDT**, sensory integration. **Classic vs current:** not inhibit-then-facilitate; handling only to set up active movement.

- **Diplegia gait (S18):** stretch and cast calves, hamstrings; strengthen hip extensors, abductors, quads, calf → AFO; abduction in standing → weight shift, wide-base stepping → bars → **posterior walker** → treadmill ± body-weight support → outdoors.
- **Athetoid: proximal stability** (prone on elbows, four-point, kneeling, standing frame, approximation, holding against resistance); midline play; slow small-range movement; seat with pelvic belt, 90–90–90.
- **Hypotonic (S22):** prone over wedge; sitting with less support; approximation, tapping, bouncing on ball; standing box → cruising → walker; rolled towel, corner seat.
- **Hemiplegia:** CIMT or bimanual, about **40 hours**. **Quadriplegia:** seating, standing, hip and spine watch, chest care.
- **Standing:** 60–90 min a day, 5 days a week (bone); 60 min in **30–60° abduction** (hips); 45–60 min (range); 30–45 min (spasticity). **Activity:** 60 min moderate-to-vigorous, 5 days a week. **Strength:** 2–4 times a week, 1–3 × 6–15 at 50–85% 1RM, 12–16 weeks. **Fitness:** 20 min, 2–3 times a week, above 60% peak HR.

Carrying (7m S18) and foot orthoses (7m S22)

Spastic, scissoring	Facing you, **legs apart** around waist or astride hip: flexion and abduction break extension
Arches back	Curled forward, all joints bent
Athetoid	Limbs together and bent, hands together, head midline
Floppy	Back to your chest, facing out, arm under hips
Hemiplegic	Affected side lengthened over your arm

Solid AFO	Ankle 90°: equinus, non-walkers, night
Hinged + PF stop	DF in stance: walkers with calf length
Ground-reaction	Pushes knee back: **crouch**
Posterior leaf spring	Lifts foot in swing: mild drop
SMO; dynamic AFO	Flat valgus feet; mild tone, standing balance

Complications and precautions

Hip displacement (GMFCS IV–V): surveillance from about age 2; refer if **migration above 30%** or abduction below 30°. Contractures (passive stretch alone not enough); scoliosis (seating); fractures; aspiration. **Take care:** seizure plan; no forceful range in non-walkers; do not force abduction; feed upright; skin under casts daily; after botulinum more practice.

Home care (W17) and prognosis

Side lying, prone over wedge, supported sitting; avoid **W-sitting**; feed upright; dress stiffer limb first; daily play; **no baby walkers**; picture home programme; school. **Prognosis:** ~90% of motor potential by 5 years; **sitting by 2 years** → walks (Molnar 1976).

Exam plan (long case)

1. Restate: age, type, birth clue, estimated GMFCS, school, family.
2. **ICF (5)** with reasoning; functional diagnosis. **Goals (3–5)** short and long.
3. **Management (7–10)**: principles; for the type; orthoses; standing, hips; home programme.

Draw: three spastic distributions; ICF; carrying. **Underline:** non-progressive, PVL, GMFCS, scissoring, fluctuating tone, proximal stability, goal-directed, hip surveillance.

Down's syndrome

3 seven-mark questions.

Full notes p. 289

Asked: 3 of 20 papers, all 7 marks: PT management (S20, W25); physical traits and neurological impairments (S23). No long case; expect genetic types or atlantoaxial instability as 3-mark notes.

Definition and types

Trisomy 21: extra chromosome 21; typical features, intellectual disability (usually **moderate, IQ 35–50**), low tone, delay, medical problems; commonest chromosomal cause of intellectual disability. India about **1 in 1000** births. Risk: **maternal age over 35**, previous child, parent with balanced translocation.

Non-disjunction	**About 96%**; fails to separate, usually in the egg; maternal age
Translocation	3–4%; parent may carry it, higher recurrence
Mosaicism	1–2%; some cells only, often milder

Why they move late

Smaller brain, cerebellum and brainstem → **low tone**, poor postural control, slow reactions and motor learning. Collagen → **lax ligaments**: hypermobility, flat feet, unstable patellae and hips, **atlantoaxial instability**. Child widens the base and locks joints. Heart defects, sleep apnoea, chest infections ↓ endurance; obesity.

Traits and impairments (7m S23)

Physical traits: brachycephaly; flat face and nasal bridge; upslanting eyes, epicanthic folds; **Brushfield spots**; small mouth, protruding tongue; small low-set ears; excess neck skin; short broad hands, **single palmar crease**, clinodactyly; **sandal gap**, flat feet; short stature; lax joints.

Neurological: hypotonia, weakness; intellectual disability, speech delay; milestones: sits ~**10 months**, crawls ~14, walks ~**26 months** (slowest ~40); poor balance, slow reactions, poor motor planning; gait wide base, out-turned feet, **knees locked**, little heel strike; hearing loss (75%), vision (60–80%), seizures (1–13%), autism (7–19%), early Alzheimer's.

Medical: heart disease **40–50%** (AV septal defect); sleep apnoea 50–79%; thyroid up to 50%; hip 2–8%; symptomatic atlantoaxial instability **1–2%**; gut atresias 12%.

Investigations and assessment

Karyotype; echo; hearing, vision, thyroid. Neck X-ray (neutral) only if symptoms: **no routine screening** (AAP 2022). PT: heart status, neck symptoms; milestones, AIMS, GMFM, Peabody; tone, **Beighton**, strength; balance, gait; feet, hips; endurance.

PT management (7m S20, W25)

- **Developmental sequence:** head and trunk control → sitting → crawling → pull to stand → cruising → walking. Infancy: tummy time, midline, weight on arms; avoid frog position.
- **Treadmill stepping** (parent supports) from sitting alone: **8 min a day, 5 days a week**; walked about **101 days earlier** (Ulrich 2001).
- **Strength:** squat to stand, step-ups, climbing daily; adolescents progressive resistance **twice a week for 10 weeks**.
- **Balance** on varied surfaces, ball play; **fitness**: swimming, cycling, dance most days; weight control.
- **Lax joints:** no end-range stretching; supramalleolar orthoses or insoles for flat feet.
- Short, frequent, play-based sessions; demonstration, simple words, praise; **parent-run home programme**.

Classic vs current: not tapping and approximation while waiting for "readiness"; **practise the skills** (treadmill, strength); facilitation only an add-on.

Precautions

Atlantoaxial instability, stop and refer: neck pain, head tilt, new weakness, change in walking, brisk reflexes, bladder or bowel change. **No trampolines**; care with contact sports, gymnastics; support the neck in handling. **Heart:** stop for breathlessness, blue lips, unusual tiredness. Face the child (hearing). **Prognosis:** almost all walk, later (~26 months).

Exam plan

PT (7): intro (1) → assessment and aims (1) → sequence, treadmill, strength, balance, fitness, orthoses, teaching style (4) → neck and heart safety, home programme (1). **S23:** traits (3) + impairments (4). **Draw:** ICF for a 2-year-old. **Underline:** trisomy 21, non-disjunction, hypotonia, ligamentous laxity, atlantoaxial instability, treadmill training, parent-led home programme.

Neural tube defects: spina bifida and hydrocephalus

11 short questions in 9 papers.

Full notes p. 298

Asked: 9 of 20 papers; no long case. **7 marks:** PT for meningomyelocele (W16, S17), for spina bifida (W21), for lumbar myelomeningocele (W22); deformities in infantile spina bifida + PT (W17, 2 + 5); PT for hydrocephalus (W21). **3 marks:** deformities (S17), types of spina bifida (S25), types and features of hydrocephalus (W19), features of hydrocephalus (S20, W20).

Definition and types (3m S25)

Neural tube defect: tube fails to close in the **first 4 weeks** of pregnancy (anencephaly, encephalocele at the head; spina bifida along the spine). **Spina bifida:** incomplete closure of the vertebral arches, with or without a sac of meninges and cord.

- **Occulta:** bony gap only, skin closed (hair tuft, dimple, fatty lump); usually no deficit.
- **Meningocele:** skin-covered sac of **meninges and CSF**; cord in the canal; few deficits.
- **Myelomeningocele: cord and roots in the sac**, often open; paralysis, sensory loss, bladder below the level; commonest serious type.
- Others: lipomyelomeningocele (fatty mass tethers the cord); myeloschisis (open flat cord plate).

Causes and prevention

Multifactorial: **low folate**, previous affected pregnancy, maternal diabetes, obesity, **valproate**, high fever in early pregnancy, genes. India about **4.5 per 1000** births. **Folic acid 400 micrograms daily** from at least a month before conception to 12 weeks; 4–5 mg on medical advice after an affected pregnancy; food fortification.

Hydrocephalus: types (3m W19)

Too much CSF in the ventricles, usually under raised pressure. **Non-communicating (obstructive):** blocked inside the ventricles (aqueduct stenosis, **Chiari II**, tumour). **Communicating:** CSF leaves the ventricles but is not absorbed (bleed in a premature baby, meningitis). Congenital or acquired; normal-pressure in older adults.

Pathophysiology to impairments

- Exposed cord damaged → **flaccid paralysis** and numb skin below the level; some spasticity.
- **Unopposed muscles** above the level → deformity: L3–L4 hip flexors and adductors → flexion–adduction, **dislocation**; tibialis anterior with paralysed calf → **calcaneus**.
- S2–S4 in almost all → **neurogenic bladder and bowel**.
- Chiari II → hydrocephalus; about **8 in 10** repaired after birth need a shunt in year 1.
- Numb skin breaks down; thin bones fracture (after casts); **tethered cord** stretches with growth → new loss.

Deformities (3m S17; 7m W17)

Spine	Scoliosis, kyphosis (congenital thoracolumbar), lordosis
Hip	Flexion contracture; **flexion–adduction with dislocation**; flexion, abduction, ER (frog) in high lesions
Knee	Flexion contracture (high); valgus (low lumbar, sacral); extension (rare)
Rotation	Femoral anteversion; external or internal tibial torsion
Foot	**Clubfoot, vertical talus** (congenital); **calcaneus** (low lumbar); cavovarus, claw toes (sacral); valgus

Hydrocephalus features (3m W19, S20, W20)

Infants: head crossing centiles; **bulging fontanelle**; widened sutures; scalp veins; **setting-sun eyes**; irritability, vomiting, poor feeding, drowsiness; high-pitched cry; stiff legs; delay; seizures. **Older:** morning headache, vomiting, drowsiness, papilloedema, diplopia, poor balance. **Normal-pressure:** walking difficulty, memory loss, incontinence.

Tethered cord signs: back pain with activity, new weakness or loss of walking, changing foot shape, worsening scoliosis, new bladder or bowel problems: report at once.

Investigations and PT assessment

Before birth: maternal serum alpha-fetoprotein, anomaly scan. After: MRI brain and spine, cranial ultrasound, **serial head circumference**, spine and hip X-rays; kidney ultrasound, urodynamics. **PT: muscle chart and sensory testing** (touch, pin) for the level, repeated to catch change; tone; hips, knees, feet, spine, limb lengths; skin; shunt signs; development and mobility; arm strength; cognition.

Level	Muscles; mobility
Thoracic, high lumbar	Hip flexors, adductors at most; **no quadriceps**; reciprocating gait orthosis (RGO) or HKAFO in childhood; most adults wheelchair
Low lumbar	**Quadriceps**, medial hamstrings; no glutei; **AFOs + crutches**; ~**80%** community walkers as adults
Sacral	Quads, gluteus medius; high: AFOs, gluteal lurch; low: near normal

Level, not the hip, decides walking (Swaroop and Dias 2009).

Medical and surgical

Back closure in the first days; prenatal repair (MOMS: shunt need **82% → 40%**, better walking at 30 months, more premature births). **VP shunt** or endoscopic third ventriculostomy (shunts block, infect, over-drain). Clean intermittent catheterisation, bowel routine. Clubfoot casting; releases; untethering; spinal fusion.

ICF (practice: 4-year-old, L4 myelomeningocele, VP shunt)

- **Structure/function:** glutei and calves paralysed; no sensation below knees; hip flexion–adduction and calcaneus risk; **heel pressure mark**; neurogenic bladder.
- **Activity:** walks short distances with AFOs and walker; floor to stand, stairs, toileting need help.
- **Participation:** keeping up at nursery and play; toileting at school.
- **Environment:** AFOs, walker, parents, nursery (facilitators); distance, rough ground, latex, no wheelchair (barriers). **Personal:** 4, keen to play.

Goals: short-term (6–8 weeks): walks at home, starts forearm crutches; full hip extension in prone; heel healed, skin checks twice daily. Long-term (6–12 months): AFOs + crutches at nursery, light wheelchair for distance; level pelvis, plantigrade feet; no sores; independent transfers.

PT for lumbar myelomeningocele (7m W16, S17, W21, W22)

- **Before closure:** prone or side lying, never press the sac; record moving muscles. **Infancy:** positioning, gentle ROM, handling, head control, rolling, sitting, arm strength.
- **Strengthen** working muscles (hip flexors, adductors, quads, trunk) and **arms and shoulders** (push-ups in sitting, crutch drills) daily.
- **Prevent deformity: prone lying daily**; stretch hip flexors, adductors, hamstrings, calf; abduction positioning; night splints; serial casts (knee flexion over 20° hinders orthoses); assist clubfoot casting.
- **Standing frame from about 12 months**; parapodium or RGO from about 2 years (high lesions).
- **Orthoses by level:** no quads → RGO or HKAFO + walker + wheelchair; weak quads → KAFO; strong quads (L4) → AFOs + crutches; sacral → AFOs or none.
- **Gait:** standing balance → walker → crutches (swing-through or four-point) → stairs, falls; wheelchair skills for distance.
- **Skin:** checks twice daily (mirror), padded shoes, no heat; bladder; latex-free; shunt and tethered cord signs.

Classic vs current: not surgical reduction of every dislocated hip; aim for a **level pelvis** and free movement.

PT for hydrocephalus (7m W21)

Support the heavy head when lifting and carrying; change head position every few hours (scalp sores, flattening), no long pressure on the shunt valve; side lying, prone on wedge; head control, rolling, sitting, reaching, standing frame in sequence; stretch stiff legs; sensory and visual play; seating with head support; teach **shunt warning signs**.

Precautions and complications

Latex-free gloves, balls, bands from birth. **Stop and refer:** shunt signs (vomiting, drowsiness, irritability, headache, bulging fontanelle, fever, fast head growth); new weakness, pain or bladder change (tethered cord). Thin bones: gentle ROM, no forceful stretch; avoid crawling for **3–4 weeks** after immobilisation. No heat on numb skin; check skin after orthoses. Complications: pressure sores, fractures, contractures, obesity.

Education and prognosis

Daily prone lying, stretches, standing; skin and orthosis care; bladder routine; latex; warning signs; **folic acid** in future pregnancy; accessible school, toilet, wheelchair. Walking depends on **level**; most survive into adulthood.

Exam plan

PT (7): define and level (1) → muscle chart, aims by level (1) → by age: positioning, development, standing frame, orthoses and gait by level, wheelchair, strength, deformity prevention (4) → skin, bladder, latex, shunt and tethered cord (1). **W17:** deformities by region (2) + PT (5).

Draw: three types of spina bifida; level–orthosis–walking table. **Underline:** neurological level, muscle charting, hip flexion–adduction, calcaneus, standing frame, RGO, KAFO, AFO, latex-free, VP shunt, tethered cord, folic acid.

Obstetric brachial plexus injury

A 6-month-old with Erb's palsy (S19).

Full notes p. 309

Asked: 1 long case (S19, 5 + 10): 6-month-old cannot move the right arm since a forceps delivery, other milestones normal: differential diagnosis, short- and long-term goals, PT. Adult plexus questions are on nerve injuries (p. 17).

Definition, groups, causes

Obstetric brachial plexus palsy: infant arm weakness from **stretching or tearing of C5–T1** at birth; about 0.4–5 per 1000 births. **Causes: shoulder dystocia** (traction on the head stretches upper roots); large baby, maternal diabetes, prolonged labour, forceps or vacuum, breech; some after normal or caesarean delivery.

Narakas I: C5–C6 (Erb's)	No shoulder abduction or elbow flexion; hand works; commonest, best outlook
II: C5–C7	+ weak wrist and finger extension (dropped wrist)
III: C5–T1	Flail arm, no Horner's
IV: C5–T1 + Horner's	T1 avulsion; worst outlook

Pathophysiology and features

Deltoid, spinati, biceps paralysed; pectoralis major, subscapularis work → **internal rotation pull** on a forming joint → **glenohumeral dysplasia**, posterior subluxation. Non-use → **developmental disregard**; shorter arm.

Erb's: arm by side, IR, elbow extended, pronated (**"waiter's tip"**). **Moro absent** on that side; grasp present in Erb's, absent in total. Phrenic (C3–C5): breathing difficulty. Later: IR and elbow flexion contracture, winging, **trumpet sign**.

Tests and differential diagnosis (S19, 5 marks)

X-ray clavicle, humerus; shoulder ultrasound (dysplasia); MRI for avulsion. **Active Movement Scale** (0–7); **Mallet score** from ~3 years. Assess passive ER at side, Moro, grasp, sensation, Horner's, milestones.

Fracture clavicle or humerus	Pain, swelling, crepitus; X-ray; moves again in 2–3 weeks
Septic shoulder	Fever, swelling, pain; raised markers; emergency
Hemiplegic CP	UMN signs, fisting, leg involved, other delays
Cervical cord injury	Both arms or legs; breathing, bladder
Arthrogryposis	Stiff joints in several limbs

Medical and surgical

Refer by 1 month, start before 3 months. Nerve surgery if **no biceps by ~3 months** (Toronto under 3.5) or fails **cookie test** by ~9 months; total + Horner's explored early. Later: internal rotator release, tendon transfer, humeral osteotomy; botulinum + casting.

ICF and goals (S19 case)

Structure/function: weak shoulder and elbow; IR posture, dysplasia risk; sensory loss, disregard. **Activity:** reach, hand to mouth, prop and crawl. **Participation:** two-handed play with family. **Environment:** parents for home programme; clinic distance, guilt. **Personal:** 6 months, otherwise normal; past the 3-month point: refer now.

Short-term (4–6 weeks): full passive ER and abduction; hand towards mouth gravity eliminated; holds toys at midline; parents do the programme. **Long-term (6–12 months):** hand to mouth and overhead reach; both hands in play, crawls; no deformity.

PT management

- **First 1–2 weeks:** gentle handling; **never pull**; dress affected arm first; no rigid splint.
- **Home ROM several times a day** (each nappy change): **ER with scapula held**, abduction, flexion, elbow extension, supination, wrist and finger extension; touch, textures.
- **Active play:** gravity eliminated (side lying, water) → against gravity; toys on the affected side; hand to mouth; prone on elbows, four-point, crawling.
- **Later:** short constraint of good arm for disregard; resting wrist splint at night (group II); elbow splint or serial casts.

Precautions and prognosis

Never force ER or abduction; do not lift by the arms; stop and refer for clunk, lost range, swelling or pain; protect a numb hand. **Good:** Erb's only, antigravity biceps by ~3 months. **Poor:** total palsy, Horner's, no biceps by 3–6 months. ER and abduction often stay limited: review until growth ends.

Exam plan (S19 long case)

DD (5) → goals (3–4) → management (6–7): refer, handling, home ROM, play, constraint and splints later, precautions, parents. **Draw:** plexus, upper roots marked; ICF. **Underline:** shoulder dystocia, Narakas, waiter's tip, refer by 1 month, biceps by 3 months, external rotation, dysplasia.

Infections of the nervous system

Kernig's and Brudzinski's signs twice.

Full notes p. 317

Asked: 2 of 20 papers, both 3 marks: Kernig's and Brudzinski's signs (W19, S21). No long case; expect TBM complications or PT after meningitis as practice questions.

Meningitis: definition and types

Inflammation of the meninges, usually infection. **Bacterial** (most serious): group B streptococcus in newborns; meningococcus, pneumococcus, Haemophilus influenzae later. **Tuberculous (TBM):** slow, over weeks; common in India; leaves **hydrocephalus, strokes, cranial nerve palsies**; stages I alert, no focal signs; II drowsy or focal; III coma. Viral (milder), fungal, parasitic.

Features and sequelae

Fever, headache, **neck stiffness**, light sensitivity, vomiting, confusion; seizures, coma, weakness if severe; non-blanching rash (meningococcal). Infants: irritability, poor feeding, drowsiness, **bulging fontanelle**.

Bacterial: about **1 in 6 die**; **1 in 5 survivors** have lasting effects: hearing loss, seizures, limb weakness, vision, speech, memory, amputations after sepsis (WHO). **TBM complications:** hydrocephalus, strokes (hemiparesis), cranial nerve palsies, vision loss, seizures, spinal arachnoiditis, cognitive and behaviour problems.

Kernig's and Brudzinski's signs (3m W19, S21)

Kernig's	Supine, **hip and knee to 90°**, try to straighten the knee → **pain and resistance** in back of thigh stop extension
Brudzinski's neck	Passive **neck flexion**, chin to chest → hips and knees flex involuntarily
Brudzinski's contralateral leg	Flex one hip and knee → other leg flexes

Meaning: meningeal irritation (meningitis, subarachnoid haemorrhage); stretch on inflamed roots. **Limit:** positive in only about **5%** of adults with meningitis (Thomas 2002): a negative sign never rules it out; neck stiffness often absent.

Tests: lumbar puncture (cells, protein, sugar, organisms), blood cultures, CT or MRI first when needed. **Hearing test for every child** after bacterial meningitis.

Medical; encephalitis; others

Urgent antibiotics (steroids in some); anti-TB drugs for months; shunt for hydrocephalus.

Encephalitis: brain inflammation, usually viral: fever, confusion, behaviour change, seizures, coma, focal weakness. **Japanese encephalitis: Culex** mosquitoes; 1 in 250 infections severe; up to 30% with symptoms die; **20–30%** of survivors keep problems (WHO 2024); vaccine in endemic districts. After-effects: parkinsonism, dystonia, spasticity, cognitive and behaviour problems, epilepsy. PT as for brain injury (p. 13).

Others: neurocysticercosis (seizures in India); brain abscess; Pott's spine, transverse myelitis (spinal cord (p. 15)); GBS, polio, leprosy, HIV.

PT after meningitis (ICF: hemiparesis, ataxia, hearing loss; school, play)

- **Acute, drowsy or ICU: head up about 30°**, turning, chest care, gentle ROM, foot and hand splints: no rise in brain pressure.
- **Weakness, spasticity:** task practice, strengthening, stretching, positioning, orthoses (as stroke or CP).
- **Lost milestones:** developmental training in sequence, play-based.
- **Ataxia:** balance and coordination training.
- **Hearing loss, attention:** face the child, gestures, demonstration, short sessions; hearing aids.

Precautions

Infection control (masks, hand hygiene) as the team directs; **avoid neck flexion and head-down** while brain pressure may be high; watch seizures; after TBM watch shunt problems and new weakness (stroke, hydrocephalus).

Exam plan

3 marks: Kernig's method + response; Brudzinski's method + response; meaning + low-sensitivity line. **Draw:** Kernig's (hip 90°, knee being straightened); Brudzinski's neck sign. **Underline:** meningeal irritation, Kernig's, Brudzinski's, low sensitivity, TBM, hydrocephalus, hearing test, Japanese encephalitis.

Post-polio residual paralysis

PPRP in 4 papers.

Full notes p. 322

Asked: 4 of 20 papers; no long case. **7 marks:** PT for PPRP (W18); define PPRP + any 3 deformities (W22); features and PT of PPRP complications (S25). **3 marks:** what is PPRP (S24).

Definition and stages (3m S24)

Polio: poliovirus destroys **anterior horn cells** → sudden, **asymmetrical flaccid paralysis**, **normal sensation**. **PPRP:** permanent weakness, wasting, deformity once recovery stops, about **2 years** after the illness. Not post-polio syndrome (new problems after **at least 15** stable years). Faecal–oral, children under 5; 1 in 200 infections paralyse. India polio-free since 2014.

Acute (weeks)	Fever, painful spasm, paralysis peaks in days → **neutral positioning, no frog position**, splints, warmth, gentle passive movement, chest care
Recovery (to 2 years)	Neurons sprout, most in 6 months → muscle chart, graded re-education, orthoses, stretching
Residual (after 2 years)	No recovery; imbalance and growth → deformity, shortening

Pathophysiology and features

Patchy: **tibialis anterior, quadriceps, deltoid, opponens** hit hardest. Sprouted large motor units may fail later. Strong vs weak muscle + gravity → contracture; flail joints; short, thin, **cold bluish** limb. Legs more than arms; reflexes lost; bladder normal. **Gaits: hand-on-thigh** (quads), Trendelenburg, backward lurch, high-stepping, short-limb.

Deformities (7m W22: any three)

Hip flexion, abduction, ER	**Tight iliotibial band (ITB)** (frog position); Thomas, Ober +; pelvic obliquity, lumbar scoliosis → prone lying, stretch, release
Knee flexion; valgus	Tight hamstrings, ITB; crouch → serial casts, release; valgus + tibial ER → Yount's release
Genu recurvatum	Weak quads, knee locked back (+ equinus); leave mild; **over about 15°** brace or osteotomy
Equinus	Weak tibialis anterior, working calf; drop, tripping → stretch, AFO, transfer
Calcaneus	Weak calf; heel walking → AFO, transfer or fusion
Scoliosis; arm	Weak trunk, obliquity; flail shoulder, no opposition

Assessment and surgery

Muscle chart (MRC 0–5, both sides, dated); Thomas, **Ober's**, popliteal angle, recurvatum angle; true and apparent length; scoliosis; gait, 6MWT. **Surgery after 2 years:** release; **tendon transfer only if donor grade 4–5** (loses about one grade); osteotomy, arthrodesis; shoe raise (ignore under 2 cm); spinal fusion.

PT for PPRP (7m W18)

- **Grade 3+:** progressive resisted, **8–12 reps, 2–3 days a week**, short of fatigue. **Grade 1–2:** assisted, gravity-eliminated, hydrotherapy.
- **Stretch** ITB, hip flexors, hamstrings, calf; **daily prone lying**; serial casts, night splints.
- **Orthoses:** AFO (leaf spring) for drop; **floor-reaction orthosis** if quads grade 3 or less, no deformity; **KAFO, drop-lock** if hip weak too; shoe raise over 2 cm; crutches.
- Gait, stairs, transfers; tendon transfer: pre-strengthen donor, then ask for its old action, biofeedback; AFO about 6 months after foot transfers.
- **Current:** leave mild recurvatum or equinus that stabilises a weak knee.

Complications (7m S25) and precautions

Contractures (stretch, casts, orthoses); instability (strengthen, floor-reaction or KAFO); length difference (shoe raise); scoliosis, obliquity (trunk work, seating, breathing); **osteoporosis, femur fractures** (weight bearing, falls prevention, gentle force); overuse pain, arthritis (joint protection, crutch height, pacing); cold limb (warmth).

Post-polio syndrome (Halstead): confirmed paralytic polio; recovery; **15+ stable years**; 2+ new problems (**new weakness**, fatigue, pain, wasting, cold intolerance); no other cause. **EFNS:** supervised aerobic and isometric training safe (Level B); warm water. **Take care:** no exercise to exhaustion (reduce if fatigue lasts to next day); skin under orthoses; refer new weakness, breathing or swallowing problems.

Exam plan

7 marks: definition and three stages (1–2) → features, deformities or complications (2–3: cause, signs, treatment) → PT: chart, strength by grade, stretch, orthoses, gait, surgery, post-polio syndrome (3–4). **Draw:** tight ITB chain; stage timeline; muscle chart. **Underline:** anterior horn cells, sensation normal, after 2 years, hand-on-thigh, floor-reaction orthosis.

Duchenne muscular dystrophy and other myopathies

6 long cases and 7 seven-mark questions.

Full notes p. 334

Asked: a DMD long case in 6 papers (ICF + PT W21, S20 with parent counselling; ICF + goals W20; home care + impairments + short-term plan W16; investigations + goals + PT S16; wheelchair prescription + goals S26). **7 marks:** PT for DMD (S16, S24); complications (S18) and their management (W24); principles of PT (W22); features and PT (S25); Gowers' sign and gait deviations (S26). **3 marks:** Gowers' sign (S16, S24).

Definition and types

DMD: X-linked recessive; **dystrophin absent**, so fibres are damaged by their own contractions and replaced by fat and fibrous tissue; progressive, mainly **proximal** weakness from early childhood, later breathing and heart. Boys, about 16–20 per 100,000 male births; signs at **2–5 years**. Exon deletion; carrier mother → half of sons affected, half of daughters carriers; some new mutations. **Maternal** cousin or uncle (W20) is a clue.

Becker	Dystrophin reduced; milder, later; many walk into adult life
Limb-girdle	Both sexes; shoulder and pelvic girdle
Facioscapulohumeral	Face, scapular winging, upper arms; asymmetrical
Myotonic	Adults; slow grip release, distal weakness, cataracts, heart conduction

Stages (2018): diagnosis → **early ambulatory** (Gowers', waddling, toe walking; climbs stairs) → late ambulatory (loses stairs, floor rise) → early non-ambulatory (self-propels; scoliosis starts) → late non-ambulatory (arm, breathing, heart).

Pathophysiology to impairments

- No dystrophin → membrane tears in contraction, especially **eccentric** → calcium floods in → fibre dies → fat and fibrosis.
- **Pseudohypertrophy:** calves big but weak (fat and fibrous tissue).
- **Proximal first** (hip extensors, quadriceps, shoulder girdle) → floor rise, stairs, running fail.
- Contractures: **plantarflexors, iliotibial band, hip flexors** first; later knees, elbows, wrists.
- After walking is lost: **scoliosis**, restrictive lung disease; **cardiomyopathy**.

Why no eccentric or heavy resistance: lengthening and high loads stress the fragile membrane most, causing more damage than repair; swimming, cycling keep fitness safely.

Gowers' sign (3m S16, S24) and gait (7m S26)

Features: late walking, falls, cannot run, jump, climb stairs; toe walking; big calves; learning or speech delay in some.

Gowers' sign: turns **face down** → hands and knees → hands and feet, **hips high** → **walks hands up the knees and thighs** → pushes trunk upright. Weak **hip and knee extensors**: proximal lower-limb weakness; typical of DMD, also other myopathies and SMA.

- **Waddling (Trendelenburg):** pelvis drops, trunk sways over stance leg (weak abductors).
- **Lumbar lordosis**, shoulders back: keeps line of gravity behind weak hip extensors.
- Wide base; **toe walking**, no heel strike (tight calves, weak DF); knee locked back; arms behind trunk; slow short steps; falls.

Investigations (S16) and PT assessment

CK very high from early life; **genetic testing** (deletion and duplication first, then sequencing); muscle biopsy (absent dystrophin) only if unclear; EMG myopathic; ECG, echo; FVC, peak cough flow, mouth pressures; spine X-ray, bone density.

PT: maternal family history, steroids, home (floor living, stairs, toilet); **Gowers' time**, 10 m walk or run, four-stair climb, 6MWT; **North Star Ambulatory Assessment**, Performance of the Upper Limb; MMT; ROM (DF knee straight, Ober, Thomas, hamstrings); lordosis, scoliosis; FVC, cough. Repeat **every 6 months** to stage.

Medical management

Steroids before major decline: prednisone or prednisolone **0.75 mg/kg a day** or deflazacort **0.9 mg/kg a day**; mean age at losing walking 10.0 → 11.2 (prednisone) → 13.9 years (deflazacort). **Never stop suddenly** (adrenal crisis); watch weight, bones, behaviour, growth. Exon skipping, gene therapy in a few countries. Achilles lengthening in selected walkers; posterior spinal fusion after walking is lost.

ICF (S20: 5-year-old, falls while playing, difficulty rising, big calves)

- **Structure/function:** proximal weakness (hip extensors, abductors, quads, shoulder girdle); calf pseudohypertrophy, tight calves and iliotibial bands; lordosis, waddling, toe walking; poor balance, endurance.
- **Activity:** falls; Gowers' to rise; cannot run, jump, climb stairs like peers.
- **Participation:** left out of active play; slower at school.
- **Environment:** concerned parents who came early, school (facilitators); stairs, floor sitting, little knowledge (barriers). **Personal:** 5-year-old who wants to play. **Early ambulatory stage.**

Goals: short-term (4–6 weeks): parents stretch 4–6 times a week, night AFOs worn; swims or cycles without next-day exhaustion; home and school safer; walks school with rests; parents understand. Long-term (6–12 months): DF at least neutral; keeps up in short play; fewer falls, no fractures; walking kept as long as possible; genetic counselling done.

Principles and PT by stage (7m S16, W22, S24, S25)

Principles: preserve walking and range; active without overworking muscle; prevent deformity, falls; support breathing; inclusion. **Classic vs current:** no progressive resistance; **submaximal aerobic** (swimming, cycling) with rests; stop before fatigue.

- **Early ambulatory:** home stretching, night AFOs; swimming, cycling; buggy or wheelchair for long distances; fall prevention; parent education.
- **Late ambulatory:** + serial casting; **KAFOs with locked knees** to prolong standing and walking; manual wheelchair; standing frame.
- **Early non-ambulatory:** power wheelchair, custom seating; supported standing; daytime AFOs; arm stretching; breathing exercises; spine checks.
- **Late non-ambulatory:** seating, pressure care; assisted cough; arm supports; hoist.

Contracture	Home stretching of ankles, knees, hips **4–6 times a week**; later wrists, hands, neck
Ankle tightness	**Night AFOs** early (prolonged low-load stretch); serial casts
Scoliosis risk	Solid seat and back, **lateral trunk supports**, hip guides, headrest
Weak cough	Lung volume recruitment at **FVC 60% or less**; assisted cough at **FVC under 50%**, peak cough flow **under 270 L/min** or expiratory pressure under 60 cmH_2O; non-invasive ventilation as advised

Wheelchair prescription (S26: 8-year-old, rural, walks only with support)

Assess: stage, arm and trunk strength, contractures, spine, breathing; home, school, rough roads, no ramps; who pushes; budget, repairs. **Choose:** light manual chair (family pushes) while he walks indoors → **power chair** when walking is lost (tilt, recline, elevating leg rests, standing support); rural: sturdy frame, large pneumatic tyres repairable locally. **Seating:** solid seat and back, lateral supports, hip guides, pommel, headrest, pressure cushion, lap belt.

Measure: seat width hip + **5 cm**; depth buttock-to-knee – **5 cm**; height so feet clear with footplate; armrest **2.5 cm** above elbow; back high enough for the trunk. Train family; review fit every few months.

Complications (7m S18, W24) and precautions

- **Contractures**; **scoliosis** (common after walking is lost: seating, spine checks, refer early, fusion).
- **Fractures:** 20–60% limb, up to 30% vertebral; can end walking: safe home, vitamin D, calcium, get him up quickly.
- **Respiratory failure:** weak cough, night hypoventilation (morning headache): thresholds above. **Cardiomyopathy:** often silent; regular review.
- Obesity on steroids; pain, low mood.

Avoid eccentric work (jumping down, downhill running, lowering loads) and heavy resistance. **Too hard:** pain or weakness into the next day, **dark urine**. No forceful stretching; never stop steroids suddenly; watch breathlessness and colour.

Parent education and home care (3m W16; S20)
Explain honestly; not their fault. Daily stretches, night splints, skin checks; swimming, cycling; no jumping games or long walks; buggy for outings. Clear floors, stair rails, western toilet or commode, chairs he can rise from. School: ground floor, rests, adapted sport. Diet and weight on steroids. **Genetic counselling** for mother and sisters; disability certificate; support groups. **Prognosis:** walking lost ~10 years without steroids; many now live into adulthood.

Exam plan (long case)

1. Restate: age, falls, rising, calves, walking, maternal history, home and school; **name the stage**.
2. **ICF (5)** with reasoning; functional diagnosis. **Goals (3–5)**: range, safe activity, parents; long term keep walking and school.
3. **Management (7–10)**: stretching and night AFOs; submaximal aerobic; no eccentric; energy conservation; falls; KAFOs, wheelchair, seating by stage; breathing; counselling; precautions.

Draw: Gowers' sign in four steps; ICF. **Underline:** dystrophin, X-linked, pseudohypertrophy, Gowers' sign, lordosis, 4–6 times a week, night AFOs, submaximal aerobic, avoid eccentric.

Spinal muscular atrophy and hereditary neuropathies

In the syllabus; not asked on its own in 2016–2026.

Full notes p. 345

Asked: not asked on its own in 2016–2026. Expect 3 marks (types of SMA; features of CMT) or 7 marks (PT for either). See also anterior horn cell (p. 30) and neuropathies (p. 19).

SMA: definition and types

Autosomal recessive loss of anterior horn cells → **symmetrical, proximal LMN weakness**. About **94%**: deletion of both **SMN1** copies (exon 7, chromosome 5); **SMN2 copies** modify severity. About 1 in 10,000 births; 1 in 40–50 carry it.

Type	Onset; best function untreated
0	Before birth; severe weakness, breathing failure
1 (Werdnig–Hoffmann)	Before 6 months; **never sits**; most died before 2 years
2	6–18 months; **sits, never walks**
3 (Kugelberg–Welander)	After 18 months; **walks**, many lose it later
4	Adult; mild proximal weakness

SMA: features

Symmetrical weakness, low tone, legs more than arms, proximal more than distal; **absent reflexes**; **tongue fasciculations**; hand tremor (minipolymyoclonus). **Normal sensation and intelligence**; face spared. Weak intercostals, working diaphragm → **bell-shaped chest**, belly breathing, weak cough, chest infections. Scoliosis, contractures, hip subluxation, feeding difficulty (type 1).

SMA: medical

Genetic test (SMN1 deletion), newborn screening. **Disease-modifying drugs:** nusinersen (intrathecal, 2016), onasemnogene abeparvovec (one-time gene therapy, 2019), risdiplam (oral, 2020). Treated babies reach new milestones, so PT now aims at **gaining skills**, not only preventing complications. Cost and access are barriers in India.

SMA: PT by functional level (2018 standards)

Non-sitters	Positioning, supported seating, gentle ROM, splints, chest physiotherapy, airway clearance, cough assist and non-invasive ventilation as advised, developmental play
Sitters (type 2)	Seating, **powered wheelchair**, spinal support, scoliosis watch, **standing frame**, orthoses, stretching hip, knee, ankle, arm function, respiratory care, aquatic play, review gains on drugs
Walkers	Strength, endurance, balance **without exhaustion**, aquatic exercise, falls prevention, orthoses, wheelchair for distance

Precautions: no exhausting exercise; watch night-time breathing and weak cough; fragile bones, gentle handling.

HSMN (Charcot–Marie–Tooth): definition and types

Commonest inherited neuropathy (about **1 in 2500**): slowly progressive damage to the **longest** motor and sensory nerves; usually **autosomal dominant**, onset in childhood or adolescence. **CMT1 (HSMN I):** demyelinating, very slow conduction; **CMT1A (PMP22 duplication)** commonest. **CMT2 (HSMN II):** axonal, near-normal velocity, small responses.

CMT: features

Distal leg weakness and wasting (**"inverted champagne bottle"**), later hands; **pes cavus**, clawed toes; **foot drop, high-stepping gait**; ankle sprains, falls; reduced or absent ankle reflexes; **glove and stocking** sensory loss; thickened nerves in some.

CMT: PT by problem

- **Foot drop, ankle instability:** AFO (posterior leaf spring or carbon fibre); high-top supportive footwear.
- **Calf and foot tightness, cavus:** daily calf and plantar fascia stretching; night splints; surgery if severe.
- **Weakness:** moderate strengthening of hip, knee, proximal muscles; do not overwork very weak distal muscles.
- **Balance:** training with vision and a stick; home safety. **Hands:** exercises, splints, built-up handles, button hooks. **Numb feet:** daily inspection, footwear, callus care.

Exam plan

3 marks: SMA types with onset and best function + SMN1, SMN2; or CMT features. **7 marks:** SMA by level (sitter for type 2) or CMT by problem. **Contrast:** SMA anterior horn cells, symmetrical proximal, sensation normal; CMT long nerves, distal, sensory loss. **Draw:** SMA types 1–3 table. **Underline:** SMN1 deletion, SMN2 copies, tongue fasciculations, non-sitter, sitter, walker, CMT1 demyelinating, pes cavus, inverted champagne bottle, AFO.

Paediatric extrapyramidal disorders

Dyskinetic and athetoid CP cases (W25).

Full notes p. 350

Asked: 2 of 20 papers, both as dyskinetic CP: **long case** W25 (7-year-old with jerky movements, fluctuating tone, truncal flexion in sitting: functional diagnosis, goals, PT); **7 marks:** posture and movement problems in athetoid CP (S26). Full answers on cerebral palsy (p. 47). Expect 3 marks on athetosis vs chorea vs dystonia, kernicterus, Wilson's.

Types of involuntary movement

Dystonia	Sustained or intermittent contractions → **twisting**, repetitive movements or abnormal postures; worse on trying to move
Athetosis	**Slow, continuous, writhing**, mostly hands and feet
Chorea	**Brief, irregular, flowing jerks** moving between body parts
Ballismus	Large flinging of a whole limb
Tremor	Rhythmic oscillation at rest, in posture or on action
Tics	Sudden repetitive movements or sounds, can be held back for a while
Myoclonus	Sudden shock-like jerks

All from disordered **basal ganglia** circuits (with cortex, cerebellum); usually **gone in sleep**, worse with effort, emotion, excitement.

Dyskinetic CP (W25; 7m S26)

Commonest childhood extrapyramidal disorder: dystonic or choreo-athetoid movements, **fluctuating tone**, poor head and trunk control, drooling, unclear speech, often normal intelligence. Linked to **severe birth asphyxia** (deep grey matter) and **kernicterus**.

Other childhood disorders

- **Kernicterus:** very high unconjugated bilirubin damages basal ganglia, brainstem nuclei, hearing pathways → **athetosis, limited upward gaze, sensorineural hearing loss, enamel defects**; intelligence often spared. Prevent: early jaundice detection, phototherapy, exchange transfusion. PT as dyskinetic CP; check hearing early; vision and gesture.
- **Wilson's disease** (hepatolenticular degeneration): autosomal recessive copper build-up in liver and basal ganglia; tremor (**"wing-beating"**), dystonia, slurred speech, drooling, parkinsonism, poor handwriting, mood change; **Kayser–Fleischer rings**; liver disease. Lifelong copper-removing drugs or zinc; transplant; early treatment improves much. PT: stable positions, weighted or built-up utensils, task practice, balance and gait, speech and swallow referral.
- **Sydenham's chorea:** rheumatic fever, weeks to months after a **streptococcal throat** infection; school age; chorea (often one side worse), low tone, **"milkmaid's" grip**, emotional lability; settles over weeks to months; long-term antibiotics. PT: safety, rest, graded return, handwriting support, **check heart** before strenuous exercise.
- **Early-onset generalised dystonia:** starts in a leg or foot, spreads; deep brain stimulation of globus pallidus. **Dopa-responsive dystonia:** leg dystonia **worse in the evening**, better after sleep; dramatic response to low-dose **levodopa** (trial in every unexplained dystonia). **Tourette:** multiple motor tics + one vocal tic over a year; PT advice only.

PT principles for involuntary movements

- **Stable base first:** seating with **pelvic belt**, foot and trunk supports; standing frame; work where steady.
- **Midline and symmetry:** hands together, head midline, toys in front.
- **Graded sustained holding**, weight bearing through arms and legs → proximal stability.
- **Task practice** in stable positions; less effort and excitement (overflow); calm surroundings.
- Wrist and thumb splints, weighted or built-up utensils, switches, communication aids.
- Team: anticholinergics, baclofen, botulinum toxin (focal dystonia); speech and feeding; family.

Exam plan

Child with involuntary movements: name the movement and likely cause → impairments (involuntary movement, fluctuating tone, poor proximal stability, speech, feeding) → PT (stable seating, midline, sustained holding, task practice, devices, team, family). **Draw:** seated child with pelvic belt, foot support, lateral supports, tray. **Underline:** basal ganglia, dystonia, athetosis, chorea, fluctuating tone, kernicterus, Kayser–Fleischer ring, dopa-responsive dystonia, proximal stability, midline control.

Answer writing for the Neuro paper

The paper, ICF in long cases, and answer skeletons.

Full notes p. 355

Asked: every paper 2016–2026 had the same shape: **80 marks in 3 hours**; Q1 any 5 of 6 × 3; Q2 any 5 of 6 × 7; Q3 and Q4 any 1 of 2 × 15 (patient cases). The printed syllabus scheme (MCQs, 5-mark questions) was never used.

Time plan

Read and choose	Tick choices, plan both cases: 5 min
Q1: 5 × 3 = 15	~6 min each (30 min)
Q2: 5 × 7 = 35	~14 min each (70 min)
Q3, Q4: 2 × 15 = 30	~30 min each (60 min)
Revise	Underline keywords; add doses, precautions: 15 min

~2 minutes a mark; never 20 min on a 3-mark answer. Slow writer: **long cases first** (30 marks safe). Choose the case you know best, not the shorter-looking one; options usually mix UMN and LMN, or adult and paediatric.

What is asked most (2016–2026)

CP (p. 47)	Long case in 13 papers + short questions
Stroke (p. 11)	18 papers: 11 long cases, 13 seven-mark
SCI, bladder, dysreflexia	18 papers: 9 long cases, a short question almost every paper
GBS	9 long cases, often ventilated or just out of ICU
Nerves and plexus	14 papers; nerve injury classification 5 times
PD, MS, DMD, brain injury	5–7 long cases each
7-mark favourites	Vestibular, approaches, tone, motor control and learning
3-mark favourites	GCS, UMN vs LMN, MAS, ATNR, nerve injury classification, bladder types, dysreflexia

Short answers: 3 and 7 marks

3 marks (half a page): one-line definition → 3–5 points, small table or labelled sketch → one line of PT relevance. *Example (dysreflexia):* sudden dangerous BP rise in SCI **at or above T6** from a stimulus below the lesion; pounding headache, flushing and sweating above, slow pulse, blocked nose; sit up, legs down, loosen clothing, check bladder then bowel, monitor BP; commonest cause a full bladder.

7 marks (1.5–2 pages): definition (2 lines) → classification, cause or "why" (flowchart or table) → **6–8 headed points**: assessment, aims, techniques with doses, by stage or problem → labelled diagram → precautions → one closing line (outcome measure or evidence).

The 15-mark case

1. Read the split (5 + 7 + 3, 7 + 8) and give space in proportion.
2. **Underline clues:** age, job, earning role, family, home (stairs, rural), time since onset, main complaint, what the patient wants.
3. Draw the **ICF model**, then the ICF table with a **clinical reasoning** column and a one-line functional diagnosis (ICF (p. 3)).
4. Problem list by priority → **SMART goals**, short and long term, with time frames.
5. Management by phase, then by impairment: **technique, dose, reason**; precautions; education, home programme.
6. Outcome measures and discharge criteria in 1–2 lines.

UMN cases (stroke, SCI, MS, CP, brain injury): tone, motor control, task practice, balance, gait, cognition, contracture prevention. **LMN cases** (GBS, nerve injury, polio, neuropathy): protect denervated muscle and numb skin, strengthen by grade, splints, breathing in GBS, re-education with reinnervation.

Diagrams and common mistakes

Draw: ICF model (every case); motor pathway with UMN and LMN; circle of Willis; incomplete cord syndromes; dysreflexia flowchart; bladder types; brachial plexus; nerve injury classification; strength-duration curve; gait cycle; iliotibial band in polio; spina bifida types; neuromuscular junction. Simple, labelled, large; never instead of text.

Mistakes: writing all you know instead of the parts asked; goals without time frames; treatment without doses; no precautions; ignoring job, family, home; **old answers** (passive stretching alone, bed rest, no exercise in MS or myasthenia).

Checklist and last week

Before handing in: question numbers and sub-letters right; keywords underlined; headings visible; every case has ICF model, functional diagnosis, timed goals, doses.

Last week: days 1–3 Tier 1 (stroke, CP, SCI, GBS, nerves, PD, MS, DMD, brain injury): ICF, goals, skeletons; days 4–5 seven-mark favourites (vestibular, approaches, tone, motor learning, Bell's palsy, spina bifida, MND); day 6 three-mark favourites; day 7 two timed long cases + five short answers.

The Neuro practical exam: long case, short case, spots and journal

Spots: EMG and nerve conduction, orthoses, prostheses, scales.

Full notes p. 358

Practical exam: 80 marks + 20 internal assessment. Long case 45, short case 2 × 10, spots 5 × 2, journal 5; marked in cognitive, psychomotor and affective domains. (Terminal and prelim: spots 5 × 3 = 15, journal not marked.)

Marking scheme

Long case 45	(a) subjective and physical examination 10; (b) evaluation and PT diagnosis (ICF) 10; (c) plan of care and goals 10; (d) **demonstrate one test and one treatment** on the patient 15
Short case 20	Two PT intervention skills, 10 each
Spots 10	5 × 2, **3 min each**: EMG and nerve conduction, orthoses, prostheses, assessment tools and scales
Journal 5	Assessment, evaluation, ICF diagnosis, prognosis, intervention, outcome

Long case proforma

Subjective: demographics, dominance, **occupation**, family, home (stairs, **toilet type**), referral; chief complaints in the patient's words; HPI (onset, progression, investigations, treatment); past, personal, family, social history; drugs; red flags; **the patient's own goals**.

Objective: vitals, GCS, posture, deformity, skin, aids; higher functions, cranial nerves, sensation, tone (MAS), power (MRC or voluntary control), reflexes, coordination, balance (Berg, TUG), gait, function (Barthel, FIM). **Condition-specific:** ISNCSCI (SCI), Hoehn and Yahr (PD), GMFCS (CP), muscle chart (LMN). **Then:** investigations → ICF table and functional diagnosis → problems → goals → plan → outcome measures.

Presenting (3–5 min)

1. One sentence: "Mr A, 54, bank manager, right MCA infarct 2 weeks ago, now..."
2. Key positive findings **in ICF order**: body functions and structures, activities, participation, environmental and personal factors.
3. Functional diagnosis in one line; 2–3 priority problems with short- and long-term goals.
4. Plan → **demonstrate** the test and treatment (part d, 15 marks): explain, position, perform, check the response.

Match them to the main problem: **MAS + sit-to-stand training** (stroke); **ISNCSCI key muscles + sliding-board transfer** (paraplegia).

Short case: 2 skills × 10

Skills: sensory testing and re-education; muscle testing or voluntary control and re-education; electrical modalities (electrotherapy (p. 34)); tone management, facilitation and inhibition (tone (p. 37)); posture; transfers; functional re-education; gait (gait (p. 40)); coordination and balance (balance (p. 38)); fitness; outcome measures, QoL questionnaires.

Steps: introduce, explain, **consent**; check precautions (BP, skin, pain) → position patient and self; drape; **guard** → clear commands, feedback, show progression → re-check response; state **dose** (sets, repetitions, frequency) and why.

Spots: what to say

EMG, nerve conduction	Normal or not; neurogenic or myopathic; demyelinating or axonal; likely condition (slow conduction, absent F-waves = GBS); PT meaning
Orthosis	Name (AFO type, KAFO, cock-up, knuckle-bender), parts, principle, indication, one precaution (orthoses (p. 43))
Prosthesis	Level; socket, suspension, pylon, foot; indication (diabetic foot); one gait problem
Tool	Reflex hammer, **128 Hz** tuning fork, two-point discriminator, goniometer: what it tests, how
Scale	Name, measures, scoring, interpretation (GCS, MAS, ASIA, Berg, Hoehn and Yahr, Barthel)

Journal, affective domain, last two weeks

Journal: 8 cases with ICF: 4 UMN (stroke, SCI, brain injury, degenerative or demyelinating), 2 LMN (nerve or plexus, GBS), 2 paediatric (CP, myopathy, meningocele); measures at start and end; neat, signed, up to date (any case may be asked).

Affective domain: introduction, consent, dignity, draping (caring (p. 42)); safety: guarding, gait belt, brakes on, BP and skin, stop for warning signs; hand hygiene; tidy equipment; thank the patient; "I would check with the team" rather than guessing.

Last two weeks: proforma on classmates in **under 20 min**; present stroke, SCI, CP aloud in ICF order; practise each skill with commands and doses aloud; spots with real reports, orthoses, scales; finish the journal.

Abbreviations

1RM: One-repetition maximum
6MWT: Six-minute walk test
ABG: Arterial blood gas
ACA: Anterior cerebral artery
AD: Autogenic drainage
ADA: American Diabetes Association
ADIP: Assistance to Disabled Persons for purchase of aids and appliances (scheme)
ADL: Activities of daily living
AF: Atrial fibrillation
AFO: Ankle–foot orthosis
AHA: American Heart Association
ALIMCO: Artificial Limbs Manufacturing Corporation of India
ALS: Amyotrophic lateral sclerosis
APTA: American Physical Therapy Association
ASA: American Stroke Association
ASIA: American Spinal Injury Association
ATNR: Asymmetrical tonic neck reflex
AV: Atrioventricular
BP: Blood pressure
BPPV: Benign paroxysmal positional vertigo
CFCS: Communication Function Classification System
CIDP: Chronic inflammatory demyelinating polyneuropathy
CIMT: Constraint-induced movement therapy
CK: Creatine kinase
CMT: Charcot–Marie–Tooth disease
CNS: Central nervous system
COMT: Catechol-O-methyltransferase
CP: Cerebral palsy
CPP: Cerebral perfusion pressure
CRPS: Complex regional pain syndrome
CSF: Cerebrospinal fluid
CT: Computed tomography
D2: Diagonal 2 (PNF pattern)
DASH: Disabilities of the Arm, Shoulder and Hand questionnaire
DGI: Dynamic Gait Index
DMD: Duchenne muscular dystrophy
DVT: Deep vein thrombosis
EAU: European Association of Urology
ECG: Electrocardiogram
EDSS: Expanded Disability Status Scale
EEG: Electroencephalography
EFNS: European Federation of Neurological Societies
EMG: Electromyography
EQ-5D: EuroQol five-dimension questionnaire
ER: External rotation
FES: Functional electrical stimulation
FIM: Functional Independence Measure
FVC: Forced vital capacity
GBS: Guillain–Barré syndrome
GCS: Glasgow Coma Scale
GMFCS: Gross Motor Function Classification System
HbA1c: Glycated haemoglobin (3-month blood sugar)
HINTS: Head impulse, nystagmus, test of skew (bedside stroke check in acute vertigo)
HIV: Human immunodeficiency virus
HKAFO: Hip–knee–ankle–foot orthosis
HR: Heart rate
HRR: Heart-rate reserve
HSMN: Hereditary sensory and motor neuropathy
ICF: International Classification of Functioning, Disability and Health
ICP: Intracranial pressure
ICU: Intensive care unit
IMT: Inspiratory muscle training
IP: Interphalangeal
IQ: Intelligence quotient
ISNCSCI: International Standards for Neurological Classification of Spinal Cord Injury
IVIG: Intravenous immunoglobulin
IWGDF: International Working Group on the Diabetic Foot
KAFO: Knee–ankle–foot orthosis
LAQ: Long-answer question
LMN: Lower motor neuron
LOPS: Loss of protective sensation
LSVT: Lee Silverman Voice Treatment
MAO-B: Monoamine oxidase B
MAP: Mean arterial pressure
MAS: Modified Ashworth Scale
MCA: Middle cerebral artery
MCID: Minimal clinically important difference
MCP: Metacarpophalangeal
MDS: International Parkinson and Movement Disorder Society
MDS-UPDRS: Movement Disorder Society Unified Parkinson's Disease Rating Scale
MGFA: Myasthenia Gravis Foundation of America
MIP: Maximal inspiratory pressure
MMSE: Mini-Mental State Examination
MND: Motor neuron disease
MoCA: Montreal Cognitive Assessment
MRC: Medical Research Council
MRI: Magnetic resonance imaging
MRP: Motor Relearning Programme
mRS: Modified Rankin Scale
MS: Multiple sclerosis
NCS: Nerve conduction studies
NCV: Nerve conduction velocity
NDT: Neurodevelopmental treatment (Bobath)
NICE: National Institute for Health and Care Excellence (UK)
NIHSS: National Institutes of Health Stroke Scale
NIV: Non-invasive ventilation
NMES: Neuromuscular electrical stimulation
PAD: Peripheral arterial disease
PCA: Posterior cerebral artery
PD: Parkinson's disease
PDQ-39: Parkinson's Disease Questionnaire (39 items)
PFMT: Pelvic floor muscle training
PNF: Proprioceptive neuromuscular facilitation
PPRP: Post-polio residual paralysis
PT: Physiotherapy
QoL: Quality of life
RHD: Rheumatic heart disease
ROM: Range of motion
RPE: Rating of perceived exertion (Borg scale)
RR: Respiratory rate
SCI: Spinal cord injury
SF-36: Short Form 36 health survey
SMA: Spinal muscular atrophy
SMART: Specific, measurable, achievable, relevant, time-bound
SOL: Space-occupying lesion
SpO_2: Oxygen saturation by pulse oximeter
STNR: Symmetrical tonic neck reflex
STREAM: Stroke Rehabilitation Assessment of Movement
TB: Tuberculosis
TBI: Traumatic brain injury
TENS: Transcutaneous electrical nerve stimulation
TIA: Transient ischaemic attack
TUG: Timed Up and Go test
UK: United Kingdom
UMN: Upper motor neuron
UPDRS: Unified Parkinson's Disease Rating Scale
UTI: Urinary tract infection
VC: Vital capacity
VP: Ventriculoperitoneal (shunt)
WHO: World Health Organization